

PRODUCTION AI DICTIONARY

Agentic AI Slang: The Complete Dictionary

**1600+ Production Terms Across LLM Optimization, Failure Modes
& Agent Systems - 2026 Edition**

A clear, interview-ready glossary for builders shipping reliable AI agents in production.



LLM OPTIMIZATION • FAILURE MODES • AGENT SYSTEMS • PRODUCTION TERMS

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Published by AI Engineering Insider.

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Preface

Shipping reliable AI in production is harder than it looks. A model can score well on every benchmark and still hallucinate in a live demo. A retrieval pipeline can return high-similarity documents that are semantically wrong. An agent can execute every tool call correctly and still fail the task. The gap between a working prototype and a production-grade system is almost always a terminology gap—engineers who cannot name a failure cannot fix it, and teams who cannot agree on definitions cannot design around them.

This reference closes that gap. It brings together three complete glossaries in a single volume: 261 LLM optimization terms covering pre-training, fine-tuning, quantization, inference, and efficient serving; more than 400 AI failure modes drawn from production incidents across data pipelines, model training, retrieval, agent systems, and deployment; and over 1,000 agentic AI terms spanning agent architectures, memory, tool use, orchestration, evaluation, observability, security, MLOps, LLMOps, NLP, search, and classical AI foundations.

Each card is structured for immediate use. Optimization terms carry a precise definition and a real-world purpose statement so you can explain the technique in a design review, not just recognize it. Failure terms include a direct solution so you can act during an incident, not just diagnose after. Agentic AI terms are organized by category so you can read a full section as a mental model or jump to a single definition as a quick reference.

The layout is built for engineers who read under pressure. Every category opens on a fresh page, the table of contents links directly to each section, and the master index at the back maps every term to its definition page. Whether you are preparing for a system design interview, debugging an agent in staging, or reviewing a colleague's architecture proposal, this volume is meant to be the reference you reach for first.

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Mixed Precision Training

PRE-TRAINING OPTIMIZATION

Definition: Uses BF16/FP16 for the forward pass and FP32 master weights for gradient updates.

Purpose: Cuts GPU memory by 40% and doubles throughput on tensor cores with negligible accuracy cost.

Gradient Checkpointing

PRE-TRAINING OPTIMIZATION

Definition: Discards intermediate activations during forward pass and recomputes them on demand during backprop.

Purpose: Trades 33% extra compute for a 4–10x reduction in activation memory. Essential for large models.

Gradient Accumulation

PRE-TRAINING OPTIMIZATION

Definition: Defers the optimizer step across N mini-batches, summing gradients before updating weights.

Purpose: Emulates large batch sizes impossible to fit in VRAM without extra hardware or communication overhead.

Gradient Clipping

PRE-TRAINING OPTIMIZATION

Definition: Rescales the gradient vector when its L2 norm exceeds a fixed threshold, typically 1.0.

Purpose: The single most reliable guard against loss spikes and NaN explosions in deep transformer training.

ZeRO Optimizer

PRE-TRAINING OPTIMIZATION

Definition: Partitions optimizer states (Stage 1), gradients (Stage 2), and parameters (Stage 3) across GPU ranks.

Purpose: A 7B model needing 112GB in single-GPU training fits on 4xA100s with ZeRO-3. Core of DeepSpeed.

Tensor Parallelism

PRE-TRAINING OPTIMIZATION

Definition: Splits weight matrices column- and row-wise across devices using Megatron-style sharding.

Purpose: Allows individual layers wider than a single GPU's memory capacity; low communication latency.

Pipeline Parallelism

PRE-TRAINING OPTIMIZATION

Definition: Slices the layer stack into stages on different GPU groups; micro-batches fill the pipeline.

Purpose: Scales model depth across many nodes. Micro-batch interleaving hides inter-stage bubble latency.

Data Parallelism

PRE-TRAINING OPTIMIZATION

Definition: Each GPU holds a full model replica and processes a distinct data shard; gradients are all-reduced.

Purpose: The default distributed training strategy for models that fit on one device. Simplest to implement.

FSDP

PRE-TRAINING OPTIMIZATION

Definition: Gathers weights for the current layer's forward/backward, then immediately shards and discards them.

Purpose: PyTorch-native ZeRO-3 without DeepSpeed dependency. Production standard for large-model training.

Flash Attention

PRE-TRAINING OPTIMIZATION

Definition: Fuses $Q \times K$ softmax and $\text{softmax} \times V$ into tiled SRAM kernels, never writing the full $N \times N$ matrix to HBM.

Purpose: Makes 100K-token training feasible on A100s. The most impactful single kernel in modern LLM training.

Flash Attention 2/3

PRE-TRAINING OPTIMIZATION

Definition: FA2 rewrites inner loops for better warp occupancy; FA3 adds FP8 support and async pipelining for H100s.

Purpose: Pushes training MFU from ~35% to 55%+. The current standard efficient attention implementation.

Ring Attention

PRE-TRAINING OPTIMIZATION

Definition: Partitions the sequence dimension across GPUs in a ring; each processes local chunks while KV blocks circulate.

Purpose: Enables million-token training contexts by distributing the attention memory across the full cluster.

Sparse Attention

PRE-TRAINING OPTIMIZATION

Definition: Restricts each token's receptive field to structured patterns: local windows, strided columns, or random blocks.

Purpose: Drops attention from $O(n^2)$ to $O(n\sqrt{n})$ or better; essential for very long sequence pre-training.

Linear Attention

PRE-TRAINING OPTIMIZATION

Definition: Replaces the softmax kernel with a feature map that factorizes attention into sequential key-value products.

Purpose: True $O(n)$ complexity. Expressiveness trade-off vs. softmax; used in hybrid architectures for long

RoPE

PRE-TRAINING OPTIMIZATION

Definition: Multiplies query and key vectors by rotation matrices whose angle is proportional to token position.

Purpose: Relative positional signal emerges from the dot product naturally; the foundation for all context extension.

ALiBi

PRE-TRAINING OPTIMIZATION

Definition: Subtracts a fixed linear bias from attention logits based on token distance; no position embedding params.

Purpose: Models trained with ALiBi extrapolate gracefully beyond training context length at zero inference cost.

AdamW

PRE-TRAINING OPTIMIZATION

Definition: Adam optimizer with decoupled L2 weight decay that correctly avoids interacting with adaptive scaling.

Purpose: The universal baseline optimizer for every LLM training run. Default in PyTorch, Transformers, and JAX.

Muon Optimizer

PRE-TRAINING OPTIMIZATION

Definition: Applies Nesterov momentum in the orthogonalized gradient space using Newton-Schulz iteration.

Purpose: Achieves faster loss decrease per step than AdamW on language model pre-training; emerging alternative.

Lion Optimizer

PRE-TRAINING OPTIMIZATION

Definition: Updates weights using only the sign of the Adam-style momentum term, discarding magnitude information.

Purpose: Requires 3x less optimizer state memory than AdamW while matching convergence on large-scale runs.

Sophia Optimizer

PRE-TRAINING OPTIMIZATION

Definition: Estimates the diagonal Hessian using Gauss-Newton-Bartlett samples to scale each parameter's update.

Purpose: Claims 2x wall-clock speedup over AdamW at equivalent loss; better curvature-aware scaling.

LR Warmup

PRE-TRAINING OPTIMIZATION

Definition: Linearly increases the learning rate from near-zero to its target value over the first 1-2% of training steps.

Purpose: Prevents large initial gradient magnitudes from destabilizing embedding and first-layer weights early on.

Cosine Annealing

PRE-TRAINING OPTIMIZATION

Definition: Decays the learning rate following a half-cosine curve from peak to a small final value over training.

Purpose: Smooth monotonic decay outperforms step schedules on final perplexity at equivalent compute budget.

Warmup-Stable-Decay (WSD)

PRE-TRAINING OPTIMIZATION

Definition: Warms up, holds constant for most of training, then rapidly decays in the final phase.

Purpose: Mid-training checkpoints can be continued cheaply without re-running the decay schedule from scratch.

Curriculum Learning

PRE-TRAINING OPTIMIZATION

Definition: Ranks training examples by difficulty and exposes the model to easier data first, harder data later.

Purpose: Improves final performance 5-15% vs. random order on the same compute budget; faster early convergence.

Data Packing

PRE-TRAINING OPTIMIZATION

Definition: Concatenates multiple documents end-to-end with separator tokens to fill each context window completely.

Purpose: Eliminates padding waste; raises effective batch token utilization from 60% to 99%. Free throughput gain.

BF16 (BF16)

PRE-TRAINING OPTIMIZATION

Definition: 16-bit float with the same 8-bit exponent as FP32 but only 7 mantissa bits; rarely overflows.

Purpose: The de-facto training dtype on A100/H100; matches FP32 loss curves without the instability of FP16.

BPE Tokenizer

PRE-TRAINING OPTIMIZATION

Definition: Byte-Pair Encoding: iteratively merges the most frequent adjacent byte pairs to build a vocabulary.

Purpose: Balances sequence length vs. vocabulary coverage. GPT-4 uses 100K BPE tokens; tokenizer choice matters.

Data Deduplication

PRE-TRAINING OPTIMIZATION

Definition: Removes near-duplicate documents using MinHash LSH or n-gram Bloom filters before any training begins.

Purpose: Prevents memorization of repeated web text; dramatically improves downstream generalization quality.

Chinchilla Scaling Laws

PRE-TRAINING OPTIMIZATION

Definition: Hoffman et al. 2022: optimal compute trains a model of N parameters on 20xN tokens minimum.

Purpose: Overturned the GPT-3 era of over-parameterized undertrained models. Now the baseline for run planning.

MuP (Maximal Update Parametrization)

PRE-TRAINING OPTIMIZATION

Definition: Rescales LR, weight init, and attention logits to remain stable as model width grows to any scale.

Purpose: Enables accurate hyperparameter transfer from a tiny proxy model to a 70B+ run; saves millions in tuning.

Weight Initialization

PRE-TRAINING OPTIMIZATION

Definition: Sets the starting values of model parameters before training; He or scaled normal init is standard.

Purpose: Poor initialization causes gradient explosion in early steps; correct init is a prerequisite for convergence.

Stochastic Depth

PRE-TRAINING OPTIMIZATION

Definition: Randomly drops entire transformer layers during training with probability increasing with layer depth.

Purpose: Acts as structural regularization; improves convergence and enables faster training for very deep models.

Sequence Length Scheduling

PRE-TRAINING OPTIMIZATION

Definition: Trains on short sequences early in training and increases length progressively as training proceeds.

Purpose: Dramatically reduces early compute cost since attention scales quadratically with sequence length.

Tokenizer Fertility

PRE-TRAINING OPTIMIZATION

Definition: Measures the average number of tokens produced per word for a given tokenizer on target-language text.

Purpose: Low fertility means longer sequences and higher inference cost. Critical metric for multilingual models.

Learning Rate Finder

PRE-TRAINING OPTIMIZATION

Definition: Runs a brief range test exponentially increasing LR while recording loss to find the optimal bracket.

Purpose: Identifies the best learning rate in minutes rather than days of expensive trial-and-error training runs.

SFT (Supervised Fine-Tuning)

FINE-TUNING OPTIMIZATION

Definition: Continues training on curated (prompt, response) pairs using cross-entropy loss on completion tokens only.

Purpose: The mandatory first step before any preference-based alignment. Teaches

RLHF

FINE-TUNING OPTIMIZATION

Definition: Trains a reward model on human pairwise preferences, then uses PPO to maximize reward within a KL budget.

Purpose: The recipe behind ChatGPT and Claude's initial alignment. Quality is capped by the reward model ceiling.

RLAIF

FINE-TUNING OPTIMIZATION

Definition: Replaces human labelers with a stronger AI judge generating preference labels at scale.

Purpose: Scales feedback data collection 100-1000x vs. human labeling. Constitutional AI uses RLAIF with self-critique.

PPO

FINE-TUNING OPTIMIZATION

Definition: Clips the policy gradient ratio within $[1-\epsilon, 1+\epsilon]$ to prevent catastrophically large policy updates.

Purpose: The stability constraint that makes billion-parameter RL-from-feedback tractable; prevents reward hacking.

DPO

FINE-TUNING OPTIMIZATION

Definition: Reformulates RLHF: the optimal policy is a closed-form solution and the reward model is implicit.

Purpose: No RL loop, no reward model training. Requires only a reference model and preference pairs; simpler pipeline.

ORPO

FINE-TUNING OPTIMIZATION

Definition: Adds an odds-ratio penalty to the SFT cross-entropy loss, combining alignment and fine-tuning in one stage.

Purpose: Eliminates the reference model and halves pipeline complexity; competitive with DPO on alignment benchmarks.

KTO

FINE-TUNING OPTIMIZATION

Definition: Derived from Kahneman-Tversky utility theory; treats chosen as gains and rejected as losses asymmetrically.

Purpose: Outperforms DPO when only binary good/bad signals are available, not ranked pairs; cheaper data collection.

GRPO

FINE-TUNING OPTIMIZATION

Definition: Samples a group of responses per prompt; uses within-group normalized rewards as the advantage signal.

Purpose: Eliminates the critic model entirely. DeepSeek-RL's alignment recipe; reduces memory and training complexity.

SimPO

FINE-TUNING OPTIMIZATION

Definition: Adds a length-normalized margin to the DPO loss and removes the reference model computation entirely.

Purpose: Trains 2x faster than standard DPO with equal or better alignment benchmark scores. Simpler to implement.

IPO

FINE-TUNING OPTIMIZATION

Definition: Replaces DPO's unbounded log-ratio objective with an L2-penalized variant.

Purpose: Provably prevents over-optimization where the model exploits the preference signal rather than improving.

LoRA

FINE-TUNING OPTIMIZATION

Definition: Freezes the base model and injects trainable low-rank matrices $A \in \mathbb{R}^{d \times r}$ and $B \in \mathbb{R}^{r \times k}$ alongside weights.

Purpose: Only $r \times (d+k)$ parameters are trained instead of $d \times k$. $r=16$ covers most tasks; the standard PEFT method.

QLoRA

FINE-TUNING OPTIMIZATION

Definition: Stores the frozen base in 4-bit NF4 with double quantization while LoRA adapters remain in BF16.

Purpose: Fine-tunes 70B models on a single A100-80GB. The breakthrough that democratized LLM customization.

DoRA

FINE-TUNING OPTIMIZATION

Definition: Decomposes each weight W into magnitude and unit direction, applying LoRA updates to the direction only.

Purpose: Closes the accuracy gap between LoRA and full fine-tuning on most tasks; drop-in replacement for LoRA.

LoRA+

FINE-TUNING OPTIMIZATION

Definition: Sets the B matrix learning rate 16x higher than A, reflecting that B starts at zero while A carries signal.

Purpose: A one-line change that consistently improves LoRA convergence speed without any quality trade-off.

VeRA

FINE-TUNING OPTIMIZATION

Definition: Shares a single frozen random Gaussian matrix pair across all layers; trains only per-layer scaling vectors.

Purpose: Under 1M trainable parameters for a 7B fine-tune. Extreme efficiency with competitive downstream results.

IA3

FINE-TUNING OPTIMIZATION

Definition: Multiplies key, value, and FFN intermediate vectors by learned scaling vectors l_k, l_v, l_{ff} .

Purpose: Only 0.01% of parameters trained. Inference overhead is zero after folding scaling vectors into weights.

Prefix Tuning

FINE-TUNING OPTIMIZATION

Definition: Prepends P trainable soft vectors to each layer's K and V matrices; the base model stays completely frozen.

Purpose: Adapters are tiny $P \times d$ tensors swapped per task. No weight modification; clean compositional fine-tuning.

Prompt Tuning

FINE-TUNING OPTIMIZATION

Definition: Learns soft token embeddings prepended to the input; every other parameter is frozen.

Purpose: Nearly as expressive as fine-tuning above 10B parameters. Lightest possible PEFT; zero architecture change.

P-Tuning v2

FINE-TUNING OPTIMIZATION

Definition: Applies independent trainable prefix vectors at every transformer layer depth, not just the input.

Purpose: Narrows the gap with full fine-tuning for NLU tasks requiring deep contextualization throughout layers.

Adapter Tuning

FINE-TUNING OPTIMIZATION

Definition: Inserts small two-layer bottleneck modules (down-project, nonlinearity, up-project) after attention and FFN.

Purpose: Modular: swap adapters at inference to switch tasks without reloading the base. Original PEFT

Full Fine-Tuning

FINE-TUNING OPTIMIZATION

Definition: Updates all parameters using the same optimizer and schedule as pre-training, on task-specific data.

Purpose: Sets the performance ceiling. Justified when dataset exceeds 100K examples and compute allows retraining.

Layer Freezing

FINE-TUNING OPTIMIZATION

Definition: Keeps the first k layers frozen and fine-tunes only later layers during the adaptation process.

Purpose: Free accuracy boost: lower layers encode syntax that rarely needs task-specific adjustment. Cuts GPU time.

Alignment Tax

FINE-TUNING OPTIMIZATION

Definition: The measurable drop in general benchmark performance that occurs after safety and helpfulness training.

Purpose: Accepted as unavoidable; the goal is to minimize it. Models with low alignment tax are production-ready.

Instruction Tuning

FINE-TUNING OPTIMIZATION

Definition: Fine-tunes on diverse NLP tasks reframed as natural language instructions and structured completions.

Purpose: The bridge from raw base model to a general-purpose assistant that follows novel zero-shot instructions.

Constitutional AI (CAI)

FINE-TUNING OPTIMIZATION

Definition: Model critiques and revises its own outputs against a written constitution; revised data trains via RLAIIF.

Purpose: Reduces dependence on human red-teams. Anthropic's primary safety training approach for Claude.

SPIN (Self-Play Fine-Tuning)

FINE-TUNING OPTIMIZATION

Definition: Current model is optimized to distinguish its outputs from the older checkpoint's on human reference data.

Purpose: Self-improving loop without new human labels. Quality improves each generation of the self-play cycle.

Self-Instruct

FINE-TUNING OPTIMIZATION

Definition: Generates instruction-response pairs from a seed set using the model itself, filtering low-quality outputs.

Purpose: Bootstraps instruction datasets at massive scale. Origin of Stanford Alpaca's 52K training examples.

Rejection Sampling FT (RFT)

FINE-TUNING OPTIMIZATION

Definition: Samples N responses per prompt, keeps those above a reward threshold, and fine-tunes on accepted outputs.

Purpose: A cheap RL alternative that often matches PPO on reasoning tasks; used in LLaMA-3 instruction tuning.

Reward Model Training

FINE-TUNING OPTIMIZATION

Definition: Trains a scalar regression head on pairwise preference data to score responses given a prompt.

Purpose: The quality of the reward model is the primary bottleneck in RLHF. Reward hacking starts here.

Safety Training

FINE-TUNING OPTIMIZATION

Definition: Fine-tunes the model to refuse harmful requests, acknowledge uncertainty, and follow behavioral policies.

Purpose: Requires red-teaming datasets and careful evaluation to avoid over-refusal, which cuts helpfulness.

Multi-task Fine-Tuning

FINE-TUNING OPTIMIZATION

Definition: Trains on diverse task datasets simultaneously using a mixture of instruction-formatted examples.

Purpose: Improves zero-shot generalization and reduces task-specific over-fitting vs. single-task fine-tuning.

Chat Templates

FINE-TUNING OPTIMIZATION

Definition: Structured markup (e.g., [INST]...[/INST] or im-start/im-end) separating system, user, and assistant turns.

Purpose: Inconsistent templates between training and inference severely degrade performance. Always match exactly.

DARE

FINE-TUNING OPTIMIZATION

Definition: Randomly zeros a fraction (typically 90%) of delta weights ($W_{ft} - W_{base}$) before merging models.

Purpose: Dropout on the delta space reduces interference between adapters. Prerequisite for clean model merging.

TIES Merging

FINE-TUNING OPTIMIZATION

Definition: Trims low-magnitude deltas, resolves sign conflicts by majority vote, and sums the surviving deltas.

Purpose: Consistently outperforms simple weight averaging when merging 3+ task-specific checkpoints into one.

SLERP

FINE-TUNING OPTIMIZATION

Definition: Interpolates two weight vectors along the shortest arc of the high-dimensional unit sphere.

Purpose: Preserves geometric structure that linear interpolation distorts; smooth blending for style mixtures.

Model Soups

FINE-TUNING OPTIMIZATION

Definition: Averages the weights of multiple SFT checkpoints trained from the same base with different hyperparameters.

Purpose: Often beats any individual checkpoint at near-zero marginal cost. Free robustness improvement.

Task Arithmetic

FINE-TUNING OPTIMIZATION

Definition: Computes task vectors $\tau = W_{ft} - W_{base}$ and adds/subtracts them from the base model.

Purpose: Negating a vector removes a capability; adding two composes skills without any retraining whatsoever.

Continual Learning

FINE-TUNING OPTIMIZATION

Definition: Trains on a sequence of tasks while explicitly preserving performance on all previous tasks.

Purpose: Key techniques: elastic weight consolidation (EWC), experience replay, and adapter-based task isolation.

Catastrophic Forgetting

FINE-TUNING OPTIMIZATION

Definition: Previously learned capabilities degrade as fine-tuning overwrites shared weights in the base model.

Purpose: The defining challenge of sequential learning. PEFT methods reduce but do not eliminate this effect.

INT8 Quantization

QUANTIZATION

Definition: Maps FP16 weights to 8-bit integers with per-channel or per-tensor scale factors applied.

Purpose: Halves model memory vs FP16 with under 1% accuracy loss when properly calibrated. First step to try.

INT4 Quantization

QUANTIZATION

Definition: Packs two 4-bit values per byte using group quantization, typically with groups of 128 weights.

Purpose: Fits a 70B model in 35GB vs. 140GB at FP16. Accuracy sensitive to group size and calibration quality.

NF4 (NormalFloat4)

QUANTIZATION

Definition: 4-bit data type with bin boundaries spaced to minimize information loss for Gaussian-distributed weights.

Purpose: Achieves strictly better quantization SNR than uniform INT4. The storage format inside QLoRA.

FP8 Quantization

QUANTIZATION

Definition: Hardware-native 8-bit float in E4M3 (inference) or E5M2 (training) formats on H100/H200 GPUs.

Purpose: Doubles effective throughput vs BF16 with nearly identical convergence. The future default training dtype.

GPTQ

QUANTIZATION

Definition: Layer-wise optimal brain compression: updates remaining weights to compensate for each quantized weight.

Purpose: State-of-the-art accuracy at 3–4 bits per weight. Widely supported by vLLM, TRT-LLM, and Ollama.

AWQ

QUANTIZATION

Definition: Identifies the 1% of salient weights by activation magnitude and protects them with higher precision.

Purpose: Outperforms GPTQ at the same bit budget and quantizes 3x faster. Hardware-efficient; runs on edge GPUs.

GGUF / GGML

QUANTIZATION

Definition: llama.cpp's self-describing binary format encoding quantized weights, tokenizer, and metadata in one file.

Purpose: Supports Q2_K through Q8_0 variants. Runs on CPU, Apple Silicon, and hybrid setups universally.

bitsandbytes

QUANTIZATION

Definition: Python library wrapping CUDA kernels for 8-bit matmul and 4-bit storage with BF16 computation.

Purpose: The de-facto standard for QLoRA via HuggingFace `load_in_4bit=True`. Zero-code quantized fine-tuning.

SmoothQuant

QUANTIZATION

Definition: Transfers quantization difficulty from outlier-heavy activations to weights by multiplying by scale s .

Purpose: Enables hardware-efficient INT8×INT8 GEMM for both weights and activations simultaneously.

QuIP#

QUANTIZATION

Definition: Incoherence processing via random orthogonal transforms plus vector quantization with Hadamard codebooks.

Purpose: Near-FP16 quality at 2 bits per weight. The frontier of ultra-low-bit quantization accuracy.

AQLM

QUANTIZATION

Definition: Additive quantization: decomposes each weight vector as a sum of M learned codebook entries.

Purpose: High accuracy at extreme compression rates; best-in-class at 2 bits where scalar quantization fails.

EXL2

QUANTIZATION

Definition: Per-layer bit allocation: assigns more bits to sensitive layers and fewer to robust ones at target BPW.

Purpose: Maximizes quality at any given memory budget on NVIDIA hardware. The format powering ExLlamaV2.

PTQ (Post-Training Quantization)

QUANTIZATION

Definition: Quantizes a trained model in minutes using a small unlabeled calibration set of 128–512 samples.

Purpose: The first thing to try before committing to QAT. Zero training cost; production standard approach.

QAT (Quantization-Aware Training)

QUANTIZATION

Definition: Inserts fake quantization operators in the forward pass so gradients account for rounding error.

Purpose: Recovers 0.5–2 PPL points lost by PTQ at the cost of a full fine-tuning run. Worth it below 4 bits.

Weight-Only Quantization

QUANTIZATION

Definition: Quantizes only the weight matrices to INT4/INT8 while dequantizing to BF16 before each matmul.

Purpose: The most practical approach for memory-bound LLM decode. No activation calibration overhead at all.

Mixed-Precision Quantization

QUANTIZATION

Definition: Assigns different bit widths layer-by-layer based on per-layer sensitivity analysis scores.

Purpose: A Pareto improvement over uniform quantization. Protects outlier-heavy layers that uniform quant damages.

KV Cache Quantization

QUANTIZATION

Definition: Stores KV cache tensors in INT8 or INT4 instead of BF16 during long-context generation.

Purpose: Doubles or quadruples effective context length for a fixed memory budget. Negligible quality impact at INT8.

Calibration Dataset

QUANTIZATION

Definition: A small (128–1024 sample) representative dataset used to compute activation statistics for PTQ.

Purpose: Quality and domain of calibration data significantly impacts final quantized model accuracy. Choose wisely.

Group Quantization

QUANTIZATION

Definition: Divides a weight vector into groups of g (e.g., 64 or 128) and applies independent scale factors per group.

Purpose: Finer granularity than per-tensor quantization. Recovers most of the accuracy lost by coarse global scaling.

Block-wise Quantization

QUANTIZATION

Definition: Extends group quantization to 2D blocks of the weight matrix with local statistics per block.

Purpose: Used in NF4/QLoRA. Allows quantization statistics to adapt to local weight structure for better accuracy.

Dynamic Quantization

QUANTIZATION

Definition: Computes activation scale factors at runtime from the current batch statistics; no pre-calibration needed.

Purpose: Zero calibration overhead. Best for CPU inference where weights are quantized but compute is flexible.

Static Quantization

QUANTIZATION

Definition: Pre-computes activation scale factors offline using a representative calibration dataset.

Purpose: Faster at runtime than dynamic. Standard for GPU inference pipelines with known input distributions.

KV Cache

INFERENCE ACCELERATION

Definition: Caches key and value projections for all past tokens to avoid recomputing them at each new decode step.

Purpose: Without KV cache, decode is $O(n^2)$; with it, each step is $O(n)$. The foundational inference optimization.

Paged Attention

INFERENCE ACCELERATION

Definition: Inspired by OS virtual memory: KV cache is divided into fixed-size pages allocated on demand via a block table.

Purpose: Eliminates memory fragmentation; enables copy-on-write KV sharing in beam search. The engine behind vLLM.

MQA (Multi-Query Attention)

INFERENCE ACCELERATION

Definition: Projects queries to H heads but collapses keys and values to a single shared head across all queries.

Purpose: Reduces KV cache by $H\times$. Dramatically speeds up memory-bound decode at a small accuracy cost.

GQA (Grouped Query Attention)

INFERENCE ACCELERATION

Definition: Generalizes MQA: G groups of query heads share one K/V head each. LLaMA-3 70B uses $G=8$.

Purpose: Maintains MHA quality while cutting KV cache 8x vs full MHA. The standard in all modern frontier models.

Speculative Decoding

INFERENCE ACCELERATION

Definition: A small draft model generates K tokens; the large model verifies all K in one parallel forward pass.

Purpose: 2–3x speedup with mathematically zero quality change for any acceptance threshold $\tau=1$.

Medusa

INFERENCE ACCELERATION

Definition: Attaches K additional LM heads to the final layer; each predicts tokens 1 to K+1 steps ahead.

Purpose: Speculative decoding without a separate model. Verified via tree attention over the candidate drafts.

EAGLE / EAGLE-2

INFERENCE ACCELERATION

Definition: Draft model predicts the next hidden state (feature vector) rather than the next token directly.

Purpose: Higher acceptance rate than token-level speculation. 3–4x speedup on NVIDIA hardware in practice.

Lookahead Decoding

INFERENCE ACCELERATION

Definition: Runs a 2D Jacobi iteration in parallel: generates draft tokens and n-grams simultaneously, then verifies.

Purpose: No draft model required. Works out-of-the-box on any LLM architecture without any additional training.

Continuous Batching

INFERENCE ACCELERATION

Definition: Inserts new requests mid-batch at token boundaries, treating the KV cache pool as a dynamic resource.

Purpose: The foundational innovation that made LLM serving economically viable. Default in vLLM and TRT-LLM.

Dynamic Batching

INFERENCE ACCELERATION

Definition: Groups concurrent requests with similar lengths to minimize padding waste in GPU kernel calls.

Purpose: Reduces effective wasted compute by 20–40%. Complements continuous batching as a preprocessing step.

Chunked Prefill

INFERENCE ACCELERATION

Definition: Splits a long prompt prefill into chunks interleaved with decode steps from other requests.

Purpose: Caps memory spikes during long-prompt prefill and reduces TTFT for queued requests simultaneously.

Prefix Caching

INFERENCE ACCELERATION

Definition: Detects identical prompt prefixes across requests and reuses their computed KV cache blocks.

Purpose: 40–80% latency reduction for system-prompt-heavy deployments. Zero cost on cache hits. Use always.

RadixAttention

INFERENCE ACCELERATION

Definition: Implements prefix caching via a radix tree where each node stores a KV block; partial prefix matches share.

Purpose: Supports thousands of concurrent requests sharing system prompts with fine-grained block-level reuse.

CUDA Graphs

INFERENCE ACCELERATION

Definition: Captures an entire decode step as a static GPU graph and replays it without per-kernel launch overhead.

Purpose: Reduces per-step latency 10–20% for small batches where kernel launch dominates. Use for real-time chat.

Kernel Fusion

INFERENCE ACCELERATION

Definition: Combines multiple elementwise operations into a single GPU kernel, avoiding repeated HBM round-trips.

Purpose: Cuts memory traffic from N kernel calls to 1. Essential for memory-bandwidth-limited operations

Triton Kernels

INFERENCE ACCELERATION

Definition: Python-embedded DSL that compiles GPU kernels via LLVM with automatic tile-size tuning.

Purpose: Flash Attention's reference implementation is in Triton. Near-CUDA performance with far simpler code.

xFormers

INFERENCE ACCELERATION

Definition: Meta's library of memory-efficient attention variants and fused transformer primitives for PyTorch.

Purpose: Drop-in swapout for standard attention. Widely used in Stable Diffusion pipelines and LLaMA stacks.

TensorRT-LLM

INFERENCE ACCELERATION

Definition: NVIDIA's production inference compiler: fuses ops, selects optimal kernel tiles, generates FP8/INT8 engines.

Purpose: Delivers 2–5x throughput vs. naive inference on the same hardware. NVIDIA's official production runtime.

vLLM

INFERENCE ACCELERATION

Definition: Open-source inference server combining PagedAttention, continuous batching, and prefix caching.

Purpose: Production standard for open-model serving. Supports 40+ architectures out of the box with one command.

SGLang

INFERENCE ACCELERATION

Definition: Structured Generation Language: Python-embedded DSL for branching, multi-call, and constrained generation.

Purpose: RadixAttention makes it fastest for complex multi-turn agentic workloads. The researchers' serving choice.

Ollama

INFERENCE ACCELERATION

Definition: Wraps llama.cpp with a Docker-like model registry and REST/gRPC API for local model serving.

Purpose: One command pulls, quantizes, and serves a model locally. The entry point for developer experimentation.

ExLlamaV2

INFERENCE ACCELERATION

Definition: Hand-tuned CUDA kernels for EXL2-quantized matmuls with fused dequantization on NVIDIA GPUs.

Purpose: Fastest local inference for quantized models on consumer GPUs (RTX 3090/4090). Maximizes VRAM efficiency.

llama.cpp

INFERENCE ACCELERATION

Definition: C/C++ inference engine with AVX2/NEON SIMD kernels, Apple Metal GPU backend, and CUDA partial offload.

Purpose: Runs on everything from Raspberry Pi to data centers. The universal runtime for quantized LLM deployment.

TTFT (Time To First Token)

INFERENCE ACCELERATION

Definition: Wall-clock time from HTTP request arrival to the first streamed token byte in the response.

Purpose: The primary latency metric for interactive chat applications. Dominated by prefill compute length.

TBT (Time Between Tokens)

INFERENCE ACCELERATION

Definition: Inverse token rate during the decode phase; determined almost entirely by KV cache memory bandwidth.

Purpose: The metric that determines whether streaming output feels fluid. Improved by MQA, quantization, batching.

TPS (Tokens Per Second)

INFERENCE ACCELERATION

Definition: Total tokens generated divided by wall time; the aggregate throughput metric for serving systems.

Purpose: Measured at peak batch size for capacity planning. Per-user TPS matters most for latency SLO compliance.

Prefill Phase

INFERENCE ACCELERATION

Definition: Processes the entire prompt in one parallel forward pass with full causal attention masking.

Purpose: Compute-bound; scales with prompt length squared. Benefits most from Flash Attention and tensor parallelism.

Decode Phase

INFERENCE ACCELERATION

Definition: Generates tokens one at a time with KV cache growing at each step; fundamentally sequential.

Purpose: Memory-bandwidth-bound. The target of speculative decoding, MQA/GQA, quantization, and KV compression.

MFU (Model FLOP Utilization)

INFERENCE ACCELERATION

Definition: Actual FLOP/s divided by the GPU's theoretical peak FLOP/s; the definitive hardware efficiency KPI.

Purpose: A100 training at 50% MFU is good. Below 30% signals a communication or memory bandwidth bottleneck.

Disaggregated Prefill

INFERENCE ACCELERATION

Definition: Routes prefill to a compute-optimized GPU pool and decode to a bandwidth-optimized pool separately.

Purpose: Prevents bursty prefill from starving decode capacity. Doubles overall cluster throughput in simulation.

Attention Sink

INFERENCE ACCELERATION

Definition: First tokens receive disproportionately high attention weights regardless of content (StreamingLLM finding).

Purpose: Keeping first 4 tokens in KV cache prevents quality collapse in streaming inference. Critical for

Prompt Caching

INFERENCE ACCELERATION

Definition: Vendor-level (Anthropic, OpenAI) feature storing computed KV states server-side; charged at reduced rate.

Purpose: Reduces API costs by up to 80% for long system-prompt workloads. Direct ROI on every repeated call.

KV Cache Sharing

INFERENCE ACCELERATION

Definition: Multiple concurrent requests with the same prefix share a single KV cache copy via copy-on-write blocks.

Purpose: PagedAttention enables zero-copy sharing at block level. Essential for multi-tenant system-prompt serving.

Token Budget

INFERENCE ACCELERATION

Definition: Maximum number of new tokens the model is allowed to generate per call, set by `max_new_tokens`.

Purpose: Controls latency and prevents runaway generation. Too low truncates responses; too high inflates serving cost.

Tensor Parallel Inference

INFERENCE ACCELERATION

Definition: Shards Q/K/V/O projections and FFN layers across GPUs, all-reducing activations at layer boundaries.

Purpose: Necessary for single-request low-latency serving of models too large for one GPU. Lowest latency at scale.

FP8 Inference

INFERENCE ACCELERATION

Definition: Uses H100 FP8 Tensor Cores for matrix multiplications with BF16 residuals and normalization layers.

Purpose: Doubles effective compute throughput vs BF16. TRT-LLM and vLLM both support FP8 serving on H100.

S-LoRA

INFERENCE ACCELERATION

Definition: Extends multi-LoRA with a unified paging system for adapter weights and custom batched LoRA kernels.

Purpose: Demonstrated serving 2000+ adapters concurrently on a single A100. The scalable adapter-serving standard.

Multi-LoRA Serving

INFERENCE ACCELERATION

Definition: Maintains a pool of LoRA delta weights in GPU memory and applies the correct adapter per request.

Purpose: Serves thousands of customers from one base model without weight switching latency. Key SaaS enabler.

DeepSpeed Inference

INFERENCE ACCELERATION

Definition: Adds tensor parallelism, INT8 quantization, and fused attention kernels on top of PyTorch.

Purpose: ZeRO-Inference offloads weights to CPU NVMe for models too large for any multi-GPU VRAM configuration.

Greedy Decoding

DECODING STRATEGIES

Definition: Always picks the single highest-probability token at each step. Fully deterministic, no randomness.

Purpose: Fastest possible decoding, but provably suboptimal: repetitive loops that sampling or beam search escape.

Beam Search

DECODING STRATEGIES

Definition: Maintains k partial sequences, expanding each by all tokens and keeping the top-k at each step.

Purpose: Used in MT and summarization where quality trumps speed. Beam width k=4 is a common production default.

Temperature Sampling

DECODING STRATEGIES

Definition: Divides all logits by T before softmax. $T < 1$ sharpens the distribution; $T > 1$ flattens it.

Purpose: $T \rightarrow 0$ converges to greedy; $T = 1$ is standard sampling. The primary creativity control knob.

Top-k Sampling

DECODING STRATEGIES

Definition: Zeros out all but the k highest-probability tokens, then samples from the normalized remainder.

Purpose: k=50 is a common default. Too small causes repetition; too large allows incoherent low-probability tokens.

Top-p (Nucleus) Sampling

DECODING STRATEGIES

Definition: Constructs the smallest token set whose cumulative probability $\geq p$, then samples from it.

Purpose: Self-calibrates vocabulary size to the model's local confidence. Generally preferred over fixed top-k.

Min-p Sampling

DECODING STRATEGIES

Definition: Filters tokens whose probability is below $p_{min} \times p_{top}$ (the top token's probability).

Purpose: Adaptive and elegant: nearly greedy at high-confidence steps, diverse at uncertain steps. Smooth tradeoff.

Mirostat

DECODING STRATEGIES

Definition: Controls output quality as a feedback loop, adjusting temperature to hit a target perplexity τ in real time.

Purpose: The only sampling method that provides a direct semantic quality knob. Consistent output across contexts.

Repetition Penalty

DECODING STRATEGIES

Definition: Divides the logit of any previously generated token by a penalty scalar greater than 1.0.

Purpose: Directly reduces loops. Too aggressive causes the model to avoid necessary common words like pronouns.

Frequency Penalty

DECODING STRATEGIES

Definition: Reduces the logit of each token by its generation count times a penalty factor.

Purpose: Proportional penalty encourages diverse vocabulary; less blunt than repetition penalty for long outputs.

Presence Penalty

DECODING STRATEGIES

Definition: Reduces the logit of each token by a fixed amount if it has appeared at least once in the output.

Purpose: Promotes topic shifts and new concepts. Weaker than frequency penalty for heavily repeated tokens.

Logit Bias

DECODING STRATEGIES

Definition: Manually increments or decrements the raw logit for specific token IDs before softmax is applied.

Purpose: Used to ban offensive tokens, force JSON brackets, or guide constrained generation in production systems.

Constrained Decoding

DECODING STRATEGIES

Definition: Masks invalid tokens at each step using a finite state machine or context-free grammar parser.

Purpose: Guarantees structurally valid JSON/SQL/YAML output without post-processing. Essential for tool-calling.

CFG (Classifier-Free Guidance)

DECODING STRATEGIES

Definition: Runs conditional and unconditional passes; extrapolates: logits = uncond + s × (cond - uncond).

Purpose: Sharpens prompt adherence at cost of 2x compute per step. Borrowed from image generation models.

Epsilon Sampling

DECODING STRATEGIES

Definition: Filters out tokens with probability below an absolute threshold ϵ (e.g., 0.0001) before sampling.

Purpose: Removes statistically improbable tokens without the sharp cutoff of top-k; complements top-p well.

Typical Sampling

DECODING STRATEGIES

Definition: Selects tokens closest to the expected entropy of the distribution rather than highest probability.

Purpose: Produces more human-like output by avoiding both too-obvious and too-surprising next tokens.

Contrastive Decoding

DECODING STRATEGIES

Definition: Boosts tokens the large expert model favors over a small amateur model at each step.

Purpose: Reduces hallucination and improves factual accuracy without a reward model. Free quality improvement.

MoE (Mixture of Experts)

ARCHITECTURE OPTIMIZATION

Definition: Routes each token to a learned subset of expert FFN layers via a trainable router network.

Purpose: Decouples parameter count from compute. A 141B MoE model uses only 39B active parameters per token.

Sparse MoE

ARCHITECTURE OPTIMIZATION

Definition: Activates only top-k (typically k=2) of E expert FFNs per token; the rest are skipped.

Purpose: Mixtral 8x7B matches LLaMA-2 70B quality at 6x less compute per token. The modern efficiency architecture.

Shared Experts

ARCHITECTURE OPTIMIZATION

Definition: Keeps always-activated experts alongside the routed sparse experts (DeepSeek-V2/V3 design).

Purpose: Provides stable shared representations that prevent over-specialization in the routed expert layers.

Expert Parallelism

ARCHITECTURE OPTIMIZATION

Definition: Places each expert on a different device; tokens are dispatched across the cluster to their assigned expert.

Purpose: The parallelism strategy that makes 1000-expert MoE training feasible at scale.

Token Routing

ARCHITECTURE OPTIMIZATION

Definition: A linear projection and softmax followed by top-k selection determines which expert processes each token.

Purpose: Router quality is critical. Collapsed routing (all tokens to one expert) is a training failure mode.

Load Balancing Loss

ARCHITECTURE OPTIMIZATION

Definition: Auxiliary loss penalizing variance in expert utilization across the batch during training.

Purpose: Without it, gradient descent collapses routing to 1-2 experts. Coefficient 0.01 requires careful tuning.

Sliding Window Attention

ARCHITECTURE OPTIMIZATION

Definition: Each token attends only to a window of W neighbors. Mistral uses W=4096 with attention sinks.

Purpose: Reduces attention from $O(n^2)$ to $O(nW)$. Local context is usually sufficient for most tokens.

Longformer Attention

ARCHITECTURE OPTIMIZATION

Definition: Combines local sliding window with global tokens (CLS, special markers) that attend to all positions.

Purpose: Designed for document-level NLP. $O(n)$ complexity for most tokens with full global coverage for key ones.

SSM (State Space Models)

ARCHITECTURE OPTIMIZATION

Definition: Models sequences as linear dynamical systems: $h_t = Ah_{t-1} + Bx_t$, $y_t = Ch_t$.

Purpose: Inference is a fixed-size hidden state regardless of sequence length. Training parallelizable via convolutions.

Mamba

ARCHITECTURE OPTIMIZATION

Definition: Adds input-dependent (selective) state transitions to SSMs: A, B, C depend on the current input x_t .

Purpose: Breaks the fixed-transition limitation that prevented SSMs from matching transformers on language tasks.

RWKV

ARCHITECTURE OPTIMIZATION

Definition: Reformulates attention as a linear RNN: WKV attention computed in parallel in training, recursively in inference.

Purpose: Linear memory at inference; no KV cache growth. Competitive with transformers up to 14B parameters.

Hyena

ARCHITECTURE OPTIMIZATION

Definition: Replaces attention with a long implicit convolution defined by a small neural network over positions.

Purpose: Sub-quadratic; reaches GPT-3 quality with 20% fewer FLOPs at sequence length 2048. Emerging alternative.

RetNet

ARCHITECTURE OPTIMIZATION

Definition: Implements retention in parallel (training), recurrent (inference), and chunkwise (long-context) modes.

Purpose: Claims to unify the best properties of Transformers, RNNs, and SSMs in one elegant formulation.

MLA (Multi-Latent Attention)

ARCHITECTURE OPTIMIZATION

Definition: Compresses KV cache via a low-rank bottleneck projection before caching keys and values.

Purpose: DeepSeek-V2 achieves 93% KV cache reduction with negligible quality loss. The key to its efficiency.

YaRN

ARCHITECTURE OPTIMIZATION

Definition: Applies NTK-aware interpolation to high-frequency RoPE dims and linear interpolation to low-frequency ones.

Purpose: Extends a 4K-trained model to 128K tokens with only 1B tokens of continued pre-training. Most used method.

LongRoPE

ARCHITECTURE OPTIMIZATION

Definition: Assigns different interpolation factors to RoPE dimensions based on empirical token distance distributions.

Purpose: Outperforms uniform interpolation on tasks requiring very long-range retrieval beyond 128K tokens.

NTK-aware Scaling

ARCHITECTURE OPTIMIZATION

Definition: Increases the RoPE base frequency to $b \times n^2$ where n is the extension ratio to preserve high-frequency signal.

Purpose: The theoretical foundation for most RoPE extensions; prevents loss of short-range positional information.

Position Interpolation (PI)

ARCHITECTURE OPTIMIZATION

Definition: Maps position indices $[0, L_{new}]$ linearly to $[0, L_{train}]$ for context extension.

Purpose: Simple and effective with brief fine-tuning. The baseline vs. which all RoPE extension methods are measured.

Weight Tying

ARCHITECTURE OPTIMIZATION

Definition: Shares the token embedding matrix $E \in \mathbb{R}^{V \times d}$ with the output projection matrix.

Purpose: Reduces parameters by $V \times d$ (e.g., 400M for 100K vocab, $d=4096$). Improves embedding consistency.

SwiGLU

ARCHITECTURE OPTIMIZATION

Definition: FFN activation: $\text{FFN}(x) = (xW_1) \odot \sigma_{\text{swish}}(xV) \cdot W_2$. Input-dependent gating.

Purpose: Consistently outperforms GELU FFNs by 0.5–1 PPL point. Standard in LLaMA, PaLM, and Mistral.

GeGLU

ARCHITECTURE OPTIMIZATION

Definition: Variant of the gated linear unit using GELU as the gate activation function in the FFN.

Purpose: Marginal differences from SwiGLU in practice. Both vastly outperform plain GELU in transformer FFNs.

RMSNorm

ARCHITECTURE OPTIMIZATION

Definition: Normalizes by $\text{RMS}(x) = \sqrt{\text{mean}(x^2)}$ without computing or subtracting the mean.

Purpose: 7% faster than full LayerNorm due to the missing mean computation. Universal in modern LLMs.

Pre-LayerNorm

ARCHITECTURE OPTIMIZATION

Definition: Applies normalization to the residual stream before feeding into attention and FFN sublayers.

Purpose: More stable training than Post-LN; gradients are better conditioned at initialization. Now universal.

Mixture of Depths (MoD)

ARCHITECTURE OPTIMIZATION

Definition: Routes each token to a subset of transformer layers, skipping the rest via a learned token router.

Purpose: Token uses only 50% of layers on average; compute per token is halved with minimal PPL increase.

Cross-Attention

ARCHITECTURE OPTIMIZATION

Definition: Attention where queries come from the decoder and keys/values from the encoder output representations.

Purpose: Essential in encoder-decoder architectures like T5 and Whisper for sequence-to-sequence generation tasks.

Encoder-Decoder Architecture

ARCHITECTURE OPTIMIZATION

Definition: Separate encoder processes input; separate decoder generates output conditioned on encoder representations.

Purpose: Natural fit for translation, summarization, and ASR. T5, BART, and Whisper all use this design.

Decoder-Only Architecture

ARCHITECTURE OPTIMIZATION

Definition: A single autoregressive transformer stack predicts the next token from all previous tokens.

Purpose: GPT, LLaMA, Claude, Gemini: every frontier model is decoder-only. Scales more efficiently than enc-dec.

Embedding Dimension

ARCHITECTURE OPTIMIZATION

Definition: The width d of all hidden state vectors throughout the transformer (d_{model}).

Purpose: Scales with model size: 768 for BERT-Base, 8192 for LLaMA-3 70B. Wider = richer representations.

Attention Head Count

ARCHITECTURE OPTIMIZATION

Definition: Number of parallel attention sub-spaces H , each with dimension d/H operating independently.

Purpose: More heads capture diverse relation types. $H=64$ at $d=8192$ gives 128-dim heads, standard for large models.

FFN Ratio

ARCHITECTURE OPTIMIZATION

Definition: Ratio of FFN intermediate dimension to embedding dimension; typically $4\times$ standard, $2.7\times$ for GLU.

Purpose: Controls what fraction of parameters live in FFN vs. attention layers. 66% of params are FFN at $4\times$ ratio.

Context Length

ARCHITECTURE OPTIMIZATION

Definition: Maximum number of tokens the model processes in a single forward pass.

Purpose: Every doubling quadruples attention memory and compute at training time. The key architectural constraint.

Depth Upscaling

ARCHITECTURE OPTIMIZATION

Definition: Initializes a deeper model by duplicating layers from a trained shallower model checkpoint.

Purpose: Solar 10.7B was built from LLaMA-2 7B via depth-upscaling then fine-tuned. Cuts pre-training cost.

Rotary Embedding Variants

ARCHITECTURE OPTIMIZATION

Definition: The family of RoPE-derived position encodings: standard, NTK-aware, YaRN, LongRoPE, LlamaRoPE.

Purpose: Choosing the right variant for context extension is one of the highest-ROI architecture decisions in production.

Deep vs. Wide Trade-off

ARCHITECTURE OPTIMIZATION

Definition: At fixed parameter count, deep networks favor compositional reasoning; wide ones favor associative recall.

Purpose: Chinchilla-optimal scaling slightly favors depth. Most frontier models use 80+ layers at production scale.

Pruning

COMPRESSION TECHNIQUES

Definition: Systematically removes parameters, heads, or layers with low importance, then optionally fine-tunes.

Purpose: Structured pruning delivers real GPU speedups; unstructured needs sparse kernels for practical benefit.

Structured Pruning

COMPRESSION TECHNIQUES

Definition: Removes entire computational units: attention heads, MLP neurons, or full transformer layers.

Purpose: Reduces the dense computational graph. Achieves real latency improvements without sparse hardware support.

Unstructured Pruning

COMPRESSION TECHNIQUES

Definition: Masks individual weights to zero, creating irregular sparsity in weight matrices.

Purpose: Requires 50–80% sparsity before practical GPU speedup. More effective on CPUs with sparse BLAS support.

Magnitude Pruning

COMPRESSION TECHNIQUES

Definition: Removes the fraction of weights with the smallest absolute values below a threshold.

Purpose: Competitive at 10–30% sparsity; degrades sharply at high sparsity without subsequent retraining.

SparseGPT

COMPRESSION TECHNIQUES

Definition: Frames pruning as a sparse regression: finds weights to zero that minimize output change via inverse Hessian.

Purpose: Achieves 50% unstructured sparsity in LLMs with under 2 PPL increase. No retraining required.

Wanda

COMPRESSION TECHNIQUES

Definition: Prunes weights by score $w_{ij} \times \|x_j\|_2$ using input activation norms as importance signals.

Purpose: Rivals SparseGPT quality in a single data pass with no Hessian computation. 10x faster to apply.

Knowledge Distillation

COMPRESSION TECHNIQUES

Definition: Minimizes KL divergence between student and teacher output distributions using soft probability targets.

Purpose: Soft labels contain dark knowledge about class similarities that hard one-hot labels completely discard.

Response Distillation

COMPRESSION TECHNIQUES

Definition: Student minimizes cross-entropy against the teacher's full probability distribution, not just the argmax.

Purpose: Temperature $T=4-8$ softens the teacher distribution and amplifies the dark knowledge training signal.

Feature Distillation

COMPRESSION TECHNIQUES

Definition: Adds L2 or cosine similarity loss between student and teacher hidden states at intermediate layers.

Purpose: Provides richer gradients than output-only distillation. Particularly effective for BERT-scale compression.

Layer Dropping

COMPRESSION TECHNIQUES

Definition: Skips transformer layers during inference using a static schedule from sensitivity analysis.

Purpose: Removes 20–30% of layers with 1–2 PPL increase. Combined with fine-tuning recovers most of the loss.

Early Exit

COMPRESSION TECHNIQUES

Definition: Attaches an exit head to each layer; a confidence classifier decides whether to generate from that layer.

Purpose: Reduces average compute per token by 30–60% on easy tokens. Hard tokens still use all layers.

Head Pruning

COMPRESSION TECHNIQUES

Definition: Removes the attention heads with lowest importance scores (attention entropy or gradient saliency).

Purpose: 30–40% of heads in BERT-scale models can be removed with near-zero downstream performance impact.

FFN Pruning

COMPRESSION TECHNIQUES

Definition: Reduces the intermediate dimension d_{ff} of FFN sublayers based on neuron importance scores.

Purpose: FFN contains 66% of transformer parameters. Reducing d_{ff} from 4d to 2d cuts params by 33%.

Matrix Decomposition (SVD)

COMPRESSION TECHNIQUES

Definition: Factorizes $W \approx U \cdot \Sigma \cdot V^T$, keeping only the top- r singular values.

Purpose: Smooth accuracy-compression tradeoff; r tuned layer-by-layer based on singular value drop-off curves.

Weight Sharing

COMPRESSION TECHNIQUES

Definition: Multiple layers or sub-networks share a single weight tensor; parameters are tied rather than independent.

Purpose: ALBERT shares all 12 BERT layers; cuts parameters 12x with surprisingly competitive GLUE benchmark scores.

Depth Pruning

COMPRESSION TECHNIQUES

Definition: Removes entire transformer blocks based on block importance scores such as input-output cosine similarity.

Purpose: ShortGPT shows later layers are often near-identity functions removable without full accuracy recovery.

Lottery Ticket Hypothesis

COMPRESSION TECHNIQUES

Definition: A dense network contains a sparse subnetwork (winning ticket) that trains in isolation to match the full network.

Purpose: Theoretical foundation for pruning. Finding winning tickets at LLM scale remains an open research challenge.

OBC (Optimal Brain Compression)

COMPRESSION TECHNIQUES

Definition: Unifying framework using the layer Hessian to find optimal weight perturbations for compression.

Purpose: GPTQ and SparseGPT are both special cases of OBC. The most principled compression framework available.

Activation Sparsity

COMPRESSION TECHNIQUES

Definition: Exploits the observation that ReLU-based models have many zero activations during forward pass.

Purpose: Adding ReLU constraints to SwiGLU models via fine-tuning recovers 50%+ activation sparsity with minimal loss.

TinyLLaMA / DistilBERT

COMPRESSION TECHNIQUES

Definition: Compact distilled models trained on massive compute budgets to match larger teachers.

Purpose: TinyLLaMA (1.1B, 3T tokens) matches much larger models; shows compute-optimal small models are powerful.

Context Window Extension

CONTEXT AND MEMORY

Definition: Enabling a model to process sequences longer than its training context using post-hoc techniques.

Purpose: The most requested production capability. Every major technique trades some accuracy or compute for length.

YaRN

CONTEXT AND MEMORY

Definition: Applies NTK-aware interpolation to high-frequency RoPE dims with attention temperature correction.

Purpose: Extends Mistral 7B to 128K tokens with only 1B tokens of fine-tuning. The most adopted method.

LongLoRA

CONTEXT AND MEMORY

Definition: Combines shifted sparse attention (S^2 -Attn) during fine-tuning with dense attention at inference.

Purpose: Trains 7B models to 100K tokens for the cost of approximately 1000 GPU hours. Highly compute-efficient.

RAG (Retrieval-Augmented Generation)

CONTEXT AND MEMORY

Definition: Retrieves the top-k most relevant documents from an index and prepends them to the context.

Purpose: Shifts factual knowledge from model weights to a searchable, updateable external store. No retraining

KV Cache Compression

CONTEXT AND MEMORY

Definition: Selectively evicts or quantizes KV cache entries to reduce memory footprint during long generation.

Purpose: Enables sequences 4–8x longer than the GPU's memory would otherwise allow. Critical for long documents.

H2O (Heavy-Hitter Oracle)

CONTEXT AND MEMORY

Definition: Retains KV entries of heavy-hitter tokens (highest cumulative attention scores) and evicts the rest.

Purpose: Preserves 80–90% of generation quality at 50% KV cache reduction. State-of-the-art eviction policy.

SnapKV

CONTEXT AND MEMORY

Definition: Uses an observation window at prompt start to identify which positions will be attended to most.

Purpose: Consistent quality with H2O but more robust to diverse prompt structures. Evicts before generation starts.

StreamingLLM

CONTEXT AND MEMORY

Definition: Retains attention sinks (first 4 tokens) and a rolling window of recent tokens; discards all middle positions.

Purpose: Enables unbounded streaming generation with a fixed memory budget. Perfect for always-on deployments.

Infini-Attention

CONTEXT AND MEMORY

Definition: Splits the sequence into local segments and maintains a compressive memory of past segments via linear attention.

Purpose: Global context without global memory. Scales to arbitrarily long documents at constant memory cost.

Memory-Efficient Attention

CONTEXT AND MEMORY

Definition: Computes attention in SRAM tiles, never writing the full $N \times N$ matrix to HBM.

Purpose: The defining technique of Flash Attention and all its variants. The prerequisite for long-context training.

CPU/NVMe Offloading

CONTEXT AND MEMORY

Definition: Pages less-recently-used KV blocks or weight shards to CPU DRAM or NVMe SSDs during inference.

Purpose: Extends effective GPU memory by 8–20x at the cost of PCIe bandwidth latency. Trade speed for capacity.

Prefix Caching

CONTEXT AND MEMORY

Definition: Stores computed KV states for shared prompt prefixes and reuses them across all requests.

Purpose: A 1000-token system prompt cached server-side saves 1000 tokens of prefill per request. ROI is immediate.

Recurrent Memory Transformer

CONTEXT AND MEMORY

Definition: Appends trainable memory tokens that persist across segments; model reads and writes them like a soft KV store.

Purpose: Hybrid architecture: transformer within segments, RNN-like memory across segments. Unbounded effective context.

Memory-Augmented LLMs

CONTEXT AND MEMORY

Definition: Equips the model with an external differentiable key-value store or vector database read via attention.

Purpose: Separates knowledge retrieval from reasoning computation. Enables explicit memory update and inspection.

RoPE Frequency Scaling

CONTEXT AND MEMORY

Definition: Multiplies the RoPE base frequency by a scalar >1 to extend context beyond training length.

Purpose: Cheapest context extension: no fine-tuning for modest 2–4x extensions. Quality drops at larger ratios.

ALiBi for Long Context

CONTEXT AND MEMORY

Definition: Extends ALiBi's linear bias to lengths beyond training without any modification or fine-tuning needed.

Purpose: Free length generalization: quality degrades gracefully rather than catastrophically past training length.

Throughput

SERVING AND SYSTEMS

Definition: Total tokens generated per second across all concurrent requests under production load.

Purpose: The primary capacity metric. Maximized by batching, quantization, and caching. Determines cost per token.

Continuous Batching

SERVING AND SYSTEMS

Definition: Injects new requests into the batch at token boundaries rather than waiting for a full batch to complete.

Purpose: The innovation that made LLM serving economically viable. Default in vLLM, TRT-LLM, and SGLang.

Model Sharding

SERVING AND SYSTEMS

Definition: Partitions model weights across GPUs using tensor, pipeline, or expert parallelism strategies.

Purpose: Required for models too large for one device. Tensor parallel is lowest latency for small batch sizes.

Load Balancing

SERVING AND SYSTEMS

Definition: Distributes requests across replicas using request-length-aware routing to minimize queue depth variance.

Purpose: Naive round-robin causes 3x latency variance. Length-aware routing cuts it to under 30%. Critical at scale.

Speculative Decoding (Serving)

SERVING AND SYSTEMS

Definition: Co-located small draft model proposes tokens; large verifier checks all in one step per request.

Purpose: 2-3x lower latency for interactive workloads. Transparent to caller; identical output distribution.

Canary Deployment

SERVING AND SYSTEMS

Definition: Routes 1-5% of traffic to a new model version while the old version serves the remainder.

Purpose: Catches regression in production metrics before full rollout. Standard ML deployment practice everywhere.

Latency

SERVING AND SYSTEMS

Definition: Total wall-clock time from request arrival to completion: decomposed as TTFT + (output_tokens × TBT).

Purpose: The primary quality-of-service metric. Different workloads optimize for TTFT vs. TBT differently.

Iteration-Level Scheduling

SERVING AND SYSTEMS

Definition: Makes scheduling decisions at every decode step rather than per-request for fine-grained control.

Purpose: Enables preemption, priority, and admission control at microsecond granularity for multi-tenant serving.

Disaggregated Prefill

SERVING AND SYSTEMS

Definition: Routes prefill to a compute-optimized GPU pool and decode to a bandwidth-optimized pool.

Purpose: Prevents bursty prefill from starving decode capacity. Doubles overall cluster throughput in simulation.

Autoscaling

SERVING AND SYSTEMS

Definition: Adds or removes serving replicas based on queue depth, GPU utilization, or predicted traffic load.

Purpose: Horizontal scaling is the escape valve for throughput. Scale-to-zero for idle models eliminates idle GPU cost.

Request Queue Management

SERVING AND SYSTEMS

Definition: Prioritizes, batches, and schedules incoming requests by deadline, SLO tier, or estimated generation length.

Purpose: Sophisticated queue management prevents head-of-line blocking and cuts P99 latency significantly.

Shadow Mode

SERVING AND SYSTEMS

Definition: Runs a new model in parallel with the production model, comparing outputs without serving them to users.

Purpose: Zero-risk validation against real production traffic before any live exposure to actual end users.

SLO (Service Level Objective)

SERVING AND SYSTEMS

Definition: Contractual targets for latency, throughput, or availability (e.g., P95 TTFT < 500ms).

Purpose: SLO violation triggers autoscaling, preemption, or degraded mode. Defines the serving system's

Preemption

SERVING AND SYSTEMS

Definition: Pauses a running request (swapping its KV cache to CPU) to immediately serve a higher-priority request.

Purpose: Prevents SLO violations for premium users while maximizing GPU utilization across the full request

Model Caching

SERVING AND SYSTEMS

Definition: Keeps model weights permanently resident in GPU HBM across all requests without eviction.

Purpose: Without it, each request reloads weights from CPU/NVMe, adding seconds per call. Always enable in serving.

FP8 Serving

SERVING AND SYSTEMS

Definition: Routes matmul operations through H100 FP8 Tensor Cores with BF16 accumulation and residuals.

Purpose: Provides 2x throughput over BF16 serving with under 1% benchmark impact. TRT-LLM default for H100.

Model Versioning

SERVING AND SYSTEMS

Definition: Maintains model weight snapshots in a registry with metadata: dataset, eval scores, safety checks.

Purpose: Essential for A/B testing, rollback, and compliance auditing. The backbone of responsible production ML.

A/B Testing Models

SERVING AND SYSTEMS

Definition: Routes traffic splits to different model versions and measures downstream business metrics.

Purpose: The gold standard for production model evaluation. Benchmark scores rarely predict business metric impact.

S-LoRA

SERVING AND SYSTEMS

Definition: Pages LoRA adapter weights into GPU memory on demand with priority-based eviction across requests.

Purpose: 2000+ concurrent adapters on 4xA100s with 4ms swap latency. Enables SaaS-scale personalization.

Punica

SERVING AND SYSTEMS

Definition: Implements a segment-level BGMV kernel for heterogeneous LoRA batches in one fused kernel call.

Purpose: Processes requests with different adapters at near-native matmul efficiency. The kernel behind S-LoRA.

vLLM / SGLang

SERVING AND SYSTEMS

Definition: Leading open-source inference servers: vLLM for compatibility breadth, SGLang for structured generation speed.

Purpose: Both are production-grade. Choose based on workload: vLLM for breadth, SGLang for agentic pipelines.

Multi-LoRA Serving

SERVING AND SYSTEMS

Definition: Serves many distinct LoRA adapter variants from one shared base model using adapter-aware kernels.

Purpose: Per-user model customization at SaaS scale. The enabling technology for personalized AI products.

Perplexity (PPL)

EVALUATION AND METRICS

Definition: Exponential of the average negative log-likelihood per token over a held-out test corpus.

Purpose: Lower is better. A 1-point PPL reduction at scale represents significant quality worth substantial compute.

Roofline Model

EVALUATION AND METRICS

Definition: Plots FLOP/s vs. arithmetic intensity bounded by compute peak and memory bandwidth on a log-log chart.

Purpose: Instantly reveals whether a workload is memory-bound (decode) or compute-bound (prefill). Essential tool.

Compute-Bound Operations

EVALUATION AND METRICS

Definition: Operations where the GPU is fully compute-saturated and memory arrives faster than consumed (e.g., prefill).

Purpose: Solutions: kernel tiling (Flash Attention), kernel fusion, and tensor cores for GEMM acceleration.

Active Parameters

EVALUATION AND METRICS

Definition: Parameters actually executed per token. Critical for MoE: 141B total but only 39B active per forward pass.

Purpose: Determines true FLOPs and memory bandwidth per token. Total params misleads; active params matter.

HumanEval

EVALUATION AND METRICS

Definition: 164 Python function synthesis problems evaluated by executing unit tests against generated code.

Purpose: Pass@k: probability that at least one of k samples passes all tests. GPT-4 reaches Pass@1 ~87%.

Elo Rating

EVALUATION AND METRICS

Definition: Tournament-style ranking where models compete pairwise and ratings update based on win/loss outcomes.

Purpose: LMSYS Chatbot Arena uses Elo over millions of human votes. The most reliable quality ranking for chat.

BPW (Bits Per Weight)

EVALUATION AND METRICS

Definition: Average bits per model parameter post-quantization. FP16=16, INT8=8, INT4=4 BPW.

Purpose: The universal compression metric. Lower BPW = more models fit in fixed VRAM; quality tradeoff must be tracked.

Arithmetic Intensity

EVALUATION AND METRICS

Definition: FLOPs performed per byte of memory traffic for a given GPU operation.

Purpose: GEMM has high intensity; elementwise ops have 1 FLOP/byte. Determines position on the roofline chart.

MFU (Model FLOP Utilization)

EVALUATION AND METRICS

Definition: Ratio of measured FLOP/s to theoretical GPU peak FLOP/s during training or inference.

Purpose: A100 training at 50% MFU is good. Below 30% signals a communication or memory bandwidth bottleneck.

KV Cache Size Formula

EVALUATION AND METRICS

Definition: Grows as $2 \times \text{layers} \times \text{heads} \times \text{head_dim} \times \text{seq_len} \times \text{batch_size}$ per element.

Purpose: 70B with GQA (8 KV heads, d=128, BF16) at seq_len=32K uses 13GB per request. Plan capacity accordingly.

GSM8K

EVALUATION AND METRICS

Definition: 8500 grade-school math word problems requiring multi-step arithmetic reasoning to solve.

Purpose: Chain-of-thought prompting dramatically improves scores. Primary benchmark for measuring emergent reasoning.

Win Rate

EVALUATION AND METRICS

Definition: Fraction of head-to-head comparisons where model A is preferred over model B by human or AI judges.

Purpose: The direct metric for model comparison studies. More actionable than benchmark scores for product decisions.

VRAM Usage

EVALUATION AND METRICS

Definition: GPU HBM consumed by model weights, activations, KV cache, and optimizer states combined.

Purpose: The hard constraint governing what is deployable. Every optimization technique ultimately targets VRAM.

Memory-Bound Operations

EVALUATION AND METRICS

Definition: GPU operations stalling on HBM data, not on compute units (e.g., decode phase).

Purpose: Solutions: quantization (fewer bytes), MQA/GQA (fewer KV bytes), and KV compression techniques.

HFU (Hardware FLOP Utilization)

EVALUATION AND METRICS

Definition: MFU adjusted to account for additional FLOPs from gradient recomputation during checkpointing.

Purpose: The honest efficiency metric when gradient checkpointing is enabled. Always report HFU for training.

MMLU

EVALUATION AND METRICS

Definition: 14K multiple-choice questions across 57 academic subjects measuring breadth of world knowledge.

Purpose: GPT-4 scores 86%; human expert baseline 89%. Standard breadth benchmark; does not measure reasoning.

MT-Bench

EVALUATION AND METRICS

Definition: 80 multi-turn conversation questions judged by GPT-4 on a 1-10 scale for quality.

Purpose: Correlates well with human preferences. The standard benchmark for evaluating chat assistant quality.

ROUGE / BLEU

EVALUATION AND METRICS

Definition: ROUGE measures n-gram recall vs. references; BLEU measures n-gram precision for translation quality.

Purpose: Widely used for summarization and MT respectively. Known to correlate poorly with human judgment at high quality.

Problem Misframing

BUSINESS AND PROBLEM DEFINITION FAILURES

Definition: The team solves the wrong problem with AI, such as building a prediction model when the real issue is a broken workflow.

Solution: Clarify the business goal, user pain point, decision process, and success criteria before modeling.

Objective Misalignment

BUSINESS AND PROBLEM DEFINITION FAILURES

Definition: The model optimizes for something different from what the business actually needs.

Solution: Align model objectives with measurable product, user, safety, and business outcomes.

Metric Misalignment

BUSINESS AND PROBLEM DEFINITION FAILURES

Definition: A metric improves, but real user experience or business value does not improve.

Solution: Choose metrics that reflect user value, quality, safety, and business impact.

Success Criteria Ambiguity

BUSINESS AND PROBLEM DEFINITION FAILURES

Definition: The team does not clearly define what “good enough” means.

Solution: Set acceptance thresholds, quality gates, and launch criteria before training or deployment.

Use Case Overreach

BUSINESS AND PROBLEM DEFINITION FAILURES

Definition: AI is used for a task where rules, workflow automation, or human review would be better.

Solution: Validate whether AI is truly needed and compare it against simpler non-AI solutions.

Stakeholder Misalignment

BUSINESS AND PROBLEM DEFINITION FAILURES

Definition: Product, data, engineering, legal, security, and leadership teams disagree on the goal.

Solution: Document ownership, requirements, risks, responsibilities, and approval flow.

Requirement Drift

BUSINESS AND PROBLEM DEFINITION FAILURES

Definition: Business requirements change, but the model or pipeline is not updated.

Solution: Schedule regular requirement reviews and connect them to model and pipeline update cycles.

Risk Misclassification

BUSINESS AND PROBLEM DEFINITION FAILURES

Definition: A high-risk AI use case is treated as low-risk, or a low-risk use case is over-controlled.

Solution: Apply risk assessments early and classify use cases by impact, safety, privacy, and compliance needs.

Automation Bias

BUSINESS AND PROBLEM DEFINITION FAILURES

Definition: Humans trust AI output too much just because it came from a model.

Solution: Show confidence, evidence, warnings, uncertainty, and human review options.

Human-in-the-Loop Failure

BUSINESS AND PROBLEM DEFINITION FAILURES

Definition: The system needs human review, but the review process is missing, weak, or ignored.

Solution: Add escalation rules, reviewer queues, override mechanisms, and clear decision ownership.

Feedback Design Failure

BUSINESS AND PROBLEM DEFINITION FAILURES

Definition: The system does not collect useful feedback for improvement.

Solution: Design feedback buttons, correction workflows, review notes, and feedback-to-evaluation loops.

Decision Boundary Confusion

BUSINESS AND PROBLEM DEFINITION FAILURES

Definition: The system does not define when AI should decide, recommend, refuse, or escalate.

Solution: Create clear decision boundaries, fallback paths, and human approval rules.

Data Unavailability

DATA COLLECTION FAILURES

Definition: The needed data does not exist or cannot be accessed.

Solution: Identify alternative sources, collect new data, redesign the use case, or narrow the scope.

Data Access Failure

DATA COLLECTION FAILURES

Definition: Permissions, APIs, governance rules, or system issues block access to data.

Solution: Create approved access paths, data-sharing agreements, and secure permission workflows.

Sampling Bias

DATA COLLECTION FAILURES

Definition: The collected data does not represent the real-world population.

Solution: Improve the sampling strategy and add missing groups, regions, languages, or scenarios.

Selection Bias

DATA COLLECTION FAILURES

Definition: Some examples are more likely to appear in the dataset than others.

Solution: Audit how data was collected and rebalance or supplement the dataset.

Survivorship Bias

DATA COLLECTION FAILURES

Definition: The dataset only includes successful or surviving examples and hides failures.

Solution: Intentionally collect failed, rejected, abandoned, or negative examples.

Historical Bias

DATA COLLECTION FAILURES

Definition: Past human or system bias is embedded in the collected data.

Solution: Audit labels, decisions, and outcomes across sensitive or important user groups.

Measurement Bias

DATA COLLECTION FAILURES

Definition: Logs, sensors, surveys, or tracking tools collect inaccurate data.

Solution: Validate measurement tools, fix instrumentation, and compare against trusted references.

Proxy Data Failure

DATA COLLECTION FAILURES

Definition: The team uses an indirect variable that poorly represents the real target.

Solution: Choose labels and features that more directly represent the desired outcome.

Low Data Coverage

DATA COLLECTION FAILURES

Definition: Important users, edge cases, languages, regions, or scenarios are missing.

Solution: Expand the dataset and add coverage tests for critical scenarios.

Data Consent Failure

DATA COLLECTION FAILURES

Definition: Data is collected or used without proper permission.

Solution: Enforce consent, purpose limitation, privacy review, and retention policies.

PII Collection Failure

DATA COLLECTION FAILURES

Definition: Personally identifiable information is collected when it is not needed.

Solution: Minimize collection and mask, tokenize, or remove sensitive fields.

Data Ownership Ambiguity

DATA COLLECTION FAILURES

Definition: No team clearly owns the data source or its quality.

Solution: Assign data owners, quality responsibilities, and escalation paths.

Event Logging Gap

DATA COLLECTION FAILURES

Definition: Important user or system events are not logged.

Solution: Define required events and telemetry before building the model or analytics pipeline.

Logging Inconsistency

DATA COLLECTION FAILURES

Definition: Different systems log the same event in different formats.

Solution: Standardize event schemas, names, timestamps, and required fields.

Data Source Instability

DATA COLLECTION FAILURES

Definition: A source system changes, breaks, or becomes unreliable.

Solution: Use data contracts, source monitoring, fallback sources, and versioned interfaces.

Synthetic Data Mismatch

DATA COLLECTION FAILURES

Definition: Synthetic data does not behave like real-world data.

Solution: Validate synthetic data against real samples and limit it to scenarios where it improves coverage.

Third-Party Data Risk

DATA COLLECTION FAILURES

Definition: External data is incomplete, biased, stale, or legally risky.

Solution: Validate vendor quality, licensing, update frequency, and compliance requirements.

Missing Values

DATA QUALITY FAILURES

Definition: Important fields are empty, null, or unavailable.

Solution: Use imputation, default handling, validation rules, or better collection processes.

Duplicate Records

DATA QUALITY FAILURES

Definition: The same entity or event appears multiple times.

Solution: Apply deduplication rules, unique identifiers, and record-linking checks.

Data Corruption

DATA QUALITY FAILURES

Definition: Data becomes malformed, damaged, unreadable, or incorrectly encoded.

Solution: Add validation checks and recover from clean backups or trusted raw sources.

Data Inconsistency

DATA QUALITY FAILURES

Definition: The same concept is recorded differently across systems.

Solution: Use standard schemas, normalization, and data contracts.

Schema Mismatch

DATA QUALITY FAILURES

Definition: Incoming data does not match the expected structure or type.

Solution: Use schema validation and producer-consumer contract testing.

Schema Drift

DATA QUALITY FAILURES

Definition: The structure of the data changes unexpectedly over time.

Solution: Monitor schema changes and use versioned schemas with compatibility checks.

Invalid Values

DATA QUALITY FAILURES

Definition: Data contains impossible, out-of-range, or wrongly formatted values.

Solution: Reject, correct, or quarantine invalid records with validation rules.

Outliers

DATA QUALITY FAILURES

Definition: Extreme values distort training, evaluation, or monitoring.

Solution: Investigate root causes and apply clipping, transformation, robust models, or special handling.

Label Noise

DATA QUALITY FAILURES

Definition: Training labels are wrong, inconsistent, or unreliable.

Solution: Improve labeling guidelines, reviewer training, consensus labeling, and label audits.

Bad Labels

DATA QUALITY FAILURES

Definition: The target values are incorrect because of human error, automation error, or unclear definitions.

Solution: Relabel samples, review edge cases, and measure inter-annotator agreement.

Incomplete Records

DATA QUALITY FAILURES

Definition: Records are missing required fields.

Solution: Enforce completeness checks before publishing data downstream.

Stale Data

DATA QUALITY FAILURES

Definition: The data is outdated and no longer reflects reality.

Solution: Refresh data more often and monitor freshness.

Data Leakage

DATA QUALITY FAILURES

Definition: Information that should not be available leaks into training or evaluation.

Solution: Review features, splits, joins, time boundaries, and target availability carefully.

Target Leakage

DATA QUALITY FAILURES

Definition: A feature accidentally contains information about the label.

Solution: Remove features that would not be available at prediction time.

Temporal Leakage

DATA QUALITY FAILURES

Definition: Future information is used to predict the past or present.

Solution: Use time-aware splits and enforce prediction-time availability rules.

Join Error

DATA QUALITY FAILURES

Definition: Tables are joined incorrectly, creating wrong relationships or duplicated rows.

Solution: Validate join keys, row counts, cardinality, and sample outputs after joins.

Entity Resolution Failure

DATA QUALITY FAILURES

Definition: Records belonging to the same person, product, company, or object are not matched correctly.

Solution: Use stronger identity matching, fuzzy matching, and manual review samples.

Unit Mismatch

DATA QUALITY FAILURES

Definition: Values use inconsistent units, such as dollars versus cents or meters versus feet.

Solution: Standardize units and add unit validation tests.

Time Zone Error

DATA QUALITY FAILURES

Definition: Events are processed or compared using incorrect time zones.

Solution: Store timestamps in UTC and convert only at the presentation layer.

Data Granularity Mismatch

DATA QUALITY FAILURES

Definition: Data from different levels is mixed incorrectly, such as user-level and transaction-level data.

Solution: Align data to the correct grain before aggregation, joining, or modeling.

Class Imbalance

DATA QUALITY FAILURES

Definition: Some classes appear much more often than others.

Solution: Use resampling, class weights, threshold tuning, better metrics, or more minority-class data.

Data Sparsity

DATA QUALITY FAILURES

Definition: There are too few useful examples for the model to learn reliable patterns.

Solution: Collect more data, simplify the model, use transfer learning, or aggregate related signals.

Dataset Bias

DATA QUALITY FAILURES

Definition: The dataset overrepresents or underrepresents certain groups, cases, or outcomes.

Solution: Run dataset audits, rebalance data, and evaluate performance across subgroups.

Data Poisoning

DATA QUALITY FAILURES

Definition: Bad or malicious data is inserted into training, retrieval, or evaluation data.

Solution: Validate sources, detect anomalies, review suspicious data, and protect ingestion pipelines.

Data Validation Failure

DATA QUALITY FAILURES

Definition: Bad data passes through because validation rules are missing or weak.

Solution: Add automated checks for schema, range, freshness, volume, uniqueness, and business rules.

Pipeline Failure

DATA PIPELINE FAILURES

Definition: A data workflow breaks before producing the expected output.

Solution: Use retries, alerts, validation checks, orchestration, and clear ownership.

ETL Failure

DATA PIPELINE FAILURES

Definition: Extract, transform, or load steps fail.

Solution: Isolate the failed step and add automated tests around extraction, transformation, and loading.

ELT Failure

DATA PIPELINE FAILURES

Definition: Data loads successfully, but warehouse or lakehouse transformations fail.

Solution: Add transformation tests, dependency checks, and release validation.

Batch Job Failure

DATA PIPELINE FAILURES

Definition: A scheduled batch process does not complete successfully.

Solution: Use job monitoring, retries, failure notifications, and runbook procedures.

Streaming Failure

DATA PIPELINE FAILURES

Definition: Real-time data ingestion stops, lags, duplicates, or loses events.

Solution: Use offset tracking, replay support, backpressure handling, and stream monitoring.

Backfill Failure

DATA PIPELINE FAILURES

Definition: Historical data is reprocessed incorrectly or incompletely.

Solution: Make jobs idempotent and validate row counts, time windows, and outputs.

Orchestration Failure

DATA PIPELINE FAILURES

Definition: Workflow tools fail to run tasks in the correct order.

Solution: Use dependency-aware orchestration, DAG validation, and failure alerts.

Dependency Failure

DATA PIPELINE FAILURES

Definition: An upstream API, database, file, or service breaks the pipeline.

Solution: Monitor dependencies and design fallbacks, retries, and graceful degradation.

Late-Arriving Data

DATA PIPELINE FAILURES

Definition: Data arrives after the processing window has already closed.

Solution: Use watermarking, delayed processing, correction jobs, or reconciliation logic.

Data Loss

DATA PIPELINE FAILURES

Definition: Records disappear during ingestion, processing, or storage.

Solution: Use checksums, reconciliation, source-to-target counts, and durable queues.

Partial Load

DATA PIPELINE FAILURES

Definition: Only part of the expected data is loaded.

Solution: Validate completeness before publishing data downstream.

Duplicate Ingestion

DATA PIPELINE FAILURES

Definition: The same event is ingested multiple times.

Solution: Use unique event IDs, deduplication, and idempotent writes.

Idempotency Failure

DATA PIPELINE FAILURES

Definition: Re-running a job creates duplicate or inconsistent results.

Solution: Design jobs so repeated runs produce the same output.

Transformation Bug

DATA PIPELINE FAILURES

Definition: Business logic inside a transformation step produces incorrect data.

Solution: Use unit tests, data tests, sample reviews, and code review.

Feature Pipeline Skew

DATA PIPELINE FAILURES

Definition: Feature generation differs between training and production.

Solution: Share feature logic through a feature store or common library.

Data Contract Violation

DATA PIPELINE FAILURES

Definition: A producer changes the data format or meaning without warning consumers.

Solution: Use data contracts, schema compatibility checks, and change notifications.

SLA Miss

DATA PIPELINE FAILURES

Definition: Data is not delivered within the expected time.

Solution: Monitor pipeline latency and plan capacity for peak loads.

Data Freshness Failure

DATA PIPELINE FAILURES

Definition: The pipeline delivers old data instead of current data.

Solution: Add freshness checks, timestamp validation, and freshness alerts.

Lineage Gap

DATA PIPELINE FAILURES

Definition: The team cannot trace where data came from or how it changed.

Solution: Use lineage tracking, metadata management, and pipeline documentation.

Silent Pipeline Failure

DATA PIPELINE FAILURES

Definition: The pipeline produces wrong output without raising an error.

Solution: Add anomaly detection, data quality checks, and sampled human review.

Monitoring Blind Spot

DATA PIPELINE FAILURES

Definition: The pipeline lacks metrics or alerts for important failures.

Solution: Create dashboards for freshness, volume, schema, quality, and error rates.

Broken Data Dependency

DATA PIPELINE FAILURES

Definition: A downstream model depends on data that no longer exists or has changed.

Solution: Track dependencies and require compatibility checks before upstream changes.

Data Warehouse Sync Failure

DATA PIPELINE FAILURES

Definition: Warehouse data is not aligned with source systems.

Solution: Run reconciliation checks and scheduled sync validation.

Lakehouse Table Corruption

DATA PIPELINE FAILURES

Definition: Tables become inconsistent because of failed writes, schema issues, or storage problems.

Solution: Use transactional table formats, checkpoints, validation, and repair procedures.

Weak Feature Signal

FEATURE ENGINEERING FAILURES

Definition: Features do not contain enough useful information for prediction.

Solution: Create stronger domain-informed features or collect better data.

Irrelevant Features

FEATURE ENGINEERING FAILURES

Definition: Features add noise without improving model performance.

Solution: Use feature selection, importance analysis, and ablation testing.

Feature Leakage

FEATURE ENGINEERING FAILURES

Definition: A feature exposes information unavailable at prediction time.

Solution: Remove future-looking, target-derived, or post-outcome features.

Feature Drift

FEATURE ENGINEERING FAILURES

Definition: A feature's distribution or meaning changes over time.

Solution: Monitor feature statistics and retrain or update features when needed.

Feature Skew

FEATURE ENGINEERING FAILURES

Definition: Training features are computed differently from production features.

Solution: Share feature logic using a feature store or common production library.

Feature Store Staleness

FEATURE ENGINEERING FAILURES

Definition: Stored features are outdated.

Solution: Add freshness checks, scheduled updates, and stale-feature alerts.

Feature Duplication

FEATURE ENGINEERING FAILURES

Definition: Multiple features represent the same signal and distort learning.

Solution: Remove redundant features and check correlation or mutual information.

Poor Encoding

FEATURE ENGINEERING FAILURES

Definition: Categorical, text, or time features are encoded in a way that loses meaning.

Solution: Use better encoding strategies such as embeddings, one-hot encoding, target encoding, or cyclical time features.

Scaling Error

FEATURE ENGINEERING FAILURES

Definition: Numerical features are normalized or standardized incorrectly.

Solution: Fit scalers only on training data and reuse the same scaler in production.

Missing Feature Handling Failure

FEATURE ENGINEERING FAILURES

Definition: The model cannot handle null or unavailable features properly.

Solution: Use imputation, default values, missingness indicators, or model-native missing handling.

High Cardinality Failure

FEATURE ENGINEERING FAILURES

Definition: A feature has too many unique values, making learning unstable.

Solution: Use embeddings, hashing, grouping, frequency thresholds, or domain grouping.

Spurious Feature

FEATURE ENGINEERING FAILURES

Definition: The model relies on a feature that correlates with the label but does not generalize.

Solution: Use stress tests, causal review, subgroup checks, and out-of-distribution evaluation.

Time Window Error

FEATURE ENGINEERING FAILURES

Definition: Aggregated features use the wrong time range.

Solution: Define lookback windows clearly and prevent future leakage.

Feature Explosion

FEATURE ENGINEERING FAILURES

Definition: Too many features increase complexity, cost, and overfitting risk.

Solution: Use feature pruning, regularization, and ablation analysis.

Feature Versioning Failure

FEATURE ENGINEERING FAILURES

Definition: Teams cannot track which feature version was used by a model.

Solution: Version feature definitions and connect them to experiment tracking.

Feature Reuse Risk

FEATURE ENGINEERING FAILURES

Definition: A feature built for one use case is reused where it behaves badly.

Solution: Validate reused features for the new use case before adoption.

Feature Availability Failure

FEATURE ENGINEERING FAILURES

Definition: A feature exists during training but is unavailable during inference.

Solution: Check online availability and prediction-time constraints before training.

Overfitting

MACHINE LEARNING TRAINING FAILURES

Definition: The model memorizes training data and performs poorly on new data.

Solution: Use more data, regularization, simpler models, dropout, cross-validation, or early stopping.

Underfitting

MACHINE LEARNING TRAINING FAILURES

Definition: The model is too simple to learn useful patterns.

Solution: Use better features, a stronger model, more training, or lower regularization.

Poor Generalization

MACHINE LEARNING TRAINING FAILURES

Definition: The model works in development but fails on real-world examples.

Solution: Use realistic validation sets, production-like test data, and edge-case coverage.

Low Generalization

MACHINE LEARNING TRAINING FAILURES

Definition: The model does not transfer well to unseen examples.

Solution: Improve data diversity and evaluate on data that better represents deployment conditions.

Data Leakage During Training

MACHINE LEARNING TRAINING FAILURES

Definition: The model learns from information it should not access.

Solution: Audit features, joins, target timing, and train-test boundaries.

Train-Test Contamination

MACHINE LEARNING TRAINING FAILURES

Definition: The same or very similar examples appear in both training and test sets.

Solution: Use duplicate detection, grouped splitting, and leakage checks.

Test Set Leakage

MACHINE LEARNING TRAINING FAILURES

Definition: Evaluation examples accidentally influence training.

Solution: Keep test sets locked, private, and separate from training decisions.

Bad Split Strategy

MACHINE LEARNING TRAINING FAILURES

Definition: Training, validation, and test sets are divided incorrectly.

Solution: Use stratified, grouped, time-based, or user-based splits depending on the problem.

Temporal Split Error

MACHINE LEARNING TRAINING FAILURES

Definition: Time-based data is split randomly, causing future information to leak.

Solution: Train on the past and validate on future time windows.

Class Imbalance Failure

MACHINE LEARNING TRAINING FAILURES

Definition: The model ignores minority classes because majority classes dominate training.

Solution: Use class weights, resampling, threshold tuning, and minority-class data collection.

Hyperparameter Instability

MACHINE LEARNING TRAINING FAILURES

Definition: Small parameter changes cause large performance differences.

Solution: Use systematic search, robust validation, and multiple random seeds.

Poor Regularization

MACHINE LEARNING TRAINING FAILURES

Definition: The model is not constrained enough, increasing overfitting risk.

Solution: Use L1/L2 regularization, dropout, pruning, simpler models, or early stopping.

Optimization Failure

MACHINE LEARNING TRAINING FAILURES

Definition: The learning algorithm fails to find a good solution.

Solution: Tune learning rate, optimizer, loss function, initialization, and preprocessing.

Local Minimum

MACHINE LEARNING TRAINING FAILURES

Definition: Training gets stuck in a poor solution.

Solution: Use better initialization, adaptive optimizers, learning-rate schedules, or restarts.

High Variance

MACHINE LEARNING TRAINING FAILURES

Definition: Model performance changes significantly across datasets or runs.

Solution: Use more data, cross-validation, ensembling, and regularization.

High Bias

MACHINE LEARNING TRAINING FAILURES

Definition: The model is too limited and consistently misses important patterns.

Solution: Use richer features, a more expressive model, or reduce excessive regularization.

Poor Calibration

MACHINE LEARNING TRAINING FAILURES

Definition: Predicted probabilities do not match real-world likelihoods.

Solution: Use calibration methods such as Platt scaling, isotonic regression, or temperature scaling.

Random Seed Instability

MACHINE LEARNING TRAINING FAILURES

Definition: Results change significantly when the random seed changes.

Solution: Run multiple seeds and report mean, variance, and confidence intervals.

Reproducibility Failure

MACHINE LEARNING TRAINING FAILURES

Definition: The team cannot reproduce training results.

Solution: Version code, data, features, configs, environments, and random seeds.

Experiment Tracking Failure

MACHINE LEARNING TRAINING FAILURES

Definition: Training parameters, datasets, metrics, and artifacts are not recorded properly.

Solution: Use experiment tracking tools such as MLflow, Weights & Biases, or a model registry.

Baseline Failure

MACHINE LEARNING TRAINING FAILURES

Definition: The team does not compare the model against a simple baseline.

Solution: Build a simple baseline first and require new models to beat it meaningfully.

Accuracy Paradox

MACHINE LEARNING TRAINING FAILURES

Definition: A model has high accuracy but performs poorly on the important class.

Solution: Use precision, recall, F1, ROC-AUC, PR-AUC, and class-specific metrics.

Spurious Correlation

MACHINE LEARNING TRAINING FAILURES

Definition: The model learns a false pattern that does not truly cause the outcome.

Solution: Run stress tests, causal review, subgroup analysis, and out-of-distribution evaluation.

Distribution Shift

MACHINE LEARNING TRAINING FAILURES

Definition: Production data differs from training data.

Solution: Monitor data distributions and retrain or adapt when shifts are detected.

Covariate Shift

MACHINE LEARNING TRAINING FAILURES

Definition: Input feature distribution changes between training and production.

Solution: Track feature distributions and update data, features, or models as needed.

Label Shift

MACHINE LEARNING TRAINING FAILURES

Definition: The distribution of output classes changes over time.

Solution: Monitor class distribution and recalibrate or retrain.

Concept Drift

MACHINE LEARNING TRAINING FAILURES

Definition: The relationship between inputs and outputs changes over time.

Solution: Detect drift and retrain with newer labeled data.

Model Decay

MACHINE LEARNING TRAINING FAILURES

Definition: Model quality gets worse as the environment changes.

Solution: Monitor performance and schedule retraining or model refreshes.

Out-of-Distribution Input

MACHINE LEARNING TRAINING FAILURES

Definition: The model receives data very different from what it saw during training.

Solution: Detect OOD inputs, reject uncertain cases, or route them to human review.

Weak Signal

MACHINE LEARNING TRAINING FAILURES

Definition: The input data does not contain enough predictive information.

Solution: Improve features, collect stronger signals, or reconsider whether the task is learnable.

Vanishing Gradient

DEEP LEARNING FAILURES

Definition: Gradients become too small, so early layers stop learning.

Solution: Use better activations, normalization, residual connections, and architecture changes.

Exploding Gradient

DEEP LEARNING FAILURES

Definition: Gradients become too large, causing unstable training.

Solution: Use gradient clipping, normalization, lower learning rates, and stable initialization.

Dead Neuron

DEEP LEARNING FAILURES

Definition: A neural unit stops activating and contributes little or nothing.

Solution: Adjust initialization, learning rate, architecture, or activation function.

Dead ReLU

DEEP LEARNING FAILURES

Definition: A ReLU neuron outputs zero for most or all inputs.

Solution: Use Leaky ReLU, GELU, better initialization, or a lower learning rate.

Saturation

DEEP LEARNING FAILURES

Definition: Activation functions enter flat regions where gradients are very small.

Solution: Use modern activations, normalization, and better initialization.

Poor Weight Initialization

DEEP LEARNING FAILURES

Definition: Initial weights make training slow, unstable, or ineffective.

Solution: Use initialization methods such as Xavier, He initialization, or pretrained weights.

Learning Rate Too High

DEEP LEARNING FAILURES

Definition: Training jumps around and fails to converge.

Solution: Lower the learning rate or use a learning-rate scheduler.

Learning Rate Too Low

DEEP LEARNING FAILURES

Definition: Training is extremely slow or gets stuck.

Solution: Increase the learning rate or use adaptive optimizers.

Batch Size Instability

DEEP LEARNING FAILURES

Definition: Batch size causes noisy gradients or poor generalization.

Solution: Tune batch size, use gradient accumulation, and scale learning rates carefully.

Mode Collapse

DEEP LEARNING FAILURES

Definition: A generative model produces limited or repetitive outputs.

Solution: Use better objectives, diversity penalties, training stabilization, and evaluation for diversity.

Catastrophic Forgetting

DEEP LEARNING FAILURES

Definition: A model forgets old knowledge when trained on new data.

Solution: Use replay data, regularization, frozen layers, or parameter-efficient fine-tuning.

Representation Collapse

DEEP LEARNING FAILURES

Definition: Learned representations become too similar and lose useful distinctions.

Solution: Use contrastive objectives, normalization, better negatives, and representation diagnostics.

Embedding Collapse

DEEP LEARNING FAILURES

Definition: Embeddings lose semantic diversity and become less useful.

Solution: Improve training objectives, data diversity, and embedding evaluation.

Gradient Noise

DEEP LEARNING FAILURES

Definition: Training updates are unstable because gradients are too noisy.

Solution: Tune batch size, optimizer, gradient accumulation, and learning rate.

Loss Plateau

DEEP LEARNING FAILURES

Definition: The loss stops improving before reaching good performance.

Solution: Adjust learning rate, architecture, data quality, optimizer, or schedule.

Convergence Failure

DEEP LEARNING FAILURES

Definition: The model never reaches a stable or useful solution.

Solution: Debug data, labels, loss function, optimizer, architecture, and preprocessing.

Attention Collapse

DEEP LEARNING FAILURES

Definition: Attention focuses on irrelevant or too narrow parts of the input.

Solution: Use better data, architecture tuning, attention diagnostics, and regularization.

Poor Transfer Learning

DEEP LEARNING FAILURES

Definition: A pretrained model does not adapt well to the target task.

Solution: Use domain data, careful fine-tuning, and task-specific validation.

Fine-Tuning Instability

DEEP LEARNING FAILURES

Definition: Fine-tuning damages useful pretrained behavior.

Solution: Use lower learning rates, LoRA or PEFT, frozen layers, and validation gates.

Catastrophic Interference

DEEP LEARNING FAILURES

Definition: New learning disrupts previously learned patterns.

Solution: Use continual-learning strategies and mixed old/new training data.

Capacity Bottleneck

DEEP LEARNING FAILURES

Definition: The model is too small or constrained for the task.

Solution: Increase model capacity, improve architecture, or simplify the task.

Overparameterization Risk

DEEP LEARNING FAILURES

Definition: The model has more parameters than needed, increasing cost and overfitting risk.

Solution: Use smaller models, regularization, pruning, distillation, or model selection.

Training Instability

DEEP LEARNING FAILURES

Definition: Loss, gradients, or metrics fluctuate unpredictably.

Solution: Inspect data, reduce learning rate, stabilize optimization, and monitor gradients.

Tokenization Error

NLP FAILURES

Definition: Text is split into tokens incorrectly, damaging meaning.

Solution: Use the right tokenizer and test domain-specific terms, names, code, and multilingual text.

Out-of-Vocabulary Failure

NLP FAILURES

Definition: The model struggles with rare words, names, slang, or new terms.

Solution: Use subword tokenization, domain data, updated embeddings, or retrieval.

Named Entity Recognition Failure

NLP FAILURES

Definition: The model misses or mislabels names, dates, places, organizations, or products.

Solution: Add better labeled examples and evaluate entity-level precision and recall.

Entity Linking Failure

NLP FAILURES

Definition: The model identifies an entity but links it to the wrong real-world concept.

Solution: Use better knowledge bases, disambiguation logic, and context-aware linking.

Coreference Failure

NLP FAILURES

Definition: The model misunderstands references such as “he,” “she,” “they,” or “it.”

Solution: Use better context modeling and coreference-specific training or evaluation data.

Negation Failure

NLP FAILURES

Definition: The model misses words like “not,” “never,” or “without.”

Solution: Add negation-heavy examples and rule-based checks for high-risk tasks.

Sarcasm Failure

NLP FAILURES

Definition: The model interprets sarcastic text literally.

Solution: Use domain-specific examples and human review for sensitive decisions.

Ambiguity Failure

NLP FAILURES

Definition: The model chooses the wrong meaning for a word or phrase.

Solution: Use more context or ask a clarifying question when the meaning is unclear.

Context Misunderstanding

NLP FAILURES

Definition: The model ignores important surrounding text.

Solution: Use better prompts, retrieval, context windows, and targeted evaluation.

Sentiment Misclassification

NLP FAILURES

Definition: The model assigns the wrong sentiment.

Solution: Use balanced labeled data, domain-specific evaluation, and error analysis.

Intent Classification Failure

NLP FAILURES

Definition: The model misunderstands what the user wants.

Solution: Clarify intent labels and train on real user examples.

Slot Filling Failure

NLP FAILURES

Definition: The system extracts the wrong values from user input.

Solution: Improve annotation rules, examples, schema validation, and extraction checks.

Translation Error

NLP FAILURES

Definition: The model mistranslates text or loses meaning.

Solution: Use domain-tuned translation and human review for high-risk content.

Summarization Hallucination

NLP FAILURES

Definition: A summary includes details not found in the source text.

Solution: Use grounding checks, citation requirements, and source-faithfulness evaluation.

Faithfulness Failure

NLP FAILURES

Definition: Generated text does not accurately reflect the input.

Solution: Use extractive checks, citations, factual consistency metrics, and verification prompts.

Toxicity Detection Failure

NLP FAILURES

Definition: Harmful content is missed or safe content is incorrectly flagged.

Solution: Use balanced safety datasets, threshold tuning, and human review for edge cases.

Language Drift

NLP FAILURES

Definition: The model performs worse on dialects, multilingual text, slang, or new terms.

Solution: Continuously evaluate language coverage and update domain data.

Prompt Sensitivity

NLP FAILURES

Definition: Small wording changes cause large output differences.

Solution: Test prompts across many examples and use robust templates.

Context Window Overflow

NLP FAILURES

Definition: The text is too long for the model to process fully.

Solution: Use chunking, summarization, retrieval, or hierarchical processing.

Long-Context Failure

NLP FAILURES

Definition: The model has access to long text but fails to use the right part.

Solution: Use retrieval, section ranking, summaries, and long-context evaluation.

Keyword Overmatching

NLP FAILURES

Definition: The system focuses on matching words instead of meaning.

Solution: Use semantic search, embeddings, hybrid retrieval, and relevance evaluation.

Semantic Misclassification

NLP FAILURES

Definition: Text is categorized incorrectly because the model misses the real meaning.

Solution: Improve labels, add representative examples, and evaluate semantic edge cases.

Entity Boundary Error

NLP FAILURES

Definition: The model selects too much or too little text as an entity.

Solution: Improve annotation guidelines and use entity-span evaluation.

Topic Drift

NLP FAILURES

Definition: Generated text slowly moves away from the original subject.

Solution: Use stronger prompts, section constraints, and validation checks.

Detection Failure

CNN AND COMPUTER VISION FAILURES

Definition: The model fails to identify an object that is present.

Solution: Add more labeled examples, tune thresholds, and improve detection architecture.

False Detection

CNN AND COMPUTER VISION FAILURES

Definition: The model detects something that is not actually there.

Solution: Add hard negative examples and tune confidence thresholds.

Misclassification

CNN AND COMPUTER VISION FAILURES

Definition: The image is assigned to the wrong class.

Solution: Improve labels, augmentation, class balance, and confusion-matrix analysis.

Localization Error

CNN AND COMPUTER VISION FAILURES

Definition: The object is detected, but the bounding box is inaccurate.

Solution: Improve annotations and use localization-focused loss functions.

Segmentation Failure

CNN AND COMPUTER VISION FAILURES

Definition: The model fails to correctly separate object regions.

Solution: Improve masks, add training examples, and evaluate segmentation metrics.

Occlusion Failure

CNN AND COMPUTER VISION FAILURES

Definition: The model fails when objects are partially hidden.

Solution: Use occlusion augmentation and realistic training data.

Scale Sensitivity

CNN AND COMPUTER VISION FAILURES

Definition: The model fails when objects are too small, too large, or at unusual distances.

Solution: Use multi-scale training and feature pyramid networks.

Rotation Sensitivity

CNN AND COMPUTER VISION FAILURES

Definition: The model fails when objects appear at different angles.

Solution: Use rotation augmentation and rotation-invariant architectures when needed.

Lighting Sensitivity

CNN AND COMPUTER VISION FAILURES

Definition: Shadows, glare, darkness, or brightness reduce performance.

Solution: Use lighting augmentation and collect diverse image data.

Blur Sensitivity

CNN AND COMPUTER VISION FAILURES

Definition: Motion blur or low image quality causes mistakes.

Solution: Use blur augmentation and image quality checks.

Background Bias

CNN AND COMPUTER VISION FAILURES

Definition: The model relies on background patterns instead of the main object.

Solution: Train with diverse backgrounds and object-focused augmentation.

Texture Bias

CNN AND COMPUTER VISION FAILURES

Definition: The CNN relies too much on texture rather than shape.

Solution: Use shape-focused augmentation, diverse data, and robustness testing.

Domain Shift in Vision

CNN AND COMPUTER VISION FAILURES

Definition: Production images differ from training images in camera, lighting, angle, or environment.

Solution: Collect production samples and use domain adaptation.

Adversarial Patch Attack

CNN AND COMPUTER VISION FAILURES

Definition: A visual pattern tricks the model into a wrong prediction.

Solution: Use adversarial testing, robust training, and input monitoring.

Small Object Failure

CNN AND COMPUTER VISION FAILURES

Definition: The model misses tiny objects in the image.

Solution: Use higher-resolution inputs, tiling, and small-object-focused training.

Class Confusion

CNN AND COMPUTER VISION FAILURES

Definition: Visually similar classes are repeatedly confused.

Solution: Collect more examples, improve labels, and analyze the confusion matrix.

Data Augmentation Failure

CNN AND COMPUTER VISION FAILURES

Definition: Augmentation creates unrealistic or harmful training examples.

Solution: Review augmentations against real-world conditions and remove harmful transforms.

Annotation Box Error

CNN AND COMPUTER VISION FAILURES

Definition: Bounding boxes or masks in training data are incorrect.

Solution: Audit labels and improve reviewer quality control.

Image Preprocessing Skew

CNN AND COMPUTER VISION FAILURES

Definition: Image resizing, cropping, or normalization differs between training and inference.

Solution: Share the same preprocessing pipeline across training and production.

OCR Failure

CNN AND COMPUTER VISION FAILURES

Definition: Text in images is read incorrectly.

Solution: Improve image quality, tune OCR models, and validate extracted text.

Pose Variation Failure

CNN AND COMPUTER VISION FAILURES

Definition: The model fails when objects or people appear in unusual positions.

Solution: Add pose-diverse data and augmentation.

Camera Quality Shift

CNN AND COMPUTER VISION FAILURES

Definition: A model trained on high-quality images fails on low-quality production images.

Solution: Train and evaluate with production-quality images.

Frame Sampling Failure

CNN AND COMPUTER VISION FAILURES

Definition: A video model misses important moments because frames are sampled poorly.

Solution: Tune sampling strategy and evaluate temporal coverage.

Poor Transcription

SPEECH AND AUDIO AI FAILURES

Definition: Speech-to-text output contains incorrect words.

Solution: Use better acoustic models, domain vocabulary, noise handling, and human review for high-risk transcripts.

Speaker Confusion

SPEECH AND AUDIO AI FAILURES

Definition: The system mixes up who is speaking.

Solution: Use diarization models and speaker verification checks.

Accent Bias

SPEECH AND AUDIO AI FAILURES

Definition: The model performs worse for certain accents.

Solution: Collect accent-diverse data and evaluate performance by accent group.

Background Noise Failure

SPEECH AND AUDIO AI FAILURES

Definition: Noise causes the system to misunderstand speech.

Solution: Use noise augmentation, denoising, and microphone quality checks.

Overlapping Speech Failure

SPEECH AND AUDIO AI FAILURES

Definition: The model struggles when multiple people speak at the same time.

Solution: Use diarization, source separation, and overlap-aware training data.

Wake Word Failure

SPEECH AND AUDIO AI FAILURES

Definition: A voice assistant fails to detect or incorrectly detects a wake word.

Solution: Tune wake-word thresholds and test across noise, accents, and devices.

Audio Quality Failure

SPEECH AND AUDIO AI FAILURES

Definition: Low-quality microphones or compression reduce performance.

Solution: Monitor audio quality and train on realistic device conditions.

Language Detection Failure

SPEECH AND AUDIO AI FAILURES

Definition: The model identifies the wrong spoken language.

Solution: Use stronger language identification and multilingual evaluation.

Intent Recognition Failure

SPEECH AND AUDIO AI FAILURES

Definition: The system transcribes speech but misunderstands the user's intent.

Solution: Improve intent labels, add real utterances, and validate downstream actions.

Speech Segmentation Error

SPEECH AND AUDIO AI FAILURES

Definition: The system splits speech into the wrong phrases or speaker turns.

Solution: Tune voice activity detection and segmentation rules.

Retrieval Failure

RAG FAILURES

Definition: The system fails to retrieve useful documents.

Solution: Improve indexing, embeddings, query rewriting, search strategy, and retrieval evaluation.

Low Recall

RAG FAILURES

Definition: The retriever misses important relevant documents.

Solution: Increase top-k, improve chunking, use hybrid search, and expand queries carefully.

Low Precision

RAG FAILURES

Definition: The retriever returns too many irrelevant documents.

Solution: Use reranking, metadata filters, better embeddings, and relevance thresholds.

Bad Chunking

RAG FAILURES

Definition: Documents are split in a way that loses meaning.

Solution: Use semantic chunking, overlap, and document-aware splitting.

Chunk Boundary Failure

RAG FAILURES

Definition: The needed answer is split across chunks and not retrieved together.

Solution: Use chunk overlap, parent-child retrieval, larger chunks, or hierarchical retrieval.

Context Pollution

RAG FAILURES

Definition: Irrelevant retrieved text distracts the model.

Solution: Use stricter retrieval filters, reranking, and context pruning.

Context Stuffing

RAG FAILURES

Definition: Too much retrieved content overwhelms the model.

Solution: Select only the most relevant evidence and summarize where appropriate.

Corpus Gap

RAG FAILURES

Definition: The answer is not present in the knowledge base.

Solution: Update the corpus or return a clear "not found" response.

Index Staleness

RAG FAILURES

Definition: The vector or search index is outdated.

Solution: Schedule reindexing and monitor document freshness.

Document Drift

RAG FAILURES

Definition: Indexed documents become outdated compared with the real world.

Solution: Track document versions, owners, expiration dates, and refresh cycles.

Wrong Document Version

RAG FAILURES

Definition: The system retrieves an old or incorrect version of a document.

Solution: Use version-aware metadata filters and deprecate outdated content.

Duplicate Retrieval

RAG FAILURES

Definition: The retriever returns repeated or near-identical chunks.

Solution: Apply deduplication and diversity-aware retrieval.

Embedding Mismatch

RAG FAILURES

Definition: The query and documents are embedded in a way that fails to capture meaning.

Solution: Use stronger or domain-specific embedding models and evaluate retrieval quality.

Semantic Drift

RAG FAILURES

Definition: Retrieved results are semantically related but not actually useful.

Solution: Use reranking, relevance labels, and task-specific retrieval evaluation.

Query Understanding Failure

RAG FAILURES

Definition: The retriever misunderstands the user's search intent.

Solution: Use intent detection, query rewriting, and clarification for ambiguous queries.

Query Rewrite Failure

RAG FAILURES

Definition: The system rewrites the user query incorrectly.

Solution: Evaluate rewrite quality and keep original query signals available.

Query Expansion Failure

RAG FAILURES

Definition: Expanded terms move retrieval away from the true intent.

Solution: Limit expansion and validate expanded queries against relevance metrics.

Metadata Filter Failure

RAG FAILURES

Definition: Incorrect metadata filters exclude relevant documents.

Solution: Validate metadata quality and test filter logic.

Missing Metadata

RAG FAILURES

Definition: Important metadata such as date, author, version, or product is unavailable.

Solution: Enrich documents during ingestion and enforce metadata requirements.

Top-K Failure

RAG FAILURES

Definition: The correct document exists but is not included in the selected top results.

Solution: Tune top-k, retrieval scoring, reranking, and hybrid search.

Reranking Failure

RAG FAILURES

Definition: The reranker fails to move the best evidence to the top.

Solution: Train or select stronger rerankers and evaluate with labeled queries.

Grounding Gap

RAG FAILURES

Definition: Relevant documents are retrieved, but the model does not use them correctly.

Solution: Use answer-evidence prompts and verification checks.

Grounding Failure

RAG FAILURES

Definition: The final answer is not supported by retrieved context.

Solution: Require citations, refuse unsupported claims, and validate claim-source alignment.

Citation Hallucination

RAG FAILURES

Definition: The model cites a source that does not support the answer.

Solution: Check each claim against its cited source before final output.

Source Attribution Failure

RAG FAILURES

Definition: The model gives an answer without showing where it came from.

Solution: Require traceable source references for factual claims.

Attribution Failure

RAG FAILURES

Definition: The answer cannot be traced back to reliable evidence.

Solution: Add source linking, evidence snippets, and audit logs.

Answer Synthesis Failure

RAG FAILURES

Definition: The model retrieves the right evidence but combines it incorrectly.

Solution: Use structured synthesis prompts, chain verification, and contradiction checks.

Multi-Hop Retrieval Failure

RAG FAILURES

Definition: The system fails when the answer requires multiple documents or reasoning steps.

Solution: Use iterative retrieval, graph retrieval, or query decomposition.

Retrieval Latency

RAG FAILURES

Definition: Search or retrieval takes too long.

Solution: Optimize indexes, cache results, reduce candidate sets, and tune infrastructure.

Vector Search Failure

RAG FAILURES

Definition: Similarity search fails to find the most useful content.

Solution: Use hybrid search, better embeddings, metadata filters, and reranking.

Hallucination

LLM AND GENERATIVE AI FAILURES

Definition: The model generates false or unsupported information.

Solution: Use grounding, retrieval, citations, uncertainty handling, and factuality evaluation.

Fabrication

LLM AND GENERATIVE AI FAILURES

Definition: The model invents facts, numbers, sources, citations, or events.

Solution: Require evidence and allow the model to say "I do not know" when information is missing.

False Confidence

LLM AND GENERATIVE AI FAILURES

Definition: The model sounds certain even when it is wrong.

Solution: Calibrate responses and require source-backed claims for factual answers.

Contradiction

LLM AND GENERATIVE AI FAILURES

Definition: The model gives answers that conflict with itself or known facts.

Solution: Use consistency checks, better context management, and verification prompts.

Ambiguous Output

LLM AND GENERATIVE AI FAILURES

Definition: The model gives an unclear answer that can be interpreted multiple ways.

Solution: Ask clarifying questions or enforce structured output.

Instruction Misfollowing

LLM AND GENERATIVE AI FAILURES

Definition: The model does not follow the user or system instruction.

Solution: Use clearer prompts, examples, schemas, and output validators.

Instruction Conflict

LLM AND GENERATIVE AI FAILURES

Definition: The model receives competing instructions and follows the wrong one.

Solution: Define instruction priority and remove contradictions.

Instruction Misinterpretation

LLM AND GENERATIVE AI FAILURES

Definition: The model misunderstands what the user actually asked.

Solution: Add task clarification, examples, and intent checks.

Context Loss

LLM AND GENERATIVE AI FAILURES

Definition: The model forgets or ignores important information from earlier context.

Solution: Use conversation summaries, memory, retrieval, and better context selection.

Answer Drift

LLM AND GENERATIVE AI FAILURES

Definition: The response slowly moves away from the original question or task.

Solution: Use tighter prompts, checkpoints, and validation against the user request.

Repetition Loop

LLM AND GENERATIVE AI FAILURES

Definition: The model repeats the same phrase, idea, or pattern.

Solution: Use decoding controls, repetition penalties, and response validation.

Mode Collapse

LLM AND GENERATIVE AI FAILURES

Definition: The model gives repetitive, generic, or overly similar answers.

Solution: Improve prompting, sampling settings, examples, and output diversity checks.

Verbosity Failure

LLM AND GENERATIVE AI FAILURES

Definition: The model gives too much detail when a concise answer is needed.

Solution: Specify length, audience, and format constraints.

Under-Answering

LLM AND GENERATIVE AI FAILURES

Definition: The model gives an incomplete answer.

Solution: Use coverage rubrics, checklists, and completeness validation.

Over-Answering

LLM AND GENERATIVE AI FAILURES

Definition: The model adds unnecessary or unsupported information.

Solution: Limit scope and require evidence for added claims.

Role Confusion

LLM AND GENERATIVE AI FAILURES

Definition: The model misunderstands which role or persona it should follow.

Solution: Use clear role instructions and periodic role reminders.

Format Failure

LLM AND GENERATIVE AI FAILURES

Definition: The output does not follow the requested format.

Solution: Use schemas, examples, structured output, and validators.

Refusal Failure

LLM AND GENERATIVE AI FAILURES

Definition: The model refuses a safe request or answers an unsafe request.

Solution: Improve safety policy interpretation and refusal evaluation.

Safety Failure

LLM AND GENERATIVE AI FAILURES

Definition: The model produces harmful, unsafe, or policy-violating content.

Solution: Use safety filters, policy checks, red-team testing, and human review for high-risk cases.

Toxic Output

LLM AND GENERATIVE AI FAILURES

Definition: The model generates offensive, abusive, hateful, or unsafe language.

Solution: Use toxicity filtering, safer training data, and moderation policies.

Speculative Answering

LLM AND GENERATIVE AI FAILURES

Definition: The model guesses instead of saying it does not know.

Solution: Instruct the model to express uncertainty and ask for missing information.

Capability Overclaiming

LLM AND GENERATIVE AI FAILURES

Definition: The model claims it can do something it cannot actually do.

Solution: Clearly define available tools, limits, and system capabilities.

Unfaithful Reasoning

LLM AND GENERATIVE AI FAILURES

Definition: The explanation does not match the real basis for the answer.

Solution: Use evidence-based explanations and separate hidden reasoning from user-facing justification.

Non-Determinism

LLM AND GENERATIVE AI FAILURES

Definition: The model produces different answers for the same input.

Solution: Lower temperature, use deterministic settings where possible, and validate outputs.

Tool-Use Hallucination

LLM AND GENERATIVE AI FAILURES

Definition: The model claims it used a tool or source when it did not.

Solution: Separate tool execution from response generation and log tool calls.

Emergent Misbehavior

LLM AND GENERATIVE AI FAILURES

Definition: Unexpected harmful or incorrect behavior appears at scale.

Solution: Use staged rollout, monitoring, red teaming, and incident response.

Alignment Failure

LLM AND GENERATIVE AI FAILURES

Definition: The AI system behaves in a way that does not match human goals, rules, or expectations.

Solution: Improve instruction tuning, policy design, evaluation, and guardrails.

Ambiguous Prompt PROMPTING FAILURES Definition: The instruction is unclear, causing poor output. Solution: Specify the goal, context, constraints, audience, and desired format.	Context Overload PROMPTING FAILURES Definition: The prompt contains too much information for the model to use effectively. Solution: Remove noise and prioritize the most relevant context.	Context Window Overflow PROMPTING FAILURES Definition: The input exceeds the model's maximum context length. Solution: Use chunking, summarization, retrieval, or hierarchical processing.
Few-Shot Failure PROMPTING FAILURES Definition: Examples in the prompt confuse the model or teach the wrong pattern. Solution: Use clear, consistent, relevant examples and test them across cases.	Instruction Dilution PROMPTING FAILURES Definition: Important instructions are weakened by too much surrounding text. Solution: Place critical rules clearly and use structured sections.	Prompt Injection PROMPTING FAILURES Definition: A user or retrieved document tries to override trusted instructions. Solution: Use instruction hierarchy, content isolation, filtering, and output validation.
Prompt Leakage PROMPTING FAILURES Definition: Hidden or private prompt instructions are exposed. Solution: Avoid sensitive prompt content and filter outputs for prompt disclosure.	Prompt Sensitivity PROMPTING FAILURES Definition: Small wording changes cause large output differences. Solution: Test prompt variants and use robust templates.	Prompt Overfitting PROMPTING FAILURES Definition: The prompt works only for narrow examples and fails on real cases. Solution: Evaluate prompts against diverse, realistic, and adversarial examples.
Prompt Contradiction PROMPTING FAILURES Definition: The prompt contains conflicting rules. Solution: Simplify the prompt and prioritize instructions clearly.	Prompt Ambiguity PROMPTING FAILURES Definition: The model cannot tell which instruction matters most. Solution: Use hierarchy, numbered requirements, and examples.	Prompt Fragility PROMPTING FAILURES Definition: The prompt works in demos but breaks with real users. Solution: Use prompt evaluation sets and continuous prompt improvement.

Planning Failure

AI AGENT FAILURES

Definition: The agent creates a poor, incomplete, or impossible plan.

Solution: Use planning constraints, examples, plan validation, and step limits.

Goal Drift

AI AGENT FAILURES

Definition: The agent shifts away from the original objective.

Solution: Restate the goal and check progress after each step.

Agent Loop

AI AGENT FAILURES

Definition: The agent keeps repeating steps without finishing the task.

Solution: Add loop limits, stopping rules, and escalation paths.

Infinite Retry Loop

AI AGENT FAILURES

Definition: The agent keeps retrying failed actions without escalation.

Solution: Use retry limits, fallback strategies, and failure escalation.

Action Failure

AI AGENT FAILURES

Definition: The agent chooses or performs the wrong action.

Solution: Add action validation, safer tool APIs, and approval gates.

Tool Selection Failure

AI AGENT FAILURES

Definition: The agent fails to choose the best available tool.

Solution: Improve tool descriptions, routing logic, examples, and tool-use evaluation.

Tool Execution Failure

AI AGENT FAILURES

Definition: A tool call fails because of bad input, timeout, or system error.

Solution: Use input validation, retries, error handling, and timeout policies.

Tool Misuse

AI AGENT FAILURES

Definition: The agent uses a tool incorrectly.

Solution: Use strict schemas, tool-specific guardrails, and test cases.

Tool Output Misinterpretation

AI AGENT FAILURES

Definition: The agent reads the tool result incorrectly.

Solution: Use structured tool outputs and verification prompts.

Tool Permission Failure

AI AGENT FAILURES

Definition: The agent tries to use a tool it is not allowed to use.

Solution: Apply least privilege and permission checks before tool calls.

Memory Corruption

AI AGENT FAILURES

Definition: Stored memory becomes inaccurate, outdated, or misleading.

Solution: Use memory validation, expiration, review, and user control.

Memory Leakage

AI AGENT FAILURES

Definition: Private or irrelevant memory appears in an answer.

Solution: Use memory access rules, privacy filters, and memory scoping.

Long-Term Memory Drift

AI AGENT FAILURES

Definition: Saved knowledge becomes less accurate over time.

Solution: Refresh, expire, or revalidate memory regularly.

Memory Retrieval Failure

AI AGENT FAILURES

Definition: The agent fails to retrieve useful memory.

Solution: Improve memory indexing, relevance scoring, and retrieval tests.

Verification Failure

AI AGENT FAILURES

Definition: The agent does not check whether its answer or action is correct.

Solution: Add verifier steps before final output or external action.

Reflection Failure

AI AGENT FAILURES

Definition: The agent reviews its work but fails to catch mistakes.

Solution: Use external checks, automated tests, or human review.

Autonomy Failure

AI AGENT FAILURES

Definition: The agent takes unsafe or unnecessary actions without supervision.

Solution: Add approval gates, scope limits, and risk-based controls.

Unbounded Autonomy

AI AGENT FAILURES

Definition: The agent has too much freedom and too few limits.

Solution: Use permission boundaries, budgets, step limits, and human approval for risky actions.

Permission Boundary Failure

AI AGENT FAILURES

Definition: The agent accesses or changes resources it should not.

Solution: Apply least-privilege access and enforce tool permissions.

State Management Failure

AI AGENT FAILURES

Definition: The agent loses track of task state, progress, or decisions.

Solution: Use explicit state tracking, checkpoints, and structured task memory.

Multi-Agent Coordination Failure

AI AGENT FAILURES

Definition: Multiple agents disagree, duplicate work, or block each other.

Solution: Define clear roles, shared state, coordination protocols, and conflict resolution.

Handoff Failure

AI AGENT FAILURES

Definition: One agent passes incomplete or incorrect context to another agent.

Solution: Use structured handoff summaries and required context fields.

Human Escalation Failure

AI AGENT FAILURES

Definition: The agent should ask for human help but continues alone.

Solution: Use uncertainty thresholds, risk rules, and escalation triggers.

Tool Chain Failure

AI AGENT FAILURES

Definition: One failed tool call breaks the entire workflow.

Solution: Design recovery paths, partial success handling, and fallback tools.

Weak Test Coverage EVALUATION FAILURES Definition: Evaluation does not cover enough real-world cases. Solution: Expand the test set with edge cases, production examples, and subgroup coverage.	Gold Dataset Weakness EVALUATION FAILURES Definition: The trusted evaluation set is incomplete or low quality. Solution: Use expert review, regular refreshes, and quality audits.	Labeling Inconsistency EVALUATION FAILURES Definition: Human evaluators label similar examples differently. Solution: Improve guidelines, reviewer training, and agreement checks.
Human Evaluation Drift EVALUATION FAILURES Definition: Reviewers become inconsistent over time. Solution: Run calibration sessions and include benchmark examples.	Evaluation Bias EVALUATION FAILURES Definition: The test set or metric favors certain outputs, users, or cases. Solution: Audit evaluation data across scenarios, languages, and user groups.	Metric Gaming EVALUATION FAILURES Definition: The model improves the metric without improving real quality. Solution: Use multiple metrics, human review, and outcome-based evaluation.
Metric Misalignment EVALUATION FAILURES Definition: The metric does not match actual user or business value. Solution: Choose metrics that reflect usefulness, safety, reliability, and business impact.	False Positive EVALUATION FAILURES Definition: The model incorrectly predicts something is present. Solution: Tune thresholds, add hard negative examples, and review precision errors.	False Negative EVALUATION FAILURES Definition: The model fails to detect something that is present. Solution: Improve recall with more positive examples, threshold tuning, and feature improvements.
Precision Failure EVALUATION FAILURES Definition: Too many predicted positives are wrong. Solution: Raise thresholds, improve features, or add stronger negative examples.	Recall Failure EVALUATION FAILURES Definition: Too many real positives are missed. Solution: Lower thresholds, add positive data, and improve retrieval or feature coverage.	F1 Misinterpretation EVALUATION FAILURES Definition: A single F1 score hides important precision-recall tradeoffs. Solution: Review precision and recall separately by class and use case.
Calibration Failure EVALUATION FAILURES Definition: Confidence scores do not reflect actual correctness. Solution: Use calibration methods and evaluate confidence reliability.	Offline-Online Gap EVALUATION FAILURES Definition: Offline evaluation results do not match production behavior. Solution: Use production-like evaluation data and online testing.	Regression EVALUATION FAILURES Definition: A new model version performs worse than the previous version. Solution: Use regression tests, release gates, and rollback plans.
A/B Test Contamination EVALUATION FAILURES Definition: Experiment groups affect each other or are not separated correctly. Solution: Use clean randomization, isolation, and experiment monitoring.	Sample Size Failure EVALUATION FAILURES Definition: The test does not include enough examples to support a conclusion. Solution: Increase sample size and check statistical power.	Benchmark Contamination EVALUATION FAILURES Definition: Benchmark or test data appears in the model's training data. Solution: Use private, fresh, or carefully controlled evaluation sets.

LLM-as-Judge Bias

EVALUATION FAILURES

Definition: An LLM evaluator favors certain writing styles, lengths, or model outputs.

Solution: Calibrate judges, use multiple judges, and validate with humans.

RAG Evaluation Failure

EVALUATION FAILURES

Definition: Evaluation checks final answers but ignores retrieval quality or citation accuracy.

Solution: Measure retrieval, grounding, answer correctness, and citation support separately.

Safety Evaluation Gap

EVALUATION FAILURES

Definition: The system is not tested against harmful, adversarial, or policy-sensitive cases.

Solution: Use red-team prompts, adversarial tests, and safety rubrics.

Edge Case Blind Spot

EVALUATION FAILURES

Definition: Rare but important cases are missing from evaluation.

Solution: Add edge-case suites and scenario-based tests.

Fairness Evaluation Gap

EVALUATION FAILURES

Definition: The system is not tested across demographic or user groups.

Solution: Measure subgroup performance and fairness metrics.

Robustness Evaluation Gap

EVALUATION FAILURES

Definition: The system is not tested against noisy, adversarial, or unusual inputs.

Solution: Add perturbation tests, adversarial inputs, and stress testing.

Explainability Gap

EVALUATION FAILURES

Definition: The team cannot clearly explain why the model made a decision.

Solution: Use interpretable features, explanations, decision records, and audit trails.

Deployment Failure

MLOps and Deployment Failures

Definition: A model release breaks or behaves incorrectly in production.

Solution: Use staging tests, canary releases, release gates, and rollback plans.

Model Versioning Failure

MLOps and Deployment Failures

Definition: The team cannot track which model version is running.

Solution: Use a model registry, version tags, and deployment metadata.

Data Versioning Failure

MLOps and Deployment Failures

Definition: The team cannot reproduce which data was used for training.

Solution: Version datasets, snapshots, transformations, and training data references.

Feature Versioning Failure

MLOps and Deployment Failures

Definition: Feature definitions are not tracked across training and serving.

Solution: Version feature definitions and connect them to model artifacts.

Training-Serving Skew

MLOps and Deployment Failures

Definition: Training logic differs from production inference logic.

Solution: Share preprocessing and feature logic between training and serving.

Environment Mismatch

MLOps and Deployment Failures

Definition: Development, staging, and production environments behave differently.

Solution: Use containers, pinned dependencies, and infrastructure-as-code.

Dependency Conflict

MLOps and Deployment Failures

Definition: Library or package versions break the ML system.

Solution: Pin dependencies and test environments before release.

Containerization Failure

MLOps and Deployment Failures

Definition: The model does not run correctly inside its deployment container.

Solution: Test containers with production-like inputs before deployment.

CI/CD Failure

MLOps and Deployment Failures

Definition: Automated testing, build, or deployment pipelines break.

Solution: Add pipeline tests, clear release gates, and rollback procedures.

Rollback Failure

MLOps and Deployment Failures

Definition: The team cannot safely return to a previous working version.

Solution: Keep versioned artifacts and automate rollback workflows.

Canary Failure

MLOps and Deployment Failures

Definition: A small rollout does not detect issues before full deployment.

Solution: Use better canary metrics, traffic segmentation, and quality monitoring.

Shadow Deployment Failure

MLOps and Deployment Failures

Definition: A shadow model is tested incorrectly or not monitored properly.

Solution: Compare shadow predictions against real outcomes and baseline models.

Model Registry Failure

MLOps and Deployment Failures

Definition: Models are not approved, tagged, or stored correctly.

Solution: Use registry governance, approval workflows, and artifact validation.

Artifact Corruption

MLOps and Deployment Failures

Definition: Model files, tokenizer files, or configuration files are damaged or mismatched.

Solution: Use checksums, artifact validation, and compatibility tests.

Inference Service Failure

MLOps and Deployment Failures

Definition: The production prediction service becomes unavailable.

Solution: Use health checks, autoscaling, failover, and incident runbooks.

API Contract Failure

MLOps and Deployment Failures

Definition: The model endpoint input or output format changes unexpectedly.

Solution: Use API contracts, backward compatibility tests, and schema validation.

Autoscaling Failure

MLOps and Deployment Failures

Definition: Infrastructure cannot scale fast enough for traffic.

Solution: Use load testing, scaling policies, queueing, and capacity planning.

Cold Start Latency

MLOps and Deployment Failures

Definition: The first request is slow because infrastructure or the model is not warmed up.

Solution: Use warm pools, caching, optimized model loading, and pre-warming.

GPU Resource Failure

MLOPS AND DEPLOYMENT FAILURES

Definition: GPU memory, scheduling, or availability problems break inference or training.

Solution: Use resource limits, monitoring, optimized batch sizes, and fallback capacity.

Cost Explosion

MLOPS AND DEPLOYMENT FAILURES

Definition: Model usage becomes too expensive because of traffic, tokens, compute, or inefficient design.

Solution: Use caching, model routing, prompt optimization, budgets, and usage limits.

Token Cost Overrun

MLOPS AND DEPLOYMENT FAILURES

Definition: Prompts or responses consume too many tokens.

Solution: Shorten prompts, filter retrieval context, set output limits, and summarize.

Latency Failure

MLOPS AND DEPLOYMENT FAILURES

Definition: The model takes too long to generate a response.

Solution: Use smaller models, caching, batching, streaming, optimized serving, or model routing.

Throughput Failure

MLOPS AND DEPLOYMENT FAILURES

Definition: The system cannot handle the required request volume.

Solution: Use batching, autoscaling, queue management, and performance testing.

SLA Violation

MLOPS AND DEPLOYMENT FAILURES

Definition: The system fails reliability, speed, or uptime requirements.

Solution: Use SLAs, error budgets, monitoring, and reliability engineering.

Release Governance Failure

MLOPS AND DEPLOYMENT FAILURES

Definition: Models are shipped without proper review, approval, or documentation.

Solution: Use release checklists, approvals, model cards, and audit trails.

Production Configuration Drift

MLOPS AND DEPLOYMENT FAILURES

Definition: Production settings become different from approved settings.

Solution: Use config versioning, drift detection, and infrastructure-as-code.

Model Drift

MONITORING AND MAINTENANCE FAILURES

Definition: Model performance degrades over time.

Solution: Monitor performance, collect labels, and retrain or refresh the model.

Concept Drift

MONITORING AND MAINTENANCE FAILURES

Definition: The relationship between inputs and labels changes.

Solution: Detect drift and retrain with newer labeled data.

Covariate Shift

MONITORING AND MAINTENANCE FAILURES

Definition: Input feature distribution changes.

Solution: Track feature distributions and adapt data, features, or model strategy.

Label Shift

MONITORING AND MAINTENANCE FAILURES

Definition: Output class distribution changes.

Solution: Monitor class distribution and recalibrate or retrain.

Prediction Drift

MONITORING AND MAINTENANCE FAILURES

Definition: The distribution of model predictions changes.

Solution: Compare prediction trends against baselines and investigate anomalies.

Feature Drift

MONITORING AND MAINTENANCE FAILURES

Definition: Input features change in meaning or distribution.

Solution: Monitor feature statistics and trigger alerts for major shifts.

Data Drift

MONITORING AND MAINTENANCE FAILURES

Definition: Production data differs from training data.

Solution: Monitor data distributions and retrain when shifts affect quality.

Silent Failure

MONITORING AND MAINTENANCE FAILURES

Definition: The system produces bad results without visible errors.

Solution: Use quality checks, anomaly alerts, sampled human review, and outcome monitoring.

Monitoring Gap

MONITORING AND MAINTENANCE FAILURES

Definition: Important model, data, or system metrics are not tracked.

Solution: Monitor data, model quality, latency, cost, safety, and business outcomes together.

Alert Fatigue

MONITORING AND MAINTENANCE FAILURES

Definition: Too many alerts cause teams to ignore important signals.

Solution: Tune alert thresholds, deduplicate alerts, and prioritize severity.

Missing Alert

MONITORING AND MAINTENANCE FAILURES

Definition: A serious issue occurs without triggering an alert.

Solution: Add alerts for critical failure modes and test alert coverage.

Observability Failure

MONITORING AND MAINTENANCE FAILURES

Definition: Logs, metrics, traces, and examples are insufficient for debugging.

Solution: Use structured logging, distributed tracing, metrics, and request-level audit records.

Logging Gap

MONITORING AND MAINTENANCE FAILURES

Definition: The system does not store enough information to investigate failures.

Solution: Log inputs, outputs, model versions, data versions, errors, and decisions safely.

Feedback Loop Failure

MONITORING AND MAINTENANCE FAILURES

Definition: User feedback is not collected or connected to improvement.

Solution: Connect feedback to evaluation, labeling, retraining, and product decisions.

Delayed Label Problem

MONITORING AND MAINTENANCE FAILURES

Definition: True labels arrive too late to monitor performance quickly.

Solution: Use proxy metrics and delayed performance tracking.

Latency Regression

MONITORING AND MAINTENANCE FAILURES

Definition: A new version makes inference slower.

Solution: Run performance tests before deployment and monitor latency after release.

Error Budget Burn

MONITORING AND MAINTENANCE FAILURES

Definition: The system consumes its allowed failure budget too quickly.

Solution: Pause risky releases and prioritize reliability fixes.

Dashboard Blind Spot

MONITORING AND MAINTENANCE FAILURES

Definition: Dashboards show system health but miss model quality issues.

Solution: Add model quality, retrieval quality, safety, and user outcome dashboards.

Data Freshness Monitoring Failure

MONITORING AND MAINTENANCE FAILURES

Definition: The system does not detect stale input data.

Solution: Add freshness metrics and stale-data alerts.

Quality Monitoring Failure

MONITORING AND MAINTENANCE FAILURES

Definition: The system monitors uptime but not answer or prediction quality.

Solution: Add quality sampling, human review, and automated evaluation.

User Behavior Drift

MONITORING AND MAINTENANCE FAILURES

Definition: Users change how they interact with the system, reducing performance.

Solution: Monitor usage patterns and update prompts, UX, or models.

Feedback Bias

MONITORING AND MAINTENANCE FAILURES

Definition: Feedback comes mostly from certain user groups, creating misleading signals.

Solution: Analyze feedback coverage and balance feedback sources.

Prompt Injection Attack

SECURITY, PRIVACY, AND GOVERNANCE FAILURES

Definition: Malicious input tries to override system behavior.

Solution: Use instruction isolation, content filtering, permission controls, and output validation.

Jailbreak

SECURITY, PRIVACY, AND GOVERNANCE FAILURES

Definition: A prompt bypasses model safety restrictions.

Solution: Use stronger safety policies, red-team testing, output moderation, and layered guardrails.

Data Poisoning

SECURITY, PRIVACY, AND GOVERNANCE FAILURES

Definition: Malicious or bad data is inserted into the system.

Solution: Use trusted sources, anomaly detection, review workflows, and secure ingestion.

Training Data Poisoning

SECURITY, PRIVACY, AND GOVERNANCE FAILURES

Definition: Harmful examples are inserted during training.

Solution: Validate datasets, verify sources, and audit suspicious samples.

Retrieval Poisoning

SECURITY, PRIVACY, AND GOVERNANCE FAILURES

Definition: Malicious content is added to the RAG knowledge base and later retrieved.

Solution: Use document trust scoring, ingestion approval, and source validation.

Model Extraction

SECURITY, PRIVACY, AND GOVERNANCE FAILURES

Definition: Attackers try to copy the model through repeated queries.

Solution: Use rate limits, monitoring, access control, and abuse detection.

Model Inversion

SECURITY, PRIVACY, AND GOVERNANCE FAILURES

Definition: Attackers try to recover private training data from outputs.

Solution: Use privacy-preserving training, regularization, access limits, and output filtering.

Membership Inference

SECURITY, PRIVACY, AND GOVERNANCE FAILURES

Definition: Attackers try to determine whether a record was in training data.

Solution: Use differential privacy, regularization, and restricted access.

Data Exfiltration

SECURITY, PRIVACY, AND GOVERNANCE FAILURES

Definition: Sensitive data is extracted through the model or connected tools.

Solution: Use data access controls, output inspection, and least-privilege tool permissions.

PII Leakage

SECURITY, PRIVACY, AND GOVERNANCE FAILURES

Definition: Personal information is exposed in outputs, logs, prompts, or datasets.

Solution: Use redaction, masking, privacy filters, and safe logging practices.

Secret Leakage

SECURITY, PRIVACY, AND GOVERNANCE FAILURES

Definition: API keys, credentials, tokens, or internal details are exposed.

Solution: Use secret scanning, vaults, key rotation, and strict logging rules.

Over-Permissioned Model

SECURITY, PRIVACY, AND GOVERNANCE FAILURES

Definition: The AI system has more access than needed.

Solution: Apply least-privilege access and scope permissions tightly.

Over-Permissioned Tool

SECURITY, PRIVACY, AND GOVERNANCE FAILURES

Definition: An agent tool has broader access than necessary.

Solution: Narrow tool permissions and require approval for risky actions.

Unauthorized Action

SECURITY, PRIVACY, AND GOVERNANCE FAILURES

Definition: The AI performs an action without proper approval.

Solution: Use confirmation steps, authorization checks, and audit logging.

Unsafe Tool Access

SECURITY, PRIVACY, AND GOVERNANCE FAILURES

Definition: An agent can call tools that modify systems without guardrails.

Solution: Use sandboxing, approvals, permission scopes, and audit logs.

Supply Chain Risk

SECURITY, PRIVACY, AND GOVERNANCE FAILURES

Definition: Models, datasets, packages, or tools introduce security vulnerabilities.

Solution: Use dependency scanning, trusted sources, model provenance, and signed artifacts.

Access Control Failure

SECURITY, PRIVACY, AND GOVERNANCE FAILURES

Definition: Users or systems access data or functions they should not.

Solution: Use authentication, authorization, policy checks, and access reviews.

Auditability Failure

SECURITY, PRIVACY, AND GOVERNANCE FAILURES

Definition: The organization cannot explain what the model did and why.

Solution: Keep logs, traces, model versions, data versions, prompts, and decision records.

Compliance Failure

SECURITY, PRIVACY, AND GOVERNANCE FAILURES

Definition: The AI system violates legal, regulatory, or internal policy requirements.

Solution: Run governance reviews, compliance testing, and documentation checks.

Data Retention Failure

SECURITY, PRIVACY, AND GOVERNANCE FAILURES

Definition: Data is stored longer than allowed.

Solution: Use retention policies, automated deletion, and storage audits.

Consent Violation

SECURITY, PRIVACY, AND GOVERNANCE FAILURES

Definition: User data is used beyond the permission originally granted.

Solution: Track consent and enforce purpose limitation.

Policy Enforcement Failure

SECURITY, PRIVACY, AND GOVERNANCE FAILURES

Definition: Governance rules exist on paper but are not enforced in the system.

Solution: Implement policy checks directly in pipelines, tools, and release gates.

Safety Guardrail Failure

SECURITY, PRIVACY, AND GOVERNANCE FAILURES

Definition: Filters, policies, or validators fail to block risky output.

Solution: Use layered guardrails, adversarial testing, and monitoring.

Insecure Output Handling

SECURITY, PRIVACY, AND GOVERNANCE FAILURES

Definition: Model output is trusted directly without validation.

Solution: Validate outputs before using them in tools, code, databases, or user-facing actions.

Low Helpfulness

USER EXPERIENCE AND PRODUCT FAILURES

Definition: The AI response does not solve the user's actual problem.

Solution: Test with real users and optimize for task completion, not only model metrics.

Poor Personalization

USER EXPERIENCE AND PRODUCT FAILURES

Definition: The answer ignores user context, preferences, or skill level.

Solution: Use safe user preferences and adapt response style without exposing private data.

Bad Refusal

USER EXPERIENCE AND PRODUCT FAILURES

Definition: The AI refuses a request it should answer.

Solution: Improve policy interpretation, refusal evaluation, and safe-completion behavior.

Unsafe Compliance

USER EXPERIENCE AND PRODUCT FAILURES

Definition: The AI answers a request it should refuse.

Solution: Use better safety classification, guardrails, and moderation.

Confusing Explanation

USER EXPERIENCE AND PRODUCT FAILURES

Definition: The answer is technically correct but hard to understand.

Solution: Use simpler language, examples, and structure.

Poor Tone

USER EXPERIENCE AND PRODUCT FAILURES

Definition: The response sounds rude, robotic, cold, or inappropriate.

Solution: Tune tone guidelines and test user perception.

Unactionable Output

USER EXPERIENCE AND PRODUCT FAILURES

Definition: The response gives theory but no clear next step.

Solution: Add concrete actions, examples, checklists, or templates.

Missing Clarification

USER EXPERIENCE AND PRODUCT FAILURES

Definition: The model should ask a question but guesses instead.

Solution: Add clarification rules for ambiguous or high-risk tasks.

Over-Clarification

USER EXPERIENCE AND PRODUCT FAILURES

Definition: The model asks too many questions instead of helping.

Solution: Make reasonable assumptions and move forward when the risk is low.

Trust Failure

USER EXPERIENCE AND PRODUCT FAILURES

Definition: Users lose confidence because the AI is wrong, vague, or inconsistent.

Solution: Improve transparency, citations, consistency, correction mechanisms, and reliability.

User Friction

USER EXPERIENCE AND PRODUCT FAILURES

Definition: The AI workflow is too slow, confusing, or hard to use.

Solution: Simplify the user journey and reduce unnecessary steps.

Bad Escalation

USER EXPERIENCE AND PRODUCT FAILURES

Definition: The system fails to route difficult cases to a human.

Solution: Define escalation thresholds and support handoff workflows.

Misleading Confidence

USER EXPERIENCE AND PRODUCT FAILURES

Definition: The UI or model makes uncertain results look reliable.

Solution: Show confidence, uncertainty, evidence, and limitations clearly.

Inconsistent Experience

USER EXPERIENCE AND PRODUCT FAILURES

Definition: The same user gets very different quality across similar tasks.

Solution: Standardize prompts, evaluation, UX flows, and response policies.

Poor Error Recovery

USER EXPERIENCE AND PRODUCT FAILURES

Definition: The system fails badly and does not help the user recover.

Solution: Provide friendly error messages, fallback options, and retry guidance.

Unclear Source Transparency

USER EXPERIENCE AND PRODUCT FAILURES

Definition: Users cannot tell where the answer came from.

Solution: Show citations, evidence panels, source summaries, or provenance details.

Agent Framework

ADDITIONAL ENTERPRISE AI TERMS

A development library or platform that provides building blocks for agents, such as tools, memory, orchestration, and tracing.

Continuous Learning

ADDITIONAL ENTERPRISE AI TERMS

Ongoing improvement of agent behavior through better prompts, data, evaluation, retrieval, or model updates.

Embeddings

ADDITIONAL ENTERPRISE AI TERMS

Vector numbers that represent meaning, allowing an AI system to compare text, images, or records by similarity instead of exact words.

First Contact Resolution (FCR)

ADDITIONAL ENTERPRISE AI TERMS

A support metric showing how often a customer problem is solved during the first interaction without follow-up.

Large Language Model (LLM)

ADDITIONAL ENTERPRISE AI TERMS

A large neural model that understands and generates language, often serving as the reasoning engine inside chatbots and agents.

Natural Language Generation (NLG)

ADDITIONAL ENTERPRISE AI TERMS

The part of AI that produces readable text or speech from structured data, retrieved context, or model reasoning.

Natural Language Understanding (NLU)

ADDITIONAL ENTERPRISE AI TERMS

The part of AI that identifies a user’s intent, entities, and meaning from natural language input.

Net Promoter Score (NPS)

ADDITIONAL ENTERPRISE AI TERMS

A customer loyalty metric based on how likely customers are to recommend a product, service, or company.

Proof of Concept (POC)

ADDITIONAL ENTERPRISE AI TERMS

A small experiment that proves an AI idea is technically possible and valuable enough to consider scaling.

Time to Value (TTV)

ADDITIONAL ENTERPRISE AI TERMS

The amount of time it takes for a user or organization to receive measurable benefit from a product or AI deployment.

Agent Evaluation

AGENT EVALUATION TERMS

The process of testing an agent’s reasoning, tool calls, final answers, safety, cost, and latency before or during production use.

Benchmarking

AGENT EVALUATION TERMS

Comparing an AI system against baselines, alternative models, datasets, or performance targets.

Confidence Scoring

AGENT EVALUATION TERMS

Estimating how reliable an output, classification, tool call, or decision is likely to be.

Cost per Task

AGENT EVALUATION TERMS

The average spend required for one completed task, including model tokens, tool calls, infrastructure, retries, and human review when needed.

Escalation Rate

AGENT EVALUATION TERMS

The percentage of tasks or conversations that must be handed to a human, specialist agent, or higher-trust workflow.

Eval Dataset

AGENT EVALUATION TERMS

A curated set of prompts, scenarios, expected behaviors, and edge cases used to test an AI system repeatedly.

Evaluation (Eval)

AGENT EVALUATION TERMS

A structured process for measuring quality, safety, correctness, and business value of AI behavior.

Failure Rate

AGENT EVALUATION TERMS

The share of tasks where the agent gives a wrong answer, calls the wrong tool, violates policy, or fails to finish.

False Negative

AGENT EVALUATION TERMS

A missed detection, such as failing to flag a risky output, bad retrieval result, policy violation, or defect.

False Positive

AGENT EVALUATION TERMS

An incorrect alert or block, such as flagging safe content, valid tool use, or correct output as unsafe.

Final Answer Evaluation

AGENT EVALUATION TERMS

Reviewing the final response for correctness, groundedness, completeness, tone, policy compliance, and usefulness to the user.

Goal Completion Rate

AGENT EVALUATION TERMS

The percentage of tasks where the agent reaches the requested outcome without unnecessary failure, escalation, or user rework.

Golden Dataset

AGENT EVALUATION TERMS

A trusted set of examples, expected outputs, or human-labeled judgments used for evaluation.

Groundedness

AGENT EVALUATION TERMS

How strongly an answer is supported by retrieved documents, verified data, tool results, or other trusted evidence.

Helpfulness

AGENT EVALUATION TERMS

How well the response solves the user’s real problem with clear, relevant, and actionable information.

Human Evaluation

AGENT EVALUATION TERMS

Quality review performed by people who judge usefulness, correctness, safety, tone, and real-world task success.

Latency per Task

AGENT EVALUATION TERMS

The total time required to finish one task, including model calls, retrieval, tool execution, retries, and final response generation.

LLM-as-Judge

AGENT EVALUATION TERMS

Using a language model to assess outputs, often with rubrics, references, or pairwise comparisons.

Multi-Turn Evaluation

AGENT EVALUATION TERMS

Testing whether an agent stays accurate, safe, and context-aware across a conversation or long-running workflow.

Pairwise Evaluation

AGENT EVALUATION TERMS

Comparing two outputs side by side so reviewers or judge models can select the better answer or behavior.

Plan Accuracy

AGENT EVALUATION TERMS

How well the generated plan matches the task requirements, dependencies, constraints, and expected order of execution.

Plan Quality

AGENT EVALUATION TERMS

The usefulness, feasibility, and ordering of an agent’s proposed steps.

RAG Evaluation

AGENT EVALUATION TERMS

Testing retrieval and answer quality together, including context relevance, citation accuracy, faithfulness, and answer completeness.

Regression Test

AGENT EVALUATION TERMS

A repeatable test that catches quality drops after model, prompt, retrieval, or tool changes.

Relevance

AGENT EVALUATION TERMS

The degree to which retrieved context, tool output, or generated text directly answers the user’s request.

Safety Score

AGENT EVALUATION TERMS

A metric summarizing whether outputs and actions comply with safety and policy expectations.

Scenario Test

AGENT EVALUATION TERMS

An evaluation case built around a realistic user situation, including inputs, constraints, expected behavior, and pass criteria.

Simulation

AGENT EVALUATION TERMS

Controlled testing of agent behavior in representative scenarios before production exposure.

Simulation Test

AGENT EVALUATION TERMS

Running agents inside controlled mock environments to test behavior before exposing them to live users or real systems.

Step-Level Evaluation

AGENT EVALUATION TERMS

Checking each intermediate plan step, tool call, observation, and decision instead of only judging the final answer.

Task Success Rate

AGENT EVALUATION TERMS

The percentage of tasks an agent completes according to predefined success criteria.

Token Usage

AGENT EVALUATION TERMS

The number of input and output tokens consumed by a request, conversation, or task, used for cost and latency control.

Tool-Call Accuracy

AGENT EVALUATION TERMS

How often an agent chooses the correct tool and passes the correct arguments.

Tool Selection Accuracy

AGENT EVALUATION TERMS

How often the agent chooses the correct tool, with the correct arguments, for the user’s intent and system constraints.

Trajectory Evaluation

AGENT EVALUATION TERMS

Assessing the full sequence of agent thoughts, tool calls, observations, and revisions.

Agent Trajectory

AGENT LOOP TERMS

The full path of an agent run, including plans, reasoning summaries, tool calls, observations, retries, and final output.

Critique Loop

AGENT LOOP TERMS

Critique Loop is a repeated control cycle for critique loop. It lets an agent adjust its next step after reasoning, taking action, observing results, or receiving critique.

Error Recovery

AGENT LOOP TERMS

The agent’s ability to detect a failure, choose a safe fallback, retry carefully, or ask for help without losing the task.

Final Response

AGENT LOOP TERMS

The user-facing answer or action summary produced after planning, retrieval, tool calls, and validation are complete.

Intermediate Step

AGENT LOOP TERMS

A single internal action in an agent run, such as planning, retrieving, calling a tool, checking output, or revising state.

Iteration Limit

AGENT LOOP TERMS

A hard cap on the number of reasoning or action cycles allowed, preventing infinite loops and runaway cost.

Loop Termination

AGENT LOOP TERMS

Loop Termination is a repeated control cycle for loop termination. It lets an agent adjust its next step after reasoning, taking action, observing results, or receiving critique.

Max Turns

AGENT LOOP TERMS

The maximum number of back-and-forth agent steps or conversation turns allowed before stopping, escalating, or returning a result.

Plan-Act-Observe

AGENT LOOP TERMS

An agent loop where the system creates a plan, executes an action, observes the result, and decides the next step.

ReAct Loop

AGENT LOOP TERMS

ReAct Loop is a repeated control cycle for react loop. It lets an agent adjust its next step after reasoning, taking action, observing results, or receiving critique.

Reason-Act-Observe

AGENT LOOP TERMS

A loop pattern where the model reasons about the next move, performs an action, then uses the observation to continue.

Reflection Loop

AGENT LOOP TERMS

Reflection Loop is a repeated control cycle for reflection loop. It lets an agent adjust its next step after reasoning, taking action, observing results, or receiving critique.

Replanning

AGENT LOOP TERMS

Updating the original plan when tools fail, new information appears, constraints change, or the current path is no longer valid.

Retry Loop

AGENT LOOP TERMS

Retry Loop is a repeated control cycle for retry loop. It lets an agent adjust its next step after reasoning, taking action, observing results, or receiving critique.

Scratchpad

AGENT LOOP TERMS

A private working area used to store intermediate notes, state, tool results, or structured reasoning summaries during a run.

Self-Correction

AGENT LOOP TERMS

The agent’s process of identifying mistakes in its own output or actions and revising them before final delivery.

Sense-Think-Act

AGENT LOOP TERMS

A classic agent loop: perceive the environment, decide what the information means, then take an appropriate action.

State Update

AGENT LOOP TERMS

A change to stored task status, memory, variables, or workflow progress after an observation or action.

Thought

AGENT LOOP TERMS

An internal reasoning unit or planning note used by some agent frameworks to organize decisions before action.

Verifier Loop

AGENT LOOP TERMS

Verifier Loop is a repeated control cycle for verifier loop. It lets an agent adjust its next step after reasoning, taking action, observing results, or receiving critique.

Working Memory

AGENT LOOP TERMS

Working Memory is the active information an agent uses while solving the current task, such as the plan, recent observations, and pending steps.

Conversation Memory

AGENT MEMORY TERMS

Conversation Memory is a memory type focused on conversation. It preserves information that may be useful for later turns, tasks, personalization, or reasoning.

Episodic Memory

AGENT MEMORY TERMS

Episodic Memory is a memory type focused on episodic. It preserves information that may be useful for later turns, tasks, personalization, or reasoning.

Graph Memory

AGENT MEMORY TERMS

Graph Memory is a memory type focused on graph. It preserves information that may be useful for later turns, tasks, personalization, or reasoning.

Knowledge Base

AGENT MEMORY TERMS

A curated repository of documents, FAQs, policies, tickets, product information, or enterprise procedures used for grounded answers.

Knowledge Graph

AGENT MEMORY TERMS

A structured network of entities and relationships that lets agents retrieve facts by meaning, connection, and context.

Knowledge Grounding Strategy

AGENT MEMORY TERMS

A repeatable approach for choosing, retrieving, validating, and citing trustworthy knowledge sources.

Long-Term Memory

AGENT MEMORY TERMS

Persistent information such as user preferences, facts, historical outcomes, and organizational knowledge.

Memory Compression

AGENT MEMORY TERMS

Reducing stored conversation or task history into concise summaries so important context fits within limited model windows.

Memory Conflict

AGENT MEMORY TERMS

A mismatch between stored memories, user updates, retrieved facts, or current instructions that must be resolved safely.

Memory Decay

AGENT MEMORY TERMS

A policy that lowers the importance of old information so stale preferences or outdated facts do not dominate future behavior.

Memory Deletion

AGENT MEMORY TERMS

Removing stored user, task, or system information to satisfy privacy requirements, correction requests, or retention policies.

Memory Freshness

AGENT MEMORY TERMS

How current and trustworthy a stored memory is compared with newer user input, system data, or verified sources.

Memory Poisoning

AGENT MEMORY TERMS

A security failure where false or malicious information is written into memory and later influences agent behavior.

Memory Privacy

AGENT MEMORY TERMS

Controls that limit what personal, sensitive, or enterprise information can be stored, retrieved, shared, or retained.

Memory Retrieval

AGENT MEMORY TERMS

Memory Retrieval is a retrieval method focused on memory. It finds useful information before generation, tool use, grounding, or answer verification.

Memory Summarization

AGENT MEMORY TERMS

Converting long interaction history into shorter notes that preserve decisions, preferences, facts, and unresolved tasks.

Memory Update

AGENT MEMORY TERMS

Changing stored information when a user corrects a fact, a task progresses, or the system learns a newer reliable state.

Memory Write

AGENT MEMORY TERMS

Saving a fact, preference, event, or task state so it can be retrieved in a later step or future session.

Procedural Memory

AGENT MEMORY TERMS

Procedural Memory is a memory type focused on procedural. It preserves information that may be useful for later turns, tasks, personalization, or reasoning.

Profile Memory

AGENT MEMORY TERMS

Profile Memory is a memory type focused on profile. It preserves information that may be useful for later turns, tasks, personalization, or reasoning.

Semantic Memory

AGENT MEMORY TERMS

Semantic Memory is a memory type focused on semantic. It preserves information that may be useful for later turns, tasks, personalization, or reasoning.

Session Memory

AGENT MEMORY TERMS

Session Memory is a memory type focused on session. It preserves information that may be useful for later turns, tasks, personalization, or reasoning.

Short-Term Memory

AGENT MEMORY TERMS

Temporary task or conversation context kept for the current session or execution loop.

Task Memory

AGENT MEMORY TERMS

Task Memory is a memory type focused on task. It preserves information that may be useful for later turns, tasks, personalization, or reasoning.

User Memory

AGENT MEMORY TERMS

User Memory is a memory type focused on user. It preserves information that may be useful for later turns, tasks, personalization, or reasoning.

Vector Memory

AGENT MEMORY TERMS

Vector Memory is a memory type focused on vector. It preserves information that may be useful for later turns, tasks, personalization, or reasoning.

Constraint Solving

AGENT PLANNING TERMS

Finding an action plan that satisfies limits such as budget, time, permissions, dependencies, policies, or resource availability.

Critic Agent

AGENT PLANNING TERMS

A Critic Agent reviews another agent’s plan, answer, or tool trajectory and points out errors, risks, or missing evidence.

Decision Tree

AGENT PLANNING TERMS

A branching structure that represents choices and outcomes, helping agents or systems select actions under different conditions.

Dependency Graph

AGENT PLANNING TERMS

A map of tasks or components that shows which items must happen before others can safely run.

Dynamic Planning

AGENT PLANNING TERMS

Planning that changes during execution as the agent receives new observations, tool results, failures, or user constraints.

Execution Trace

AGENT PLANNING TERMS

A recorded timeline of execution activity that helps engineers debug behavior and understand execution history.

Executor Agent

AGENT PLANNING TERMS

An Executor Agent carries out the steps created by a planner, usually by calling tools, APIs, or workflow actions.

Goal Decomposition

AGENT PLANNING TERMS

Breaking a broad objective into smaller subgoals that can be planned, assigned, executed, and evaluated.

Graph of Thoughts

AGENT PLANNING TERMS

A reasoning approach that explores connected ideas or partial solutions as a graph instead of a single linear chain.

Hierarchical Planning

AGENT PLANNING TERMS

Planning in layers, where a high-level goal is divided into phases, tasks, and concrete execution steps.

Long-Horizon Planning

AGENT PLANNING TERMS

Planning across many steps, tools, or time periods where memory, checkpoints, and recovery become important.

Milestone

AGENT PLANNING TERMS

A meaningful checkpoint that shows progress toward a larger goal and helps decide whether to continue, revise, or stop.

Plan-and-Execute

AGENT PLANNING TERMS

A pattern where one component creates a plan and another component follows the plan while reporting progress and failures.

Plan Execution

AGENT PLANNING TERMS

Carrying out planned steps through model calls, tools, APIs, retrieval, human approvals, or workflow actions.

Plan Revision

AGENT PLANNING TERMS

Changing a plan after discovering missing information, failed assumptions, blocked tools, or better available options.

Plan Validation

AGENT PLANNING TERMS

Checking that a plan is feasible, safe, complete, and aligned with the user’s goal before execution.

Planner Agent

AGENT PLANNING TERMS

A Planner Agent breaks a high-level goal into ordered steps, dependencies, and checkpoints before execution begins.

Search-Based Planning

AGENT PLANNING TERMS

Using search over possible states or actions to find a path from the current situation to a desired goal.

Short-Horizon Planning

AGENT PLANNING TERMS

Planning only the next few steps, useful when the environment changes quickly or uncertainty is high.

Step Planning

AGENT PLANNING TERMS

Turning a task into clear ordered steps that can be executed, checked, retried, or delegated.

Subtask

AGENT PLANNING TERMS

A smaller unit of work inside a larger task, often assigned to a tool, worker agent, or workflow step.

Task Decomposition

AGENT PLANNING TERMS

Splitting a complex request into manageable subtasks with clear order, dependencies, and completion criteria.

Tree of Thoughts

AGENT PLANNING TERMS

A reasoning approach that explores multiple possible solution paths and chooses the most promising branch.

Verifier Agent

AGENT PLANNING TERMS

A Verifier Agent checks whether an output, tool result, or completed task satisfies the requirements and quality criteria.

Backoff

AGENT RELIABILITY TERMS

A retry strategy that waits longer after repeated failures, reducing pressure on rate-limited or unstable services.

Checkpoint Recovery

AGENT RELIABILITY TERMS

Restoring a workflow from a saved state so an agent can resume after failure without starting over.

Concurrency

AGENT RELIABILITY TERMS

The ability to process multiple sessions, tasks, or tool calls at the same time.

Dead Letter Queue

AGENT RELIABILITY TERMS

A queue that stores failed messages or tasks for later inspection, repair, replay, or manual handling.

Degraded Mode

AGENT RELIABILITY TERMS

A reduced-capability operating state used when a model, tool, data source, or service is unavailable.

Determinism

AGENT RELIABILITY TERMS

The degree to which the same input and configuration produce the same result, important for testing and auditability.

Error Handling

AGENT RELIABILITY TERMS

Logic that detects exceptions, communicates failures clearly, retries when safe, and prevents uncontrolled agent behavior.

Exception Handling

AGENT RELIABILITY TERMS

Code-level handling of unexpected runtime problems such as bad inputs, unavailable APIs, or invalid tool outputs.

Failure Mode

AGENT RELIABILITY TERMS

A specific way an agent or system can fail, such as hallucination, wrong tool use, timeout, or unsafe escalation.

Fallback

AGENT RELIABILITY TERMS

An alternate path used when the preferred model, tool, retrieval source, or workflow cannot complete safely.

Fault Tolerance

AGENT RELIABILITY TERMS

The ability to continue operating safely when components fail, slow down, or return partial results.

Graceful Degradation

AGENT RELIABILITY TERMS

Reducing functionality in a controlled way instead of failing completely when resources or services are limited.

Idempotency

AGENT RELIABILITY TERMS

A design property where repeating the same action does not create duplicate or harmful side effects.

Non-Determinism

AGENT RELIABILITY TERMS

Behavior where the same request may produce different outputs because of sampling, model updates, tool state, or timing.

Rate Limit Handling

AGENT RELIABILITY TERMS

Managing provider or API limits through throttling, queuing, backoff, caching, or routing to alternatives.

Recovery Strategy

AGENT RELIABILITY TERMS

Recovery Strategy is a planned approach for recovery. It guides design, deployment, measurement, governance, and continuous improvement.

Reliability

AGENT RELIABILITY TERMS

The ability of an AI system to deliver correct, safe, and available behavior across normal and edge-case conditions.

Reproducibility

AGENT RELIABILITY TERMS

The ability to rerun an evaluation or agent trace and obtain comparable behavior for debugging and compliance.

Resilience

AGENT RELIABILITY TERMS

The system’s ability to absorb failures, adapt, and keep delivering acceptable service.

Retry

AGENT RELIABILITY TERMS

Repeating a failed request or tool call under controlled limits, often with backoff and idempotency protection.

Robustness

AGENT RELIABILITY TERMS

The ability to handle noisy inputs, edge cases, bad retrieval, ambiguous requests, or tool errors without breaking.

Root Cause Analysis

AGENT RELIABILITY TERMS

Investigating the underlying reason for a failure, such as bad data, prompt design, tool errors, or infrastructure limits.

Safe Failure

AGENT RELIABILITY TERMS

A failure state that avoids harm, data leakage, unsafe actions, or misleading output even when the task cannot finish.

Scalability

AGENT RELIABILITY TERMS

The ability to maintain quality, reliability, and cost control as usage grows.

Self-Healing

AGENT RELIABILITY TERMS

Automatic detection, retry, repair, or rerouting after an execution failure.

Session Management

AGENT RELIABILITY TERMS

Securely managing the lifecycle and context of individual user or task sessions.

State Recovery

AGENT RELIABILITY TERMS

Restoring task variables, memory, progress, and context after interruption or system failure.

Timeout

AGENT RELIABILITY TERMS

A maximum wait time for a model, tool, or workflow step before the system stops, retries, or falls back.

Agent Graph

AGENT WORKFLOW PATTERNS

Agent Graph is a workflow pattern for agent graph. It coordinates agent steps, tools, state, approvals, handoffs, or recovery paths.

Agentic Workflow

AGENT WORKFLOW PATTERNS

Agentic Workflow is a workflow pattern for agentic workflow. It coordinates agent steps, tools, state, approvals, handoffs, or recovery paths.

Checkpointing

AGENT WORKFLOW PATTERNS

Checkpointing is a workflow pattern for checkpointing. It coordinates agent steps, tools, state, approvals, handoffs, or recovery paths.

Critic-Reviewer

AGENT WORKFLOW PATTERNS

Critic-Reviewer is a workflow pattern for critic-reviewer. It coordinates agent steps, tools, state, approvals, handoffs, or recovery paths.

DAG

AGENT WORKFLOW PATTERNS

DAG is a workflow pattern for dag. It coordinates agent steps, tools, state, approvals, handoffs, or recovery paths.

Directed Acyclic Graph

AGENT WORKFLOW PATTERNS

Directed Acyclic Graph is a workflow pattern for directed acyclic graph. It coordinates agent steps, tools, state, approvals, handoffs, or recovery paths.

Durable Execution

AGENT WORKFLOW PATTERNS

Durable Execution is a workflow pattern for durable execution. It coordinates agent steps, tools, state, approvals, handoffs, or recovery paths.

Escalation Flow

AGENT WORKFLOW PATTERNS

Escalation Flow is a workflow pattern for escalation flow. It coordinates agent steps, tools, state, approvals, handoffs, or recovery paths.

Evaluator-Optimizer

AGENT WORKFLOW PATTERNS

Evaluator-Optimizer is a workflow pattern for evaluator-optimizer. It coordinates agent steps, tools, state, approvals, handoffs, or recovery paths.

Event-Driven Architecture

AGENT WORKFLOW PATTERNS

A design where systems react to events, triggers, messages, or state changes rather than only direct user requests.

Event-Driven Workflow

AGENT WORKFLOW PATTERNS

Event-Driven Workflow is a workflow pattern for event-driven workflow. It coordinates agent steps, tools, state, approvals, handoffs, or recovery paths.

Human Approval Flow

AGENT WORKFLOW PATTERNS

Human Approval Flow is a workflow pattern for human approval flow. It coordinates agent steps, tools, state, approvals, handoffs, or recovery paths.

Multi-Step Workflow

AGENT WORKFLOW PATTERNS

Multi-Step Workflow is a workflow pattern for multi-step workflow. It coordinates agent steps, tools, state, approvals, handoffs, or recovery paths.

Orchestration

AGENT WORKFLOW PATTERNS

Coordinating models, tools, agents, state, policies, and workflows into one reliable execution path.

Orchestrator-Worker

AGENT WORKFLOW PATTERNS

Orchestrator-Worker is a workflow pattern for orchestrator-worker. It coordinates agent steps, tools, state, approvals, handoffs, or recovery paths.

Parallelization

AGENT WORKFLOW PATTERNS

Parallelization is a workflow pattern for parallelization. It coordinates agent steps, tools, state, approvals, handoffs, or recovery paths.

Planner-Executor

AGENT WORKFLOW PATTERNS

Planner-Executor is a workflow pattern for planner-executor. It coordinates agent steps, tools, state, approvals, handoffs, or recovery paths.

RAG Workflow

AGENT WORKFLOW PATTERNS

RAG Workflow is a workflow pattern for rag workflow. It coordinates agent steps, tools, state, approvals, handoffs, or recovery paths.

Routing

AGENT WORKFLOW PATTERNS

Routing is a workflow pattern for routing. It coordinates agent steps, tools, state, approvals, handoffs, or recovery paths.

State Machine

AGENT WORKFLOW PATTERNS

State Machine is a workflow pattern for state machine. It coordinates agent steps, tools, state, approvals, handoffs, or recovery paths.

Supervisor-Worker

AGENT WORKFLOW PATTERNS

Supervisor-Worker is a workflow pattern for supervisor-worker. It coordinates agent steps, tools, state, approvals, handoffs, or recovery paths.

Tool-Use Flow

AGENT WORKFLOW PATTERNS

Tool-Use Flow is a workflow pattern for tool-use flow. It coordinates agent steps, tools, state, approvals, handoffs, or recovery paths.

Workflow

AGENT WORKFLOW PATTERNS

Workflow is a workflow pattern for workflow. It coordinates agent steps, tools, state, approvals, handoffs, or recovery paths.

Workflow Automation

AGENT WORKFLOW PATTERNS

Automating a multi-step business or technical process through tools, rules, and agent decisions.

Workflow Orchestration

AGENT WORKFLOW PATTERNS

Workflow Orchestration is a workflow pattern for workflow orchestration. It coordinates agent steps, tools, state, approvals, handoffs, or recovery paths.

Acting

AGENTIC AI FOUNDATIONS

The execution part of an agent loop where the agent does something, such as calling a tool or sending a response.

Action

AGENTIC AI FOUNDATIONS

The execution part of an agent loop where the agent does something, such as calling a tool or sending a response In this book, it is defined specifically for Action within Agentic AI Foundations.

Action Space

AGENTIC AI FOUNDATIONS

The set of actions, tools, APIs, or workflows an agent is allowed to execute.

Agency

AGENTIC AI FOUNDATIONS

Agency is a foundation concept in agentic AI. It explains how agents handle agency while pursuing goals, making decisions, acting, or staying controlled.

Agent Loop

AGENTIC AI FOUNDATIONS

Agent Loop is a foundation concept in agentic AI. It explains how agents handle agent loop while pursuing goals, making decisions, acting, or staying controlled.

Agentic AI

AGENTIC AI FOUNDATIONS

AI systems that pursue goals through planning, tool use, memory, observation, and iterative action rather than only generating one response.

AI Agent

AGENTIC AI FOUNDATIONS

A software component powered by an AI model that can interpret context, choose actions, call tools, and work toward a defined objective.

Autonomous Agent

AGENTIC AI FOUNDATIONS

An Autonomous Agent can decide and act across multiple steps with limited human intervention while staying inside defined boundaries.

Autonomy

AGENTIC AI FOUNDATIONS

The degree of independence an agent has when deciding, acting, and recovering without direct human intervention.

Completion Criteria

AGENTIC AI FOUNDATIONS

Completion Criteria is a foundation concept in agentic AI. It explains how agents handle completion criteria while pursuing goals, making decisions, acting, or staying controlled.

Environment

AGENTIC AI FOUNDATIONS

Environment is a foundation concept in agentic AI. It explains how agents handle environment while pursuing goals, making decisions, acting, or staying controlled.

Executor

AGENTIC AI FOUNDATIONS

The component or agent role that performs planned steps by calling tools, APIs, or internal workflows.

Feedback

AGENTIC AI FOUNDATIONS

Feedback is a foundation concept in agentic AI. It explains how agents handle feedback while pursuing goals, making decisions, acting, or staying controlled.

Goal

AGENTIC AI FOUNDATIONS

The target outcome an agent is trying to reach through planning and action.

Goal-Oriented Agent

AGENTIC AI FOUNDATIONS

A Goal-Oriented Agent works toward a defined outcome instead of only answering one isolated prompt.

Goal-Oriented Behavior

AGENTIC AI FOUNDATIONS

The ability to optimize actions around an outcome instead of treating every request as an isolated message.

Human-in-the-Loop

AGENTIC AI FOUNDATIONS

A pattern where humans review, approve, correct, or complete parts of an agent workflow.

Human-on-the-Loop

AGENTIC AI FOUNDATIONS

A supervision pattern where humans monitor agent behavior and intervene when thresholds or risks appear.

Human-out-of-the-Loop

AGENTIC AI FOUNDATIONS

Human-out-of-the-Loop is a foundation concept in agentic AI. It explains how agents handle human-out-of-the-loop while pursuing goals, making decisions, acting, or staying controlled.

Memory

AGENTIC AI FOUNDATIONS

The storage and retrieval layer that lets an agent preserve useful information across turns, sessions, or tasks.

Multi-Agent System

AGENTIC AI FOUNDATIONS

A design where multiple specialized agents coordinate, delegate, review, or debate to complete a complex task.

Observation

AGENTIC AI FOUNDATIONS

Information returned after an agent acts, such as a tool result, error message, retrieved document, or user response.

Perception

AGENTIC AI FOUNDATIONS

The process of receiving and interpreting input from users, tools, documents, events, or the environment.

Planner

AGENTIC AI FOUNDATIONS

The agent component that decomposes a goal into steps, dependencies, and execution order.

Planning

AGENTIC AI FOUNDATIONS

Planning is a foundation concept in agentic AI. It explains how agents handle planning while pursuing goals, making decisions, acting, or staying controlled.

Policy

AGENTIC AI FOUNDATIONS

Rules and constraints that guide how an agent should decide, respond, escalate, or use tools.

Reasoning

AGENTIC AI FOUNDATIONS

Reasoning is a foundation concept in agentic AI. It explains how agents handle reasoning while pursuing goals, making decisions, acting, or staying controlled.

Reasoning Engine

AGENTIC AI FOUNDATIONS

The decision layer that evaluates context, constraints, options, and evidence before choosing the next step.

Reflection

AGENTIC AI FOUNDATIONS

An agent pattern where the system reviews its own plan, answer, or tool trajectory and attempts improvement.

Semi-Autonomous Agent

AGENTIC AI FOUNDATIONS

A Semi-Autonomous Agent handles routine steps by itself but asks for human approval when risk, uncertainty, or policy limits appear.

Single-Agent System

AGENTIC AI FOUNDATIONS

Single-Agent System is a foundation concept in agentic AI. It explains how agents handle single-agent system while pursuing goals, making decisions, acting, or staying controlled.

State

AGENTIC AI FOUNDATIONS

The current snapshot of an agent workflow, including context, progress, memory, and pending decisions.

Step

AGENTIC AI FOUNDATIONS

One unit of work inside a larger agent plan or execution sequence.

Task

AGENTIC AI FOUNDATIONS

A specific piece of work the agent is asked to complete or contribute to.

Task-Oriented Agent

AGENTIC AI FOUNDATIONS

A Task-Oriented Agent is specialized for a specific type of work, such as scheduling, support, testing, coding, or data analysis.

Tool Use

AGENTIC AI FOUNDATIONS

Tool Use is a foundation concept in agentic AI. It explains how agents handle tool use while pursuing goals, making decisions, acting, or staying controlled.

Bug Triage

AI CODING AGENT TERMS

Bug Triage sorts reported issues by severity, owner, root cause, reproducibility, and release impact.

Build Agent

AI CODING AGENT TERMS

A Build Agent compiles, packages, and validates software changes, often inside CI/CD pipelines.

CI/CD Agent

AI CODING AGENT TERMS

A CI/CD Agent monitors pipeline runs, explains failures, suggests fixes, and helps automate release steps.

Code Generation

AI CODING AGENT TERMS

Code Generation uses AI to create source code from requirements, examples, comments, tickets, or design descriptions.

Code Review

AI CODING AGENT TERMS

Code Review uses AI or humans to inspect code for correctness, readability, security, maintainability, and test coverage.

Code Search

AI CODING AGENT TERMS

Code Search finds relevant files, functions, classes, or examples inside a codebase using keywords, embeddings, or semantic search.

Codebase RAG

AI CODING AGENT TERMS

Codebase RAG retrieves relevant repository files, symbols, and documentation so an AI assistant can answer code questions accurately.

DevOps Agent

AI CODING AGENT TERMS

A DevOps Agent assists with deployments, infrastructure checks, logs, incidents, and operational workflows.

Documentation Agent

AI CODING AGENT TERMS

A Documentation Agent creates, updates, or explains documentation from code, tickets, designs, or product behavior.

Dynamic Analysis

AI CODING AGENT TERMS

Dynamic Analysis examines software while it runs, using runtime behavior, logs, inputs, or traces to find issues.

Incident Response Agent

AI CODING AGENT TERMS

An Incident Response Agent helps investigate production incidents, connect symptoms to causes, and recommend containment or recovery steps.

Integration Test Generation

AI CODING AGENT TERMS

Integration Test Generation creates tests that verify multiple services, modules, or APIs work correctly together.

Linting

AI CODING AGENT TERMS

Linting means using AI to support linting in software engineering. It helps developers or agents write, review, test, debug, document, or release code.

Log Analysis Agent

AI CODING AGENT TERMS

A Log Analysis Agent searches and summarizes logs to identify errors, incidents, trends, and likely root causes.

Pull Request Review

AI CODING AGENT TERMS

Pull Request Review inspects proposed code changes before merge, checking quality, tests, risk, and alignment with requirements.

Refactoring

AI CODING AGENT TERMS

Refactoring changes code structure to improve readability, maintainability, or design without changing external behavior.

Release Agent

AI CODING AGENT TERMS

A Release Agent helps prepare, validate, communicate, and monitor a software release.

Repo-Aware Agent

AI CODING AGENT TERMS

A Repo-Aware Agent understands repository structure, dependencies, files, and conventions before making code suggestions.

Root Cause Agent

AI CODING AGENT TERMS

A Root Cause Agent investigates logs, traces, code changes, and symptoms to identify why a defect or incident happened.

Software Engineering Agent

AI CODING AGENT TERMS

A Software Engineering Agent helps plan, implement, test, review, and maintain software using tools and codebase context.

Static Analysis

AI CODING AGENT TERMS

Static Analysis inspects code without running it to find bugs, security risks, style issues, or maintainability problems.

Test Repair

AI CODING AGENT TERMS

Test Repair updates failing or outdated tests when application behavior, selectors, data, or APIs change.

Unit Test Generation

AI CODING AGENT TERMS

Unit Test Generation creates small tests for individual functions, methods, or classes to verify expected behavior.

API

AI ENGINEER BACKEND TERMS

An application programming interface that lets software systems exchange requests, responses, data, or actions.

API Gateway

AI ENGINEER BACKEND TERMS

API Gateway is a backend component related to api gateway. AI engineers use it to connect applications with users, services, workers, or data systems.

Authentication

AI ENGINEER BACKEND TERMS

Authentication is a backend component related to authentication. AI engineers use it to connect applications with users, services, workers, or data systems.

Authorization

AI ENGINEER BACKEND TERMS

Authorization is a backend component related to authorization. AI engineers use it to connect applications with users, services, workers, or data systems.

Background Job

AI ENGINEER BACKEND TERMS

Background Job is a backend component related to background job. AI engineers use it to connect applications with users, services, workers, or data systems.

Celery

AI ENGINEER BACKEND TERMS

A Python task queue used to run background jobs, scheduled work, and asynchronous processing.

Circuit Breaker

AI ENGINEER BACKEND TERMS

Circuit Breaker is a backend component related to circuit breaker. AI engineers use it to connect applications with users, services, workers, or data systems.

Cron Job

AI ENGINEER BACKEND TERMS

Cron Job is a backend component related to cron job. AI engineers use it to connect applications with users, services, workers, or data systems.

Django

AI ENGINEER BACKEND TERMS

A full-featured Python web framework used to build database-backed web applications and admin systems.

Express

AI ENGINEER BACKEND TERMS

A Node.js web framework used to build APIs, web servers, and middleware services.

FastAPI

AI ENGINEER BACKEND TERMS

A Python web framework often used to build fast, typed APIs for AI applications and model services.

Flask

AI ENGINEER BACKEND TERMS

A lightweight Python web framework commonly used for small APIs, prototypes, and internal AI tools.

GraphQL

AI ENGINEER BACKEND TERMS

An API query language that lets clients request exactly the fields they need from a service.

gRPC

AI ENGINEER BACKEND TERMS

A high-performance API protocol that uses typed service definitions and binary messages for fast system-to-system communication.

JWT

AI ENGINEER BACKEND TERMS

A signed JSON Web Token used to prove identity or authorization claims between services.

Kafka

AI ENGINEER BACKEND TERMS

A distributed event streaming platform used to move high-volume messages between systems.

Message Broker

AI ENGINEER BACKEND TERMS

Message Broker is a backend component related to message broker. AI engineers use it to connect applications with users, services, workers, or data systems.

Node.js

AI ENGINEER BACKEND TERMS

A JavaScript runtime used to build server-side applications, APIs, real-time services, and tool backends.

OAuth

AI ENGINEER BACKEND TERMS

An authorization standard that lets applications access resources on behalf of a user without sharing the user's password.

Queue

AI ENGINEER BACKEND TERMS

Queue is a backend component related to queue. AI engineers use it to connect applications with users, services, workers, or data systems.

RabbitMQ

AI ENGINEER BACKEND TERMS

A message broker used to route tasks or events through queues between applications.

Rate Limiting

AI ENGINEER BACKEND TERMS

Rate Limiting is a backend component related to rate limiting. AI engineers use it to connect applications with users, services, workers, or data systems.

Redis Queue

AI ENGINEER BACKEND TERMS

Redis Queue is a backend component related to redis queue. AI engineers use it to connect applications with users, services, workers, or data systems.

Request Timeout

AI ENGINEER BACKEND TERMS

Request Timeout is a backend component related to request timeout. AI engineers use it to connect applications with users, services, workers, or data systems.

REST API

AI ENGINEER BACKEND TERMS

An API style that uses standard HTTP methods such as GET, POST, PUT, and DELETE to access resources.

Retry Policy

AI ENGINEER BACKEND TERMS

Retry Policy is a backend component related to retry policy. AI engineers use it to connect applications with users, services, workers, or data systems.

Server-Sent Events

AI ENGINEER BACKEND TERMS

A web streaming method where a server pushes one-way updates to a browser or client over HTTP.

Spring Boot

AI ENGINEER BACKEND TERMS

A Java framework that simplifies building production-ready APIs, services, and enterprise applications.

Streaming Response

AI ENGINEER BACKEND TERMS

Streaming Response is a backend component related to streaming response. AI engineers use it to connect applications with users, services, workers, or data systems.

Temporal

AI ENGINEER BACKEND TERMS

A durable workflow platform used to run long-lived processes with retries, state, and recovery.

Webhook

AI ENGINEER BACKEND TERMS

An HTTP callback that lets one system notify another system when an event occurs.

WebSocket

AI ENGINEER BACKEND TERMS

A persistent two-way connection that allows servers and clients to exchange messages in real time.

Worker

AI ENGINEER BACKEND TERMS

Worker is a backend component related to worker. AI engineers use it to connect applications with users, services, workers, or data systems.

API Design

AI ENGINEER SYSTEM DESIGN TERMS

API Design means designing the structure, data flow, controls, and trade-offs for api. It turns requirements into an architecture that can be built and operated.

Architecture

AI ENGINEER SYSTEM DESIGN TERMS

The high-level structure of components, data flows, models, tools, storage, and controls in a system.

Availability

AI ENGINEER SYSTEM DESIGN TERMS

The percentage of time an AI service is usable and able to respond within expected reliability targets.

Best-of-Breed

AI ENGINEER SYSTEM DESIGN TERMS

Using specialized tools for each layer instead of one integrated platform.

Caching Strategy

AI ENGINEER SYSTEM DESIGN TERMS

Caching Strategy is a planned approach for caching. It guides design, deployment, measurement, governance, and continuous improvement.

Control Flow

AI ENGINEER SYSTEM DESIGN TERMS

The order of decisions and actions that determines which model, tool, branch, or human approval path runs next.

Cost Budget

AI ENGINEER SYSTEM DESIGN TERMS

Cost Budget sets a limit or target for cost. Engineers use it to make practical trade-offs during system design and operations.

Data Flow

AI ENGINEER SYSTEM DESIGN TERMS

The path data follows through ingestion, retrieval, model calls, tools, storage, logging, and output.

Database Design

AI ENGINEER SYSTEM DESIGN TERMS

Database Design means designing the structure, data flow, controls, and trade-offs for database. It turns requirements into an architecture that can be built and operated.

Deployment Design

AI ENGINEER SYSTEM DESIGN TERMS

Deployment Design means designing the structure, data flow, controls, and trade-offs for deployment. It turns requirements into an architecture that can be built and operated.

Enterprise-Grade

AI ENGINEER SYSTEM DESIGN TERMS

Built to satisfy enterprise expectations for security, governance, scale, support, compliance, and reliability.

Evaluation Design

AI ENGINEER SYSTEM DESIGN TERMS

Evaluation Design means designing the structure, data flow, controls, and trade-offs for evaluation. It turns requirements into an architecture that can be built and operated.

Failure Mode Analysis

AI ENGINEER SYSTEM DESIGN TERMS

A review of possible failure points and their impact so teams can design tests, guardrails, and recovery paths.

Functional Requirement

AI ENGINEER SYSTEM DESIGN TERMS

A behavior the system must provide, such as answering questions, calling tools, creating tickets, or generating reports.

Guardrail Design

AI ENGINEER SYSTEM DESIGN TERMS

Guardrail Design means designing the structure, data flow, controls, and trade-offs for guardrail. It turns requirements into an architecture that can be built and operated.

High-Level Design

AI ENGINEER SYSTEM DESIGN TERMS

High-Level Design means designing the structure, data flow, controls, and trade-offs for high-level. It turns requirements into an architecture that can be built and operated.

Latency Budget

AI ENGINEER SYSTEM DESIGN TERMS

Latency Budget sets a limit or target for latency. Engineers use it to make practical trade-offs during system design and operations.

Low-Level Design

AI ENGINEER SYSTEM DESIGN TERMS

Low-Level Design means designing the structure, data flow, controls, and trade-offs for low-level. It turns requirements into an architecture that can be built and operated.

Maintainability

AI ENGINEER SYSTEM DESIGN TERMS

How easy the system is to update, debug, test, document, and operate as requirements change.

Memory Design

AI ENGINEER SYSTEM DESIGN TERMS

Memory Design means designing the structure, data flow, controls, and trade-offs for memory. It turns requirements into an architecture that can be built and operated.

Non-Functional Requirement

AI ENGINEER SYSTEM DESIGN TERMS

A quality constraint such as latency, availability, security, privacy, cost, scalability, or maintainability.

Observability Design

AI ENGINEER SYSTEM DESIGN TERMS

Observability Design means designing the structure, data flow, controls, and trade-offs for observability. It turns requirements into an architecture that can be built and operated.

Platform Approach

AI ENGINEER SYSTEM DESIGN TERMS

Using an integrated platform for knowledge, actions, orchestration, monitoring, and governance rather than many disconnected tools.

Platform Strategy

AI ENGINEER SYSTEM DESIGN TERMS

Platform Strategy is a planned approach for platform. It guides design, deployment, measurement, governance, and continuous improvement.

POC

AI ENGINEER SYSTEM DESIGN TERMS

Proof of concept: a small validation project before larger rollout.

Privacy

AI ENGINEER SYSTEM DESIGN TERMS

Design practices that limit collection, exposure, retention, and misuse of personal or sensitive data.

Production Readiness

AI ENGINEER SYSTEM DESIGN TERMS

The level of reliability, security, observability, scalability, and support needed for live use.

Proof of Concept

AI ENGINEER SYSTEM DESIGN TERMS

A limited implementation used to validate feasibility, value, risk, and stakeholder confidence.

Queue Design

AI ENGINEER SYSTEM DESIGN TERMS

Queue Design means designing the structure, data flow, controls, and trade-offs for queue. It turns requirements into an architecture that can be built and operated.

RAG Design

AI ENGINEER SYSTEM DESIGN TERMS

RAG Design means designing the structure, data flow, controls, and trade-offs for rag. It turns requirements into an architecture that can be built and operated.

Requirement Gathering

AI ENGINEER SYSTEM DESIGN TERMS

The process of discovering user goals, constraints, data sources, risks, and success metrics before designing a system.

Roadmap

AI ENGINEER SYSTEM DESIGN TERMS

A planned evolution of capabilities, milestones, and product direction.

Tool Design

AI ENGINEER SYSTEM DESIGN TERMS

Tool Design means designing the structure, data flow, controls, and trade-offs for tool. It turns requirements into an architecture that can be built and operated.

Trade-Off Analysis

AI ENGINEER SYSTEM DESIGN TERMS

Comparing design choices such as speed versus accuracy, cost versus quality, or autonomy versus safety.

Vendor Lock-In

AI ENGINEER SYSTEM DESIGN TERMS

Risk that a product architecture becomes difficult to migrate away from due to proprietary dependencies.

Activations

AI ENGINEERING FOUNDATIONS

Intermediate values produced inside a neural network after applying transformations to inputs.

Artificial Intelligence

AI ENGINEERING FOUNDATIONS

The field of building systems that can perceive, reason, learn, generate, or act in ways associated with human intelligence.

Attention

AI ENGINEERING FOUNDATIONS

A mechanism that lets a model focus on the most relevant parts of input context.

Backpropagation

AI ENGINEERING FOUNDATIONS

The algorithm used to compute gradients through a neural network during training.

Bias

AI ENGINEERING FOUNDATIONS

Systematic error, preference, or imbalance in model behavior, data, or evaluation.

Decoder

AI ENGINEERING FOUNDATIONS

A model component that generates output tokens from internal representations and prior tokens.

Deep Learning

AI ENGINEERING FOUNDATIONS

Machine learning based on multi-layer neural networks that learn hierarchical representations from data.

Encoder

AI ENGINEERING FOUNDATIONS

A model component that converts input into internal representations.

Evaluation

AI ENGINEERING FOUNDATIONS

Measuring model or system quality against desired criteria such as correctness, safety, and business value.

Fine-Tuning

AI ENGINEERING FOUNDATIONS

Additional training that adapts a model to a task, domain, style, or preference set.

Foundation Model

AI ENGINEERING FOUNDATIONS

A general-purpose model trained at scale and adapted to tasks through prompting, retrieval, fine-tuning, or tool use.

Generalization

AI ENGINEERING FOUNDATIONS

A model’s ability to perform well on new examples beyond its training data.

Generative AI

AI ENGINEERING FOUNDATIONS

AI that creates new content such as text, code, images, audio, plans, or structured data.

Gradient Descent

AI ENGINEERING FOUNDATIONS

An optimization method that updates parameters in the direction that reduces loss.

Inference

AI ENGINEERING FOUNDATIONS

Running a trained model to produce predictions, answers, actions, or generated content.

Large Language Model

AI ENGINEERING FOUNDATIONS

A neural language model trained on large text datasets and used to understand, generate, summarize, reason, and follow instructions.

Loss Function

AI ENGINEERING FOUNDATIONS

A mathematical objective that measures how wrong a model is during training.

Machine Learning

AI ENGINEERING FOUNDATIONS

A branch of AI where models learn patterns from data instead of being explicitly programmed for every rule.

Multimodal Model

AI ENGINEERING FOUNDATIONS

Multimodal Model is a model type or component focused on multimodal. It helps produce, transform, rank, interpret, or evaluate AI outputs.

Neural Network

AI ENGINEERING FOUNDATIONS

A model composed of connected layers that transform input data into predictions or generated outputs.

Optimization

AI ENGINEERING FOUNDATIONS

Improving a model, prompt, workflow, or system to meet accuracy, cost, speed, or reliability goals.

Overfitting

AI ENGINEERING FOUNDATIONS

When a model memorizes training data patterns and performs poorly on new data.

Parameters

AI ENGINEERING FOUNDATIONS

Learned numerical values inside a model that determine how it transforms inputs into outputs.

Pretraining

AI ENGINEERING FOUNDATIONS

Large-scale initial model training on broad data before task-specific adaptation.

Self-Attention

AI ENGINEERING FOUNDATIONS

Attention applied within the same sequence so each token can relate to other tokens.

Sequence-to-Sequence Model

AI ENGINEERING FOUNDATIONS

Sequence-to-Sequence Model is a model type or component focused on sequence-to-sequence. It helps produce, transform, rank, interpret, or evaluate AI outputs.

Small Language Model

AI ENGINEERING FOUNDATIONS

Small Language Model is a model type or component focused on small language. It helps produce, transform, rank, interpret, or evaluate AI outputs.

Training

AI ENGINEERING FOUNDATIONS

The process of updating model parameters using data and an optimization objective.

Transformer

AI ENGINEERING FOUNDATIONS

A neural architecture based on attention mechanisms that powers most modern LLMs.

Underfitting

AI ENGINEERING FOUNDATIONS

When a model is too simple or poorly trained to capture useful patterns.

Variance

AI ENGINEERING FOUNDATIONS

Sensitivity of a model to changes in training data or input conditions.

Weights

AI ENGINEERING FOUNDATIONS

Model parameters that control the strength of connections between neural network components.

Adoption

AI PRODUCT AND UX TERMS

How often and how successfully users start using a product or feature.

Adoption Metrics

AI PRODUCT AND UX TERMS

Measurements showing usage, activation, retention, acceptance, and engagement with an AI system.

AI Assistant

AI PRODUCT AND UX TERMS

An AI experience that helps users complete tasks through conversation or guided actions.

AI Copilot

AI PRODUCT AND UX TERMS

An AI assistant that works alongside a user instead of fully replacing the user.

AI Product

AI PRODUCT AND UX TERMS

A product that uses AI to create value for users or automate part of a workflow.

Build vs. Buy

AI PRODUCT AND UX TERMS

The strategic choice between developing an AI capability internally or purchasing a platform or vendor solution.

Business Value

AI PRODUCT AND UX TERMS

The measurable benefit a product creates, such as revenue, savings, speed, or quality.

Change Management

AI PRODUCT AND UX TERMS

Preparing people, processes, communication, and governance so teams adopt AI safely.

Citation

AI PRODUCT AND UX TERMS

A reference to the source used to support an AI answer.

Confidence Score

AI PRODUCT AND UX TERMS

A number or label that estimates how reliable an answer, prediction, or decision is.

Confirmation Step

AI PRODUCT AND UX TERMS

A user approval step before an AI system performs an important or risky action.

Enablement

AI PRODUCT AND UX TERMS

Training and supporting users, operators, developers, and stakeholders to work effectively with AI systems.

Feedback Button

AI PRODUCT AND UX TERMS

A UI control that lets users rate, correct, or comment on an AI response.

High-Risk Action

AI PRODUCT AND UX TERMS

An action that can cause meaningful harm, loss, compliance issues, or security exposure.

Human Approval

AI PRODUCT AND UX TERMS

A design pattern where a person approves an AI action before it is completed.

Low-Risk Action

AI PRODUCT AND UX TERMS

An action with limited impact that can usually be automated with fewer approvals.

Maturity Model

AI PRODUCT AND UX TERMS

A staged framework for assessing current AI capability and planning improvement.

Onboarding

AI PRODUCT AND UX TERMS

The process of helping users understand and start using an AI product effectively.

Personalization

AI PRODUCT AND UX TERMS

Adapting an AI experience based on user context, history, role, preferences, or account data.

Risky Action

AI PRODUCT AND UX TERMS

An AI action that should be checked because it may affect money, data, safety, or access.

ROI

AI PRODUCT AND UX TERMS

Return on investment: value gained relative to cost.

Source Attribution

AI PRODUCT AND UX TERMS

Showing where information came from so users can verify the answer.

Time to Value

AI PRODUCT AND UX TERMS

The time required before users or the organization experience measurable benefit.

Transparency

AI PRODUCT AND UX TERMS

Making AI behavior, limitations, sources, and actions understandable to users.

Trust

AI PRODUCT AND UX TERMS

User confidence that the AI system is useful, safe, honest, and controllable.

Undo Action

AI PRODUCT AND UX TERMS

A way for users to reverse or recover from an AI action.

User Control

AI PRODUCT AND UX TERMS

Giving users clear choices, approvals, limits, and ways to correct the AI system.

User Goal

AI PRODUCT AND UX TERMS

The outcome the user wants to achieve.

User Intent

AI PRODUCT AND UX TERMS

The meaning or purpose behind a user request.

User Journey

AI PRODUCT AND UX TERMS

The path a user follows from need to successful outcome.

User Preference

AI PRODUCT AND UX TERMS

A user setting or habit that helps personalize the AI experience.

ABAC

AI SAFETY AND SECURITY TERMS

Attribute-based access control, where permissions depend on attributes such as user role, resource type, location, or risk.

Abuse Testing

AI SAFETY AND SECURITY TERMS

Abuse Testing validates abuse behavior through examples, scenarios, checks, or automated test runs. It helps catch issues before production.

Access Control

AI SAFETY AND SECURITY TERMS

Permission management that determines who or what can access data, tools, actions, and administrative functions.

AI Red Teaming

AI SAFETY AND SECURITY TERMS

AI Red Teaming describes a safety or security concern around ai red teaming. Teams use it to reduce misuse, data exposure, unauthorized access, or harmful AI behavior.

AI Safety

AI SAFETY AND SECURITY TERMS

AI Safety describes a safety or security concern around ai safety. Teams use it to reduce misuse, data exposure, unauthorized access, or harmful AI behavior.

AI Security

AI SAFETY AND SECURITY TERMS

AI Security describes a safety or security concern around ai security. Teams use it to reduce misuse, data exposure, unauthorized access, or harmful AI behavior.

Compliance

AI SAFETY AND SECURITY TERMS

Conformance with legal, regulatory, contractual, and organizational requirements.

Credential Leakage

AI SAFETY AND SECURITY TERMS

Credential Leakage describes a safety or security concern around credential leakage. Teams use it to reduce misuse, data exposure, unauthorized access, or harmful AI behavior.

Data Exfiltration

AI SAFETY AND SECURITY TERMS

Data Exfiltration describes a safety or security concern around data exfiltration. Teams use it to reduce misuse, data exposure, unauthorized access, or harmful AI behavior.

Data Privacy

AI SAFETY AND SECURITY TERMS

Protection of personal, sensitive, or confidential data throughout AI processing and storage.

Excessive Agency

AI SAFETY AND SECURITY TERMS

Excessive Agency describes a safety or security concern around excessive agency. Teams use it to reduce misuse, data exposure, unauthorized access, or harmful AI behavior.

Explainability

AI SAFETY AND SECURITY TERMS

The ability to describe why an agent produced an output or chose an action in terms understandable to humans.

Governance

AI SAFETY AND SECURITY TERMS

Policies, ownership, processes, and controls that make AI systems accountable and responsible.

Indirect Prompt Injection

AI SAFETY AND SECURITY TERMS

Indirect Prompt Injection describes a safety or security concern around indirect prompt injection. Teams use it to reduce misuse, data exposure, unauthorized access, or harmful AI behavior.

Insecure Output Handling

AI SAFETY AND SECURITY TERMS

Insecure Output Handling describes a safety or security concern around insecure output handling. Teams use it to reduce misuse, data exposure, unauthorized access, or harmful AI behavior.

Jailbreak

AI SAFETY AND SECURITY TERMS

Jailbreak describes a safety or security concern around jailbreak. Teams use it to reduce misuse, data exposure, unauthorized access, or harmful AI behavior.

Least Privilege

AI SAFETY AND SECURITY TERMS

Least Privilege describes a safety or security concern around least privilege. Teams use it to reduce misuse, data exposure, unauthorized access, or harmful AI behavior.

LLM Security

AI SAFETY AND SECURITY TERMS

LLM Security describes a safety or security concern around llm security. Teams use it to reduce misuse, data exposure, unauthorized access, or harmful AI behavior.

Model Denial of Service

AI SAFETY AND SECURITY TERMS

Model Denial of Service describes a safety or security concern around model denial of service. Teams use it to reduce misuse, data exposure, unauthorized access, or harmful AI behavior.

Policy Violation

AI SAFETY AND SECURITY TERMS

Policy Violation describes a safety or security concern around policy violation. Teams use it to reduce misuse, data exposure, unauthorized access, or harmful AI behavior.

Privilege Escalation

AI SAFETY AND SECURITY TERMS

Privilege Escalation describes a safety or security concern around privilege escalation. Teams use it to reduce misuse, data exposure, unauthorized access, or harmful AI behavior.

RAG Poisoning

AI SAFETY AND SECURITY TERMS

RAG Poisoning describes a safety or security concern around rag poisoning. Teams use it to reduce misuse, data exposure, unauthorized access, or harmful AI behavior.

RBAC

AI SAFETY AND SECURITY TERMS

Role-based access control, where permissions are assigned to roles and users inherit access from their role.

Red Teaming

AI SAFETY AND SECURITY TERMS

Red Teaming describes a safety or security concern around red teaming. Teams use it to reduce misuse, data exposure, unauthorized access, or harmful AI behavior.

Sandboxing

AI SAFETY AND SECURITY TERMS

Sandboxing describes a safety or security concern around sandboxing. Teams use it to reduce misuse, data exposure, unauthorized access, or harmful AI behavior.

Secrets Leakage

AI SAFETY AND SECURITY TERMS

Secrets Leakage describes a safety or security concern around secrets leakage. Teams use it to reduce misuse, data exposure, unauthorized access, or harmful AI behavior.

Secure Logging

AI SAFETY AND SECURITY TERMS

Secure Logging describes a safety or security concern around secure logging. Teams use it to reduce misuse, data exposure, unauthorized access, or harmful AI behavior.

Security

AI SAFETY AND SECURITY TERMS

Controls that protect AI systems, data, models, tools, and users from misuse or unauthorized access.

Supply Chain Vulnerability

AI SAFETY AND SECURITY TERMS

Supply Chain Vulnerability describes a safety or security concern around supply chain vulnerability. Teams use it to reduce misuse, data exposure, unauthorized access, or harmful AI behavior.

Threat Modeling

AI SAFETY AND SECURITY TERMS

Threat Modeling describes a safety or security concern around threat modeling. Teams use it to reduce misuse, data exposure, unauthorized access, or harmful AI behavior.

Tool Misuse

AI SAFETY AND SECURITY TERMS

Tool Misuse describes a safety or security concern around tool misuse. Teams use it to reduce misuse, data exposure, unauthorized access, or harmful AI behavior.

Training Data Poisoning

AI SAFETY AND SECURITY TERMS

Training Data Poisoning describes a safety or security concern around training data poisoning. Teams use it to reduce misuse, data exposure, unauthorized access, or harmful AI behavior.

Unauthorized Tool Access

AI SAFETY AND SECURITY TERMS

Unauthorized Tool Access describes a safety or security concern around unauthorized tool access. Teams use it to reduce misuse, data exposure, unauthorized access, or harmful AI behavior.

Amazon Bedrock

CLOUD AND INFRASTRUCTURE TERMS

An AWS service for building generative AI applications with managed foundation models and enterprise controls.

Anthropic API

CLOUD AND INFRASTRUCTURE TERMS

The API used to access Claude models and related capabilities such as tool use and structured outputs.

Autoscaling

CLOUD AND INFRASTRUCTURE TERMS

Automatically adding or removing compute capacity based on traffic, queue depth, latency, or resource usage.

AWS Lambda

CLOUD AND INFRASTRUCTURE TERMS

A serverless compute service that runs code in response to events without managing servers.

Azure AI Foundry

CLOUD AND INFRASTRUCTURE TERMS

Microsoft's platform for building, evaluating, deploying, and managing AI applications and agents.

Azure Blob Storage

CLOUD AND INFRASTRUCTURE TERMS

Microsoft cloud object storage for files, documents, images, logs, and other unstructured data.

Azure Functions

CLOUD AND INFRASTRUCTURE TERMS

A serverless compute service in Azure for running event-driven code.

Container

CLOUD AND INFRASTRUCTURE TERMS

A portable package that bundles application code, dependencies, and configuration so it runs consistently across environments.

CUDA

CLOUD AND INFRASTRUCTURE TERMS

NVIDIA's parallel computing platform that lets software use GPUs for accelerated model training and inference.

Deployment

CLOUD AND INFRASTRUCTURE TERMS

The process of releasing application, model, or agent changes into an environment where users or systems can access them.

Docker

CLOUD AND INFRASTRUCTURE TERMS

A container platform commonly used to package AI services, APIs, workers, and supporting infrastructure.

Environment Variables

CLOUD AND INFRASTRUCTURE TERMS

External configuration values used to set API keys, endpoints, feature flags, and runtime settings without changing code.

GCS

CLOUD AND INFRASTRUCTURE TERMS

Google Cloud Storage, an object storage service for files, model artifacts, datasets, and logs.

Google Cloud Run

CLOUD AND INFRASTRUCTURE TERMS

A managed serverless platform for running containerized services that scale based on request traffic.

GPU

CLOUD AND INFRASTRUCTURE TERMS

A graphics processing unit used to accelerate matrix operations for model training and inference.

Helm

CLOUD AND INFRASTRUCTURE TERMS

A Kubernetes package manager used to install and configure applications with reusable charts.

Image

CLOUD AND INFRASTRUCTURE TERMS

A versioned container template containing code, dependencies, and runtime instructions used to start containers.

Infrastructure as Code

CLOUD AND INFRASTRUCTURE TERMS

Managing cloud resources with versioned configuration files so environments can be reviewed, reproduced, and automated.

Kubernetes

CLOUD AND INFRASTRUCTURE TERMS

A container orchestration system for deploying, scaling, and managing services across clusters.

Load Balancer

CLOUD AND INFRASTRUCTURE TERMS

A component that distributes traffic across service instances to improve availability, throughput, and fault tolerance.

Model Serving

CLOUD AND INFRASTRUCTURE TERMS

Hosting a model behind an API or inference endpoint so applications can send inputs and receive predictions.

Object Storage

CLOUD AND INFRASTRUCTURE TERMS

Durable storage for large files such as documents, logs, datasets, embeddings, exports, and model artifacts.

Ollama

CLOUD AND INFRASTRUCTURE TERMS

A local runtime for downloading and running open-source language models on a developer machine.

OpenAI API

CLOUD AND INFRASTRUCTURE TERMS

The API used to access OpenAI models, embeddings, tools, structured outputs, and agent capabilities.

Pod

CLOUD AND INFRASTRUCTURE TERMS

The smallest deployable unit in Kubernetes, usually containing one or more containers that run together.

S3

CLOUD AND INFRASTRUCTURE TERMS

Amazon Simple Storage Service, an object store often used for datasets, logs, artifacts, and document repositories.

Secrets Manager

CLOUD AND INFRASTRUCTURE TERMS

A secure service for storing and rotating API keys, database passwords, tokens, and certificates.

Serverless

CLOUD AND INFRASTRUCTURE TERMS

A deployment model where the cloud provider manages servers and automatically scales execution for events or requests.

Service

CLOUD AND INFRASTRUCTURE TERMS

A network-accessible application component, often exposing an API for agents, tools, users, or other systems.

Terraform

CLOUD AND INFRASTRUCTURE TERMS

An infrastructure-as-code tool used to define and provision cloud resources through configuration files.

TGI

CLOUD AND INFRASTRUCTURE TERMS

Text Generation Inference, a server for deploying and serving transformer-based language models.

TPU

CLOUD AND INFRASTRUCTURE TERMS

A tensor processing unit, Google's accelerator hardware for machine learning workloads.

Vertex AI

CLOUD AND INFRASTRUCTURE TERMS

Google Cloud's managed platform for building, training, deploying, and monitoring ML and generative AI systems.

vLLM

CLOUD AND INFRASTRUCTURE TERMS

A high-throughput inference engine for serving large language models efficiently.

AutoGen

COMMON FRAMEWORK AND TOOL TERMS

A framework for building applications where multiple AI agents communicate, collaborate, and execute tasks.

Claude Agent SDK

COMMON FRAMEWORK AND TOOL TERMS

An SDK for building agent experiences around Claude models, tools, workflows, and application integrations.

Claude API

COMMON FRAMEWORK AND TOOL TERMS

The API used to access Anthropic Claude models for language, reasoning, and tool-use tasks.

CrewAI

COMMON FRAMEWORK AND TOOL TERMS

A framework for creating role-based teams of agents that collaborate on structured tasks.

DSPy

COMMON FRAMEWORK AND TOOL TERMS

A framework for programming and optimizing language-model pipelines through modules and evaluators.

Google ADK

COMMON FRAMEWORK AND TOOL TERMS

Google’s agent development kit for building and orchestrating AI agents in Google’s ecosystem.

Gradio

COMMON FRAMEWORK AND TOOL TERMS

A Python library for quickly creating web interfaces for AI models, demos, and internal tools.

Haystack

COMMON FRAMEWORK AND TOOL TERMS

An open-source framework for search, RAG, question answering, and NLP pipelines.

Hugging Face Transformers

COMMON FRAMEWORK AND TOOL TERMS

A popular library that provides pretrained transformer models, tokenizers, and training utilities.

LangChain

COMMON FRAMEWORK AND TOOL TERMS

A framework for building LLM applications with prompts, chains, tools, agents, memory, and retrieval.

LangChain4j

COMMON FRAMEWORK AND TOOL TERMS

A Java library for building LLM and agent applications with tools, memory, and RAG integrations.

LangGraph

COMMON FRAMEWORK AND TOOL TERMS

A graph-based framework for building stateful, controllable, multi-step agent workflows.

LlamaIndex

COMMON FRAMEWORK AND TOOL TERMS

A data framework that connects LLMs to documents, indexes, databases, and retrieval pipelines.

OpenAI Agents SDK

COMMON FRAMEWORK AND TOOL TERMS

A software development kit for building agent workflows with tools, handoffs, tracing, and guardrails.

OpenAI Responses API

COMMON FRAMEWORK AND TOOL TERMS

An OpenAI API pattern for generating model responses with tools, structured outputs, and multimodal inputs.

PydanticAI

COMMON FRAMEWORK AND TOOL TERMS

A Python agent framework that uses Pydantic-style types and validation for structured AI applications.

Semantic Kernel

COMMON FRAMEWORK AND TOOL TERMS

A Microsoft framework for combining prompts, plugins, planners, memory, and native code in AI apps.

Spring AI

COMMON FRAMEWORK AND TOOL TERMS

A Spring ecosystem project that brings AI model, embedding, vector store, and tool integrations to Java applications.

Streamlit

COMMON FRAMEWORK AND TOOL TERMS

A Python library for building simple interactive data and AI apps with minimal front-end code.

Vertex AI Agent Builder

COMMON FRAMEWORK AND TOOL TERMS

A Google Cloud product for building enterprise agents connected to data, search, and workflows.

Context-Aware Agent

CONTEXT ENGINEERING TERMS

A Context-Aware Agent uses user, task, memory, tool, and retrieved context to make responses and actions more relevant.

Context Budget

CONTEXT ENGINEERING TERMS

Context Budget sets a limit or target for context. Engineers use it to make practical trade-offs during system design and operations.

Context Drift

CONTEXT ENGINEERING TERMS

Context Drift is a context-engineering action for context drift. It decides how information is added, removed, ordered, refreshed, separated, or prioritized for the model.

Context Engineering

CONTEXT ENGINEERING TERMS

Context Engineering is a context-engineering action for context engineering. It decides how information is added, removed, ordered, refreshed, separated, or prioritized for the model.

Context Grounding

CONTEXT ENGINEERING TERMS

Context Grounding is a context-engineering action for context grounding. It decides how information is added, removed, ordered, refreshed, separated, or prioritized for the model.

Context Injection

CONTEXT ENGINEERING TERMS

Context Injection is a context-engineering action for context injection. It decides how information is added, removed, ordered, refreshed, separated, or prioritized for the model.

Context Isolation

CONTEXT ENGINEERING TERMS

Context Isolation is a context-engineering action for context isolation. It decides how information is added, removed, ordered, refreshed, separated, or prioritized for the model.

Context Pollution

CONTEXT ENGINEERING TERMS

Context Pollution is a context-engineering action for context pollution. It decides how information is added, removed, ordered, refreshed, separated, or prioritized for the model.

Context Pruning

CONTEXT ENGINEERING TERMS

Context Pruning is a context-engineering action for context pruning. It decides how information is added, removed, ordered, refreshed, separated, or prioritized for the model.

Context Ranking

CONTEXT ENGINEERING TERMS

Context Ranking is a context-engineering action for context ranking. It decides how information is added, removed, ordered, refreshed, separated, or prioritized for the model.

Context Refresh

CONTEXT ENGINEERING TERMS

Context Refresh is a context-engineering action for context refresh. It decides how information is added, removed, ordered, refreshed, separated, or prioritized for the model.

Context Retrieval

CONTEXT ENGINEERING TERMS

Context Retrieval is a retrieval method focused on context. It finds useful information before generation, tool use, grounding, or answer verification.

Context Selection

CONTEXT ENGINEERING TERMS

Context Selection is a context-engineering action for context selection. It decides how information is added, removed, ordered, refreshed, separated, or prioritized for the model.

Contextual Awareness

CONTEXT ENGINEERING TERMS

Understanding the situation, user role, history, policy, and environment beyond the raw text input.

Conversation State

CONTEXT ENGINEERING TERMS

A context-engineering term for conversation state, guiding what information a model receives and how that information is controlled.

Instruction Hierarchy

CONTEXT ENGINEERING TERMS

A context-engineering term for instruction hierarchy, guiding what information a model receives and how that information is controlled.

Memory Context

CONTEXT ENGINEERING TERMS

Memory Context is the context related to memory. Supplying it helps the model interpret the request, choose actions, and avoid missing important information.

Retrieved Context

CONTEXT ENGINEERING TERMS

Retrieved Context is the context related to retrieved. Supplying it helps the model interpret the request, choose actions, and avoid missing important information.

State Management

CONTEXT ENGINEERING TERMS

Tracking the current context, progress, memory, pending actions, and decisions across an agent workflow.

State Persistence

CONTEXT ENGINEERING TERMS

A context-engineering term for state persistence, guiding what information a model receives and how that information is controlled.

System Context

CONTEXT ENGINEERING TERMS

System Context is the context related to system. Supplying it helps the model interpret the request, choose actions, and avoid missing important information.

Task Context

CONTEXT ENGINEERING TERMS

Task Context is the context related to task. Supplying it helps the model interpret the request, choose actions, and avoid missing important information.

Tool Context

CONTEXT ENGINEERING TERMS

Tool Context is the context related to tool. Supplying it helps the model interpret the request, choose actions, and avoid missing important information.

User Context

CONTEXT ENGINEERING TERMS

User Context is the context related to user. Supplying it helps the model interpret the request, choose actions, and avoid missing important information.

Async Execution

COST AND PERFORMANCE TERMS

Running work without blocking the main request, useful for slow tools, background tasks, and parallel operations.

Batching

COST AND PERFORMANCE TERMS

Grouping multiple requests or tasks together to improve throughput and reduce overhead.

Caching

COST AND PERFORMANCE TERMS

Saving reusable results, prompts, embeddings, retrieval outputs, or responses to reduce latency and cost.

Cost Optimization

COST AND PERFORMANCE TERMS

Reducing model, infrastructure, token, storage, and operations cost while preserving quality.

Cost per Request

COST AND PERFORMANCE TERMS

The average spend for one user or API request, including model usage, retrieval, tools, and infrastructure.

Cost per User

COST AND PERFORMANCE TERMS

The average AI system cost attributable to one user over a period, useful for pricing and capacity planning.

Distillation

COST AND PERFORMANCE TERMS

Training a smaller model to imitate a larger model so inference becomes cheaper or faster.

Inference Cost

COST AND PERFORMANCE TERMS

The compute and provider cost required to run a model and generate outputs after training is complete.

Large Model Escalation

COST AND PERFORMANCE TERMS

Routing difficult, risky, or low-confidence tasks from a smaller model to a stronger model.

Load Shedding

COST AND PERFORMANCE TERMS

Temporarily rejecting or delaying lower-priority work to protect system stability during overload.

Parallel Execution

COST AND PERFORMANCE TERMS

Running independent steps at the same time to reduce total task latency.

Prompt Cache

COST AND PERFORMANCE TERMS

Prompt Cache stores reusable prompt results. It reduces repeated work, lowers cost, and improves speed when similar requests appear again.

Response Cache

COST AND PERFORMANCE TERMS

Response Cache stores reusable response results. It reduces repeated work, lowers cost, and improves speed when similar requests appear again.

Retrieval Cache

COST AND PERFORMANCE TERMS

Retrieval Cache stores reusable retrieval results. It reduces repeated work, lowers cost, and improves speed when similar requests appear again.

Small Model Routing

COST AND PERFORMANCE TERMS

Sending simple or low-risk tasks to cheaper smaller models while reserving larger models for harder cases.

Streaming

COST AND PERFORMANCE TERMS

Delivering output incrementally as it is generated, improving perceived responsiveness for users.

Token Cost

COST AND PERFORMANCE TERMS

The price associated with processing input and output tokens during model inference.

Token Management

COST AND PERFORMANCE TERMS

Controlling prompt size, retrieved context, output length, and caching to manage cost and latency.

Tokens per Second

COST AND PERFORMANCE TERMS

A throughput metric showing how quickly a model produces or processes tokens.

Autonomous Enterprise

CX, AUTOMATION, AND BUSINESS OUTCOME TERMS

An organization where agents handle many repeatable tasks within guardrails while humans focus on judgment, exceptions, and relationships.

Autonomous Resolution

CX, AUTOMATION, AND BUSINESS OUTCOME TERMS

Completing a user or customer task without human intervention while staying within policy.

Case Deflection

CX, AUTOMATION, AND BUSINESS OUTCOME TERMS

Resolving a request before it becomes a support ticket or human case.

Conversational AI

CX, AUTOMATION, AND BUSINESS OUTCOME TERMS

AI systems that interact through natural language in chat, voice, or messaging experiences.

CSAT

CX, AUTOMATION, AND BUSINESS OUTCOME TERMS

Customer Satisfaction Score, a metric that measures how satisfied customers are after an interaction or experience.

CX Automation

CX, AUTOMATION, AND BUSINESS OUTCOME TERMS

Using AI agents and workflows to automate customer experience interactions and related back-office actions.

Decision Support

CX, AUTOMATION, AND BUSINESS OUTCOME TERMS

AI assistance that helps humans analyze options without fully automating the final judgment.

FCR

CX, AUTOMATION, AND BUSINESS OUTCOME TERMS

First Contact Resolution, the percentage of support issues solved in the first interaction.

First Contact Resolution

CX, AUTOMATION, AND BUSINESS OUTCOME TERMS

Resolving a customer issue during the first interaction without follow-up handoffs.

Intent Classification

CX, AUTOMATION, AND BUSINESS OUTCOME TERMS

Assigning a user request to a known category so it can be routed or automated.

Natural Language Generation

CX, AUTOMATION, AND BUSINESS OUTCOME TERMS

Producing clear, coherent, context-appropriate language as an output.

Natural Language Understanding

CX, AUTOMATION, AND BUSINESS OUTCOME TERMS

The process of interpreting intent, entities, meaning, and context from human language.

Net Promoter Score

CX, AUTOMATION, AND BUSINESS OUTCOME TERMS

A loyalty metric based on how likely customers are to recommend a product or company.

NLG

CX, AUTOMATION, AND BUSINESS OUTCOME TERMS

Natural Language Generation, the capability of producing fluent text or speech from context or data.

NLU

CX, AUTOMATION, AND BUSINESS OUTCOME TERMS

Natural Language Understanding, the capability of extracting intent, entities, and meaning from user language.

NPS

CX, AUTOMATION, AND BUSINESS OUTCOME TERMS

Net Promoter Score, a measure of customer loyalty based on willingness to recommend.

Operational AI

CX, AUTOMATION, AND BUSINESS OUTCOME TERMS

AI embedded directly inside recurring business operations and workflows.

Proactivity

CX, AUTOMATION, AND BUSINESS OUTCOME TERMS

The capability of an agent to initiate helpful actions based on triggers, context, or predicted needs.

Recommendation Engine

CX, AUTOMATION, AND BUSINESS OUTCOME TERMS

A system that suggests next-best content, actions, products, or decisions from context and data.

Sentiment Analysis

CX, AUTOMATION, AND BUSINESS OUTCOME TERMS

Detecting emotional tone, urgency, satisfaction, or frustration in user language.

Slot Filling

CX, AUTOMATION, AND BUSINESS OUTCOME TERMS

Extracting structured fields such as date, product, account, location, or issue type from unstructured input.

TTV

CX, AUTOMATION, AND BUSINESS OUTCOME TERMS

Time to Value, the time between adoption and the first meaningful business or user benefit.

Alert Triage

CYBERSECURITY AGENT TERMS

Reviewing security alerts to decide urgency, likely cause, and next action.

Attack Path Analysis

CYBERSECURITY AGENT TERMS

Mapping how an attacker could move through systems to reach a target.

Blue Team Agent

CYBERSECURITY AGENT TERMS

An agent that helps defenders monitor, detect, investigate, and respond to threats.

Compliance Agent

CYBERSECURITY AGENT TERMS

An agent that checks whether systems or processes follow security and regulatory rules.

Detection Engineering Agent

CYBERSECURITY AGENT TERMS

An agent that helps create, test, and improve security detection rules.

Log Investigation

CYBERSECURITY AGENT TERMS

Analyzing logs to understand suspicious activity, errors, or security events.

Malware Analysis Agent

CYBERSECURITY AGENT TERMS

An agent that helps inspect suspicious files, behaviors, indicators, and malware reports.

MITRE ATLAS

CYBERSECURITY AGENT TERMS

A knowledge base that describes threats, techniques, and defenses for AI systems.

MITRE ATT&CK

CYBERSECURITY AGENT TERMS

A public knowledge base that describes attacker tactics, techniques, and procedures.

Phishing Analysis Agent

CYBERSECURITY AGENT TERMS

An agent that helps inspect suspicious emails, links, attachments, and sender behavior.

Purple Team Agent

CYBERSECURITY AGENT TERMS

An agent that helps connect offensive testing with defensive improvement.

Red Team Agent

CYBERSECURITY AGENT TERMS

An agent that helps simulate attacks to test security controls and find weaknesses.

Risk Scoring

CYBERSECURITY AGENT TERMS

Assigning a severity or priority score to a threat, vulnerability, or finding.

Security Agent

CYBERSECURITY AGENT TERMS

An agent that assists with monitoring, investigation, response, or security decision support.

Security Automation

CYBERSECURITY AGENT TERMS

Using automation and AI to speed up security operations and reduce manual work.

Security Policy Agent

CYBERSECURITY AGENT TERMS

An agent that checks actions, configurations, or content against security policies.

SIEM Agent

CYBERSECURITY AGENT TERMS

An agent that works with SIEM data to search, summarize, and investigate security events.

SOAR Agent

CYBERSECURITY AGENT TERMS

An agent that supports security orchestration, automation, and response workflows.

SOC Agent

CYBERSECURITY AGENT TERMS

An agent that helps a security operations center triage, investigate, and respond to alerts.

Threat Intelligence Agent

CYBERSECURITY AGENT TERMS

An agent that gathers and summarizes threat indicators, actor behavior, and risk context.

Vulnerability Agent

CYBERSECURITY AGENT TERMS

An agent that helps find, prioritize, and explain security vulnerabilities.

Anomaly Detection

DATA AND ANALYTICS AGENT TERMS

Finding unusual patterns or values that may indicate errors, fraud, outages, or business changes.

BI Agent

DATA AND ANALYTICS AGENT TERMS

An agent that helps answer business intelligence questions and explain dashboard metrics.

BigQuery Agent

DATA AND ANALYTICS AGENT TERMS

An agent that queries, explains, or manages data workflows in Google BigQuery.

Dashboard Agent

DATA AND ANALYTICS AGENT TERMS

An agent that creates, explains, or monitors dashboards and business metrics.

Data Agent

DATA AND ANALYTICS AGENT TERMS

An agent that retrieves, validates, transforms, or explains data for users or workflows.

Data Analyst Agent

DATA AND ANALYTICS AGENT TERMS

An agent that answers business questions by querying data and explaining insights.

Data Cleaning Agent

DATA AND ANALYTICS AGENT TERMS

An agent that detects and fixes missing, duplicate, invalid, or inconsistent data.

Data Pipeline Agent

DATA AND ANALYTICS AGENT TERMS

An agent that monitors, explains, or helps repair data movement and transformation workflows.

Data Quality Agent

DATA AND ANALYTICS AGENT TERMS

An agent that checks data accuracy, completeness, freshness, consistency, and validity.

Databricks Agent

DATA AND ANALYTICS AGENT TERMS

An agent that works with Databricks notebooks, jobs, data, and lakehouse workflows.

ELT Agent

DATA AND ANALYTICS AGENT TERMS

An agent that helps load data first and transform it later inside the data platform.

ETL Agent

DATA AND ANALYTICS AGENT TERMS

An agent that helps extract, transform, and load data across systems.

Forecasting Agent

DATA AND ANALYTICS AGENT TERMS

An agent that predicts future values such as demand, revenue, traffic, or risk.

Insight Generation

DATA AND ANALYTICS AGENT TERMS

Turning data patterns into useful explanations, trends, and recommended actions.

Metric Explanation

DATA AND ANALYTICS AGENT TERMS

Explaining what a metric means, why it changed, and what may have caused the change.

Notebook Agent

DATA AND ANALYTICS AGENT TERMS

An agent that assists with analysis, code, charts, and explanations inside notebooks.

Pandas Agent

DATA AND ANALYTICS AGENT TERMS

An agent that uses pandas-style data operations to clean, analyze, and summarize datasets.

Query Agent

DATA AND ANALYTICS AGENT TERMS

An agent that creates, validates, executes, or explains database queries.

Query Validation

DATA AND ANALYTICS AGENT TERMS

Checking that a generated query is safe, correct, efficient, and aligned with user intent.

Report Generation

DATA AND ANALYTICS AGENT TERMS

Creating written summaries, tables, charts, or business reports from data.

Safe SQL Execution

DATA AND ANALYTICS AGENT TERMS

Running SQL with protections such as read-only access, limits, validation, and approval.

Snowflake Agent

DATA AND ANALYTICS AGENT TERMS

An agent that queries, explains, or manages workflows in Snowflake.

SQL Agent

DATA AND ANALYTICS AGENT TERMS

An SQL Agent turns natural-language questions into safe database queries, validates the SQL, runs approved queries, and explains the result.

Text-to-SQL

DATA AND ANALYTICS AGENT TERMS

Text-to-SQL converts a user question into a structured SQL query so databases can be searched without manually writing SQL.

Adapter Tuning

FINE-TUNING TERMS

A fine-tuning method that adds small trainable adapter layers while keeping most original model weights frozen.

Batch Size

FINE-TUNING TERMS

The number of training examples processed together before updating model parameters.

Catastrophic Forgetting

FINE-TUNING TERMS

When fine-tuning causes a model to lose useful abilities it learned during earlier training.

Checkpoint

FINE-TUNING TERMS

A saved model state that can be reused, evaluated, restored, or continued during training.

Data Cleaning

FINE-TUNING TERMS

Removing errors, duplicates, unsafe content, or irrelevant examples from training or evaluation data.

Data Curation

FINE-TUNING TERMS

Selecting and organizing high-quality examples that teach the model the desired behavior.

DPO

FINE-TUNING TERMS

Direct Preference Optimization, a preference-tuning method that trains a model to favor preferred responses without a separate reward model.

Epoch

FINE-TUNING TERMS

One full pass through the training dataset during model training.

Fine-Tuning Evaluation

FINE-TUNING TERMS

Testing a tuned model against base-model behavior, target tasks, safety cases, and regression benchmarks.

Full Fine-Tuning

FINE-TUNING TERMS

Updating all or most model weights during fine-tuning instead of using small adapters.

Human Preference Data

FINE-TUNING TERMS

Examples where people compare or rate outputs to teach the model preferred behavior.

Hyperparameter

FINE-TUNING TERMS

A training configuration value, such as learning rate or batch size, chosen before training begins.

Instruction Tuning

FINE-TUNING TERMS

Fine-tuning a model on instruction-response examples so it follows user requests better.

Learning Rate

FINE-TUNING TERMS

Learning Rate is a training hyperparameter that controls how large each model-weight update is during optimization.

LoRA

FINE-TUNING TERMS

Low-Rank Adaptation, a parameter-efficient fine-tuning method that trains small adapter matrices instead of all model weights.

Model Merge

FINE-TUNING TERMS

Combining weights or adapters from multiple models to create a new model variant.

PEFT

FINE-TUNING TERMS

Parameter-efficient fine-tuning, a family of methods that adapt models by training only a small number of parameters.

PPO

FINE-TUNING TERMS

Proximal Policy Optimization, a reinforcement learning algorithm often discussed in RLHF training pipelines.

Preference Dataset

FINE-TUNING TERMS

A dataset containing preferred and rejected outputs used to train alignment behavior.

Prefix Tuning

FINE-TUNING TERMS

A parameter-efficient method that trains prefix vectors added to the model input.

Prompt Tuning

FINE-TUNING TERMS

A parameter-efficient method that learns soft prompt embeddings instead of changing the full model.

QLoRA

FINE-TUNING TERMS

Quantized LoRA, a memory-efficient fine-tuning method that combines quantization with LoRA adapters.

Reward Model

FINE-TUNING TERMS

Reward Model is a model trained to score outputs so a fine-tuning or alignment process can prefer better responses.

RLAIF

FINE-TUNING TERMS

Reinforcement Learning from AI Feedback, where AI-generated preferences help guide model alignment.

RLHF

FINE-TUNING TERMS

Reinforcement Learning from Human Feedback, where human preferences guide model behavior after pretraining.

SFT

FINE-TUNING TERMS

Supervised Fine-Tuning, where a model is trained on labeled examples of desired instructions and responses.

Supervised Fine-Tuning

FINE-TUNING TERMS

Training a pretrained model on labeled prompt-response examples.

Synthetic Data

FINE-TUNING TERMS

Artificially generated examples used for testing, evaluation, training, simulation, or privacy-preserving development.

Test Dataset

FINE-TUNING TERMS

Held-out data used for final evaluation after training decisions are complete.

Training Dataset

FINE-TUNING TERMS

The examples used to update model parameters during training.

Validation Dataset

FINE-TUNING TERMS

Held-out data used during development to tune settings and detect overfitting.

Approval Gate

GUARDRAILS TERMS

A required human or policy checkpoint before the agent performs a risky, expensive, or irreversible action.

Auditability

GUARDRAILS TERMS

The ability to reconstruct what the agent saw, decided, called, and produced for review or compliance.

Compliance Check

GUARDRAILS TERMS

A validation step that verifies output or action meets legal, regulatory, contractual, or internal policy requirements.

Content Filter

GUARDRAILS TERMS

A rule or model that blocks, labels, or redirects content that violates safety, quality, or policy standards.

Data Loss Prevention

GUARDRAILS TERMS

Controls that detect and prevent sensitive data from being exposed, copied, logged, or sent to unsafe destinations.

Escalation

GUARDRAILS TERMS

Routing a case to a human, specialist agent, or safer workflow when confidence, risk, or complexity requires it.

Escalation Path

GUARDRAILS TERMS

A route from the agent to a human, specialist, or safer workflow when automation should not continue alone.

Fallback Response

GUARDRAILS TERMS

A safe alternative answer used when the model is uncertain, retrieval fails, tools are unavailable, or policy blocks completion.

Fallback Strategy

GUARDRAILS TERMS

A predefined alternative path when an agent has low confidence, fails, times out, or reaches a safety boundary.

Guardrail

GUARDRAILS TERMS

A rule, check, model, or workflow constraint that keeps AI behavior safe, compliant, and aligned with expectations.

Guardrails

GUARDRAILS TERMS

Rules, checks, filters, permissions, and approval gates that keep agent behavior safe and compliant.

Human Review

GUARDRAILS TERMS

Manual inspection of an AI output, decision, or planned action before it is approved, revised, or rejected.

Input Guardrail

GUARDRAILS TERMS

A check applied to user input before model processing, often detecting harmful requests, prompt injection, or sensitive data.

Jailbreak Detection

GUARDRAILS TERMS

The ability to identify jailbreak signals in inputs, outputs, logs, retrieved content, or system behavior.

Moderation

GUARDRAILS TERMS

Classifying content for safety categories so the system can allow, block, transform, or escalate it appropriately.

Output Guardrail

GUARDRAILS TERMS

A check applied after generation to catch unsafe, incorrect, private, or non-compliant output before delivery.

PII Detection

GUARDRAILS TERMS

The ability to identify pii signals in inputs, outputs, logs, retrieved content, or system behavior.

PII Redaction

GUARDRAILS TERMS

Removing or masking personally identifiable information so it is not exposed to users, logs, models, or downstream tools.

Policy-as-Code

GUARDRAILS TERMS

Representing rules and compliance logic in executable configuration so checks are consistent and auditable.

Policy Check

GUARDRAILS TERMS

A validation step that compares a request, plan, tool call, or answer against approved rules.

Prompt Injection Detection

GUARDRAILS TERMS

The ability to identify prompt injection signals in inputs, outputs, logs, retrieved content, or system behavior.

Refusal

GUARDRAILS TERMS

A safe response that declines to help with disallowed or harmful requests while keeping the tone professional.

Risk Score

GUARDRAILS TERMS

A numeric or labeled measure that estimates risk for an output, action, user experience, or workflow result.

Safe Completion

GUARDRAILS TERMS

A response that answers within allowed boundaries while avoiding unsafe instructions, private data, or unsupported claims.

Safety Check

GUARDRAILS TERMS

A pre- or post-processing validation that looks for harm, misuse, policy violation, or high-risk behavior.

Tool Permission Check

GUARDRAILS TERMS

A verification that the agent is allowed to use a specific tool, data source, action, or permission scope.

Agent State

INTERVIEW-SPECIFIC AGENTIC AI TERMS

Agent State is an interview concept about agent state. Candidates use it to explain agent architecture, trade-offs, safety controls, evaluation, or production readiness.

Agent vs Chatbot

INTERVIEW-SPECIFIC AGENTIC AI TERMS

Agent vs Chatbot is an interview concept about agent vs chatbot. Candidates use it to explain agent architecture, trade-offs, safety controls, evaluation, or production readiness.

Agent vs Workflow

INTERVIEW-SPECIFIC AGENTIC AI TERMS

Agent vs Workflow is an interview concept about agent vs workflow. Candidates use it to explain agent architecture, trade-offs, safety controls, evaluation, or production readiness.

AI Governance

INTERVIEW-SPECIFIC AGENTIC AI TERMS

AI Governance is an interview concept about ai governance. Candidates use it to explain agent architecture, trade-offs, safety controls, evaluation, or production readiness.

Evaluation Metrics

INTERVIEW-SPECIFIC AGENTIC AI TERMS

Evaluation Metrics is an interview concept about evaluation metrics. Candidates use it to explain agent architecture, trade-offs, safety controls, evaluation, or production readiness.

Failure Handling

INTERVIEW-SPECIFIC AGENTIC AI TERMS

Failure Handling is an interview concept about failure handling. Candidates use it to explain agent architecture, trade-offs, safety controls, evaluation, or production readiness.

Latency Optimization

INTERVIEW-SPECIFIC AGENTIC AI TERMS

Improving latency so the AI system becomes faster, cheaper, more accurate, or more useful.

Multi-Agent Architecture

INTERVIEW-SPECIFIC AGENTIC AI TERMS

Multi-Agent Architecture is an interview concept about multi-agent architecture. Candidates use it to explain agent architecture, trade-offs, safety controls, evaluation, or production readiness.

Planner-Executor Pattern

INTERVIEW-SPECIFIC AGENTIC AI TERMS

Planner-Executor Pattern is an interview concept about planner-executor pattern. Candidates use it to explain agent architecture, trade-offs, safety controls, evaluation, or production readiness.

Prompt Injection Defense

INTERVIEW-SPECIFIC AGENTIC AI TERMS

Prompt Injection Defense is an interview concept about prompt injection defense. Candidates use it to explain agent architecture, trade-offs, safety controls, evaluation, or production readiness.

RAG Architecture

INTERVIEW-SPECIFIC AGENTIC AI TERMS

RAG Architecture is an interview concept about rag architecture. Candidates use it to explain agent architecture, trade-offs, safety controls, evaluation, or production readiness.

ReAct Pattern

INTERVIEW-SPECIFIC AGENTIC AI TERMS

ReAct Pattern is an interview concept about react pattern. Candidates use it to explain agent architecture, trade-offs, safety controls, evaluation, or production readiness.

Responsible AI

INTERVIEW-SPECIFIC AGENTIC AI TERMS

Responsible AI is an interview concept about responsible ai. Candidates use it to explain agent architecture, trade-offs, safety controls, evaluation, or production readiness.

Tool Misuse Prevention

INTERVIEW-SPECIFIC AGENTIC AI TERMS

Tool Misuse Prevention is an interview concept about tool misuse prevention. Candidates use it to explain agent architecture, trade-offs, safety controls, evaluation, or production readiness.

Alignment

LLM CORE TERMS

The process of making model behavior better match human goals, instructions, and safety expectations.

Assistant Message

LLM CORE TERMS

A model-generated response stored as part of the conversation context.

Audio Model

LLM CORE TERMS

Audio Model is a model type or component focused on audio. It helps produce, transform, rank, interpret, or evaluate AI outputs.

Chat Model

LLM CORE TERMS

A model optimized for multi-turn conversational interaction.

Completion

LLM CORE TERMS

The model-generated continuation, answer, or output produced from a prompt.

Completion Model

LLM CORE TERMS

Completion Model is a model type or component focused on completion. It helps produce, transform, rank, interpret, or evaluate AI outputs.

Context Window

LLM CORE TERMS

The amount of text, messages, tool outputs, and retrieved information a model can consider in one request.

Embedding Model

LLM CORE TERMS

A model that converts data into vectors for search, clustering, memory, and similarity comparison.

Grounding

LLM CORE TERMS

Connecting an AI answer to verified evidence, retrieved context, citations, or system-of-record data.

Hallucination

LLM CORE TERMS

An incorrect or unsupported model output that sounds plausible but is not grounded in reliable evidence.

Instruction Following

LLM CORE TERMS

A model's ability to obey user, system, and developer instructions accurately.

Logprobs

LLM CORE TERMS

Probability scores for generated tokens, useful for confidence analysis and debugging.

Long-Context Model

LLM CORE TERMS

Long-Context Model is a model type or component focused on long-context. It helps produce, transform, rank, interpret, or evaluate AI outputs.

Max Tokens

LLM CORE TERMS

The maximum number of tokens a model is allowed to generate or process in a request.

Model Cascading

LLM CORE TERMS

Using a sequence of models, often starting with cheaper models and escalating to stronger ones only when needed.

Model Distillation

LLM CORE TERMS

Compressing knowledge from a larger model into a smaller model for faster or cheaper inference.

Model Routing

LLM CORE TERMS

Selecting the best model for a request based on difficulty, cost, latency, risk, or required capability.

Model Selection

LLM CORE TERMS

Choosing a model by matching task needs to accuracy, context length, modality, cost, latency, and safety requirements.

Multimodal Input

LLM CORE TERMS

Input that includes more than one type of data, such as text plus images or audio.

Prompt

LLM CORE TERMS

The input instructions, context, examples, or user request sent to a model.

Quantization

LLM CORE TERMS

A compression technique that lowers model numeric precision to reduce memory and improve speed.

Reasoning Model

LLM CORE TERMS

A model optimized to solve tasks that require multi-step analysis, planning, or problem solving.

Reranker Model

LLM CORE TERMS

A model that reorders retrieved candidates by relevance to improve final context quality.

Speech-to-Text

LLM CORE TERMS

A capability that converts spoken audio into written text.

Stop Sequence

LLM CORE TERMS

A string or token pattern that tells generation when to stop.

System Prompt

LLM CORE TERMS

High-priority instructions that define the model or agent role, boundaries, style, policies, and objectives.

Temperature

LLM CORE TERMS

A sampling parameter that controls randomness and creativity in model outputs.

Text-to-Speech

LLM CORE TERMS

A capability that converts written text into spoken audio.

Token

LLM CORE TERMS

A unit of text processed by a language model, such as a word piece, character group, or symbol.

Tokenizer

LLM CORE TERMS

The component that converts text into tokens and tokens back into text.

Top-k

LLM CORE TERMS

A sampling method that limits token choices to the k most likely next tokens.

Top-p

LLM CORE TERMS

A sampling method that limits token choices to the smallest set whose probabilities reach a threshold.

User Prompt

LLM CORE TERMS

The user-provided message or request that initiates or guides the model response.

Vision-Language Model

LLM CORE TERMS

Vision-Language Model is a model type or component focused on vision-language. It helps produce, transform, rank, interpret, or evaluate AI outputs.

Agent Connector

MCP TERMS

Agent Connector belongs to the MCP ecosystem and relates to agent connector. It helps agents discover, access, or govern external tools, resources, prompts, and context.

Agent Skill

MCP TERMS

Agent Skill belongs to the MCP ecosystem and relates to agent skill. It helps agents discover, access, or govern external tools, resources, prompts, and context.

Context Provider

MCP TERMS

Context Provider belongs to the MCP ecosystem and relates to context provider. It helps agents discover, access, or govern external tools, resources, prompts, and context.

Enterprise Tool Integration

MCP TERMS

Enterprise Tool Integration belongs to the MCP ecosystem and relates to enterprise tool integration. It helps agents discover, access, or govern external tools, resources, prompts, and context.

External Tooling

MCP TERMS

External Tooling belongs to the MCP ecosystem and relates to external tooling. It helps agents discover, access, or govern external tools, resources, prompts, and context.

Interoperability

MCP TERMS

The ability for systems, tools, agents, and vendors to exchange data and work together.

Local MCP Server

MCP TERMS

Local MCP Server belongs to the MCP ecosystem and relates to local mcp server. It helps agents discover, access, or govern external tools, resources, prompts, and context.

MCP

MCP TERMS

Model Context Protocol, a standard pattern for connecting models and agents to external tools, resources, and prompts.

MCP Client

MCP TERMS

The client role or object inside Model Context Protocol, used to expose tools, resources, prompts, or connections to agents.

MCP Host

MCP TERMS

The host role or object inside Model Context Protocol, used to expose tools, resources, prompts, or connections to agents.

MCP Prompt

MCP TERMS

A prompt written for mcp behavior, helping the model follow the right role, format, constraint, or task step.

MCP Resource

MCP TERMS

The resource role or object inside Model Context Protocol, used to expose tools, resources, prompts, or connections to agents.

MCP Server

MCP TERMS

The server role or object inside Model Context Protocol, used to expose tools, resources, prompts, or connections to agents.

MCP Tool

MCP TERMS

MCP Tool gives an agent controlled access to mcp capability. It is usually exposed through a function, API, service, or permissioned tool schema.

Model Context Protocol

MCP TERMS

A protocol that standardizes how AI applications expose tools, resources, and prompts to models or agents.

Open Standards

MCP TERMS

Shared protocols and specifications that improve portability and integration.

Permission Boundary

MCP TERMS

Permission Boundary belongs to the MCP ecosystem and relates to permission boundary. It helps agents discover, access, or govern external tools, resources, prompts, and context.

Remote MCP Server

MCP TERMS

Remote MCP Server belongs to the MCP ecosystem and relates to remote mcp server. It helps agents discover, access, or govern external tools, resources, prompts, and context.

Resource Discovery

MCP TERMS

Resource Discovery belongs to the MCP ecosystem and relates to resource discovery. It helps agents discover, access, or govern external tools, resources, prompts, and context.

Secure Tool Access

MCP TERMS

Secure Tool Access belongs to the MCP ecosystem and relates to secure tool access. It helps agents discover, access, or govern external tools, resources, prompts, and context.

Tool Manifest

MCP TERMS

Tool Manifest belongs to the MCP ecosystem and relates to tool manifest. It helps agents discover, access, or govern external tools, resources, prompts, and context.

A/B Testing

MLOps and LLMOps Terms

Testing different model, prompt, workflow, or UX variants against real or representative traffic.

AI Audit Trail

MLOps and LLMOps Terms

AI Audit Trail is an AI operations practice for ai audit trail. It supports versioning, deployment, evaluation, monitoring, governance, or rollback after launch.

Batch Inference

MLOps and LLMOps Terms

Batch Inference is an AI operations practice for batch inference. It supports versioning, deployment, evaluation, monitoring, governance, or rollback after launch.

Blue-Green Deployment

MLOps and LLMOps Terms

Blue-Green Deployment is an AI operations practice for blue-green deployment. It supports versioning, deployment, evaluation, monitoring, governance, or rollback after launch.

Canary Deployment

MLOps and LLMOps Terms

Canary Deployment is an AI operations practice for canary deployment. It supports versioning, deployment, evaluation, monitoring, governance, or rollback after launch.

CI/CD for ML

MLOps and LLMOps Terms

CI/CD for ML is an AI operations practice for ci/cd for ml. It supports versioning, deployment, evaluation, monitoring, governance, or rollback after launch.

Concept Drift

MLOps and LLMOps Terms

Concept Drift is an AI operations practice for concept drift. It supports versioning, deployment, evaluation, monitoring, governance, or rollback after launch.

Continuous Evaluation

MLOps and LLMOps Terms

Ongoing testing of models, prompts, retrieval, and agents as data, users, and behavior change over time.

Continuous Monitoring

MLOps and LLMOps Terms

Continuous Monitoring is an AI operations practice for continuous monitoring. It supports versioning, deployment, evaluation, monitoring, governance, or rollback after launch.

Data Card

MLOps and LLMOps Terms

Data Card is an AI operations practice for data card. It supports versioning, deployment, evaluation, monitoring, governance, or rollback after launch.

Data Drift

MLOps and LLMOps Terms

Data Drift is an AI operations practice for data drift. It supports versioning, deployment, evaluation, monitoring, governance, or rollback after launch.

Dataset Versioning

MLOps and LLMOps Terms

Dataset Versioning is an AI operations practice for dataset versioning. It supports versioning, deployment, evaluation, monitoring, governance, or rollback after launch.

Drift Detection

MLOps and LLMOps Terms

The ability to identify drift signals in inputs, outputs, logs, retrieved content, or system behavior.

Experiment Tracking

MLOps and LLMOps Terms

Experiment Tracking is an AI operations practice for experiment tracking. It supports versioning, deployment, evaluation, monitoring, governance, or rollback after launch.

Feature Store

MLOps and LLMOps Terms

A storage system for feature data that an AI application can save, query, or retrieve during execution.

Inference Endpoint

MLOps and LLMOps Terms

Inference Endpoint is an AI operations practice for inference endpoint. It supports versioning, deployment, evaluation, monitoring, governance, or rollback after launch.

LLMOps

MLOps and LLMOps Terms

LLMOps is an AI operations practice for llmops. It supports versioning, deployment, evaluation, monitoring, governance, or rollback after launch.

MLOps

MLOps and LLMOps Terms

MLOps is an AI operations practice for mlops. It supports versioning, deployment, evaluation, monitoring, governance, or rollback after launch.

Model Card

MLOPS AND LLMOPS TERMS

Model Card is an AI operations practice for model card. It supports versioning, deployment, evaluation, monitoring, governance, or rollback after launch.

Model Drift

MLOPS AND LLMOPS TERMS

Model Drift is an AI operations practice for model drift. It supports versioning, deployment, evaluation, monitoring, governance, or rollback after launch.

Model Governance

MLOPS AND LLMOPS TERMS

Model Governance is an AI operations practice for model governance. It supports versioning, deployment, evaluation, monitoring, governance, or rollback after launch.

Model Monitoring

MLOPS AND LLMOPS TERMS

Model Monitoring is an AI operations practice for model monitoring. It supports versioning, deployment, evaluation, monitoring, governance, or rollback after launch.

Model Registry

MLOPS AND LLMOPS TERMS

Model Registry is an AI operations practice for model registry. It supports versioning, deployment, evaluation, monitoring, governance, or rollback after launch.

Model Versioning

MLOPS AND LLMOPS TERMS

Model Versioning is an AI operations practice for model versioning. It supports versioning, deployment, evaluation, monitoring, governance, or rollback after launch.

Offline Evaluation

MLOPS AND LLMOPS TERMS

Testing AI behavior on saved datasets or scenarios before release, without affecting live users.

Online Inference

MLOPS AND LLMOPS TERMS

Online Inference is an AI operations practice for online inference. It supports versioning, deployment, evaluation, monitoring, governance, or rollback after launch.

Rollback

MLOPS AND LLMOPS TERMS

Rollback is an AI operations practice for rollback. It supports versioning, deployment, evaluation, monitoring, governance, or rollback after launch.

Shadow Deployment

MLOPS AND LLMOPS TERMS

Shadow Deployment is an AI operations practice for shadow deployment. It supports versioning, deployment, evaluation, monitoring, governance, or rollback after launch.

Agent Collaboration

MULTI-AGENT TERMS

A multi-agent behavior involving collaboration, used when agents need to share work, decisions, or responsibilities.

Agent Communication

MULTI-AGENT TERMS

A multi-agent behavior involving communication, used when agents need to share work, decisions, or responsibilities.

Agent Consensus

MULTI-AGENT TERMS

A multi-agent behavior involving consensus, used when agents need to share work, decisions, or responsibilities.

Agent Coordination

MULTI-AGENT TERMS

A multi-agent behavior involving coordination, used when agents need to share work, decisions, or responsibilities.

Agent Delegation

MULTI-AGENT TERMS

A multi-agent behavior involving delegation, used when agents need to share work, decisions, or responsibilities.

Agent Handoff

MULTI-AGENT TERMS

A multi-agent behavior involving handoff, used when agents need to share work, decisions, or responsibilities.

Agent Team

MULTI-AGENT TERMS

A multi-agent behavior involving team, used when agents need to share work, decisions, or responsibilities.

Agent Voting

MULTI-AGENT TERMS

A multi-agent behavior involving voting, used when agents need to share work, decisions, or responsibilities.

Coding Agent

MULTI-AGENT TERMS

A Coding Agent helps create, modify, explain, test, or review software using repository context and developer tools.

Debate Agent

MULTI-AGENT TERMS

A Debate Agent participates in structured disagreement so different answers, plans, or assumptions can be compared before a decision.

Evaluator Agent

MULTI-AGENT TERMS

An Evaluator Agent scores answers, plans, tool calls, or task outcomes against criteria such as correctness, safety, and usefulness.

Memory Agent

MULTI-AGENT TERMS

A Memory Agent manages what information should be stored, retrieved, updated, summarized, or forgotten for future use.

Multi-Agent Orchestration

MULTI-AGENT TERMS

Coordinating multiple specialized agents, their responsibilities, handoffs, shared tools, and shared state.

Multi-Agent System (MAS)

MULTI-AGENT TERMS

A multi-agent architecture in which several agents collaborate or specialize to solve problems that are too broad for one agent role.

Research Agent

MULTI-AGENT TERMS

A Research Agent gathers evidence, compares sources, summarizes findings, and prepares grounded answers or reports.

Reviewer Agent

MULTI-AGENT TERMS

A Reviewer Agent inspects another agent’s work for quality, completeness, factual support, and policy compliance.

Router Agent

MULTI-AGENT TERMS

A Router Agent sends a request to the best model, tool, workflow, or specialist agent based on intent and context.

Shared Memory

MULTI-AGENT TERMS

Shared Memory is a memory type focused on shared. It preserves information that may be useful for later turns, tasks, personalization, or reasoning.

Shared Tooling

MULTI-AGENT TERMS

A common set of tools available to multiple agents, usually controlled by permissions, schemas, and audit logs.

Specialist Agent

MULTI-AGENT TERMS

A Specialist Agent focuses on one domain or skill area, such as security, finance, testing, legal review, or data analysis.

Supervisor Agent

MULTI-AGENT TERMS

A Supervisor Agent coordinates worker agents, assigns tasks, monitors progress, and decides when to escalate or revise the plan.

Swarm Agent

MULTI-AGENT TERMS

A Swarm Agent is part of a large group of simple agents that coordinate to explore options, solve subtasks, or reach consensus.

Testing Agent

MULTI-AGENT TERMS

A Testing Agent creates, runs, repairs, or analyzes tests to validate software behavior and release quality.

Tool Agent

MULTI-AGENT TERMS

A Tool Agent specializes in selecting, calling, validating, and recovering from external tool or API actions.

Worker Agent

MULTI-AGENT TERMS

A Worker Agent performs a specific subtask assigned by a supervisor, planner, or orchestration workflow.

Agent Trace

OBSERVABILITY TERMS

A record of the agent’s steps, tool calls, observations, and decisions during execution.

Arize Phoenix

OBSERVABILITY TERMS

An observability term for arize phoenix, used to inspect agent behavior, debug failures, monitor production health, or analyze cost.

Audit Log

OBSERVABILITY TERMS

An observability term for audit log, used to inspect agent behavior, debug failures, monitor production health, or analyze cost.

Audit Logs

OBSERVABILITY TERMS

Durable records of prompts, decisions, tool calls, data access, outputs, and human approvals.

Cost Monitoring

OBSERVABILITY TERMS

An observability term for cost monitoring, used to inspect agent behavior, debug failures, monitor production health, or analyze cost.

Error Rate

OBSERVABILITY TERMS

The percentage of requests, tool calls, or workflow steps that fail or return invalid results.

Evaluation Dashboard

OBSERVABILITY TERMS

An observability term for evaluation dashboard, used to inspect agent behavior, debug failures, monitor production health, or analyze cost.

Feedback Loop

OBSERVABILITY TERMS

A mechanism for using user feedback, human review, logs, and metrics to improve future behavior.

Humanloop

OBSERVABILITY TERMS

An observability term for humanloop, used to inspect agent behavior, debug failures, monitor production health, or analyze cost.

LangSmith

OBSERVABILITY TERMS

An observability term for langsmith, used to inspect agent behavior, debug failures, monitor production health, or analyze cost.

Latency

OBSERVABILITY TERMS

The elapsed time between a request and an agent response or action.

Logging

OBSERVABILITY TERMS

An observability term for logging, used to inspect agent behavior, debug failures, monitor production health, or analyze cost.

Metrics

OBSERVABILITY TERMS

An observability term for metrics, used to inspect agent behavior, debug failures, monitor production health, or analyze cost.

Observability

OBSERVABILITY TERMS

Visibility into prompts, model calls, retrieval, tool calls, traces, metrics, errors, and outcomes.

OpenTelemetry

OBSERVABILITY TERMS

An observability term for opentelemetry, used to inspect agent behavior, debug failures, monitor production health, or analyze cost.

Prompt Trace

OBSERVABILITY TERMS

A record of prompt templates, context, model parameters, and generated outputs for debugging.

Retrieval Trace

OBSERVABILITY TERMS

A recorded timeline of retrieval activity that helps engineers debug behavior and understand execution history.

Retry Count

OBSERVABILITY TERMS

An observability term for retry count, used to inspect agent behavior, debug failures, monitor production health, or analyze cost.

Session ID

OBSERVABILITY TERMS

An observability term for session id, used to inspect agent behavior, debug failures, monitor production health, or analyze cost.

Session Replay

OBSERVABILITY TERMS

An observability term for session replay, used to inspect agent behavior, debug failures, monitor production health, or analyze cost.

Span

OBSERVABILITY TERMS

A timed unit of work inside a trace, such as retrieval, model call, or tool execution.

Telemetry

OBSERVABILITY TERMS

Operational data emitted by systems for monitoring, debugging, evaluation, and cost analysis.

Throughput

OBSERVABILITY TERMS

The amount of work an AI system can process over a period of time.

Time to First Token

OBSERVABILITY TERMS

An observability term for time to first token, used to inspect agent behavior, debug failures, monitor production health, or analyze cost.

Tool Monitoring

OBSERVABILITY TERMS

An observability term for tool monitoring, used to inspect agent behavior, debug failures, monitor production health, or analyze cost.

Tool Trace

OBSERVABILITY TERMS

A record of tool invocations, inputs, outputs, latency, errors, and retries.

Trace ID

OBSERVABILITY TERMS

A unique identifier that connects logs and spans for one request or workflow.

Tracing

OBSERVABILITY TERMS

An observability term for tracing, used to inspect agent behavior, debug failures, monitor production health, or analyze cost.

TruLens

OBSERVABILITY TERMS

An observability term for trulens, used to inspect agent behavior, debug failures, monitor production health, or analyze cost.

User Feedback

OBSERVABILITY TERMS

An observability term for user feedback, used to inspect agent behavior, debug failures, monitor production health, or analyze cost.

Weights & Biases

OBSERVABILITY TERMS

An observability term for weights & biases, used to inspect agent behavior, debug failures, monitor production health, or analyze cost.

Chain-of-Thought Prompting

PROMPT ENGINEERING TERMS

A prompting approach that encourages step-by-step reasoning before the final answer.

Constraint Prompting

PROMPT ENGINEERING TERMS

Constraint Prompting is a prompting technique for constraint prompting. It shapes the model's instructions, examples, constraints, or output format for more reliable responses.

Contextual Prompting

PROMPT ENGINEERING TERMS

Contextual Prompting is a prompting technique for contextual prompting. It shapes the model's instructions, examples, constraints, or output format for more reliable responses.

Critic Prompt

PROMPT ENGINEERING TERMS

A prompt written for critic behavior, helping the model follow the right role, format, constraint, or task step.

Developer Instruction

PROMPT ENGINEERING TERMS

Developer Instruction is a prompting technique for developer instruction. It shapes the model's instructions, examples, constraints, or output format for more reliable responses.

Few-Shot Prompting

PROMPT ENGINEERING TERMS

Prompting a model with a few examples so it can imitate the desired pattern.

Guardrail Prompt

PROMPT ENGINEERING TERMS

A prompt written for guardrail behavior, helping the model follow the right role, format, constraint, or task step.

JSON Output Prompt

PROMPT ENGINEERING TERMS

A prompt written for json output behavior, helping the model follow the right role, format, constraint, or task step.

Meta-Prompting

PROMPT ENGINEERING TERMS

Meta-Prompting is a prompting technique for meta-prompting. It shapes the model's instructions, examples, constraints, or output format for more reliable responses.

Negative Instruction

PROMPT ENGINEERING TERMS

Negative Instruction is a prompting technique for negative instruction. It shapes the model's instructions, examples, constraints, or output format for more reliable responses.

Output Schema Prompt

PROMPT ENGINEERING TERMS

A prompt written for output schema behavior, helping the model follow the right role, format, constraint, or task step.

Planning Prompt

PROMPT ENGINEERING TERMS

A prompt written for planning behavior, helping the model follow the right role, format, constraint, or task step.

Prompt Chaining

PROMPT ENGINEERING TERMS

Prompt Chaining is a prompting technique for prompt chaining. It shapes the model's instructions, examples, constraints, or output format for more reliable responses.

Prompt Compression

PROMPT ENGINEERING TERMS

Prompt Compression is a prompting technique for prompt compression. It shapes the model's instructions, examples, constraints, or output format for more reliable responses.

Prompt Engineering

PROMPT ENGINEERING TERMS

Designing instructions, examples, context, and constraints to guide model behavior reliably.

Prompt Injection

PROMPT ENGINEERING TERMS

Prompt Injection is a prompting technique for prompt injection. It shapes the model's instructions, examples, constraints, or output format for more reliable responses.

Prompt Leakage

PROMPT ENGINEERING TERMS

Prompt Leakage is a prompting technique for prompt leakage. It shapes the model's instructions, examples, constraints, or output format for more reliable responses.

Prompt Optimization

PROMPT ENGINEERING TERMS

Improving prompt so the AI system becomes faster, cheaper, more accurate, or more useful.

Prompt Template

PROMPT ENGINEERING TERMS

Prompt Template is a prompting technique for prompt template. It shapes the model's instructions, examples, constraints, or output format for more reliable responses.

Prompt Versioning

PROMPT ENGINEERING TERMS

Prompt Versioning is a prompting technique for prompt versioning. It shapes the model's instructions, examples, constraints, or output format for more reliable responses.

ReAct Prompting

PROMPT ENGINEERING TERMS

A prompting pattern where the model alternates between reasoning, taking actions, and observing results.

Reflection Prompt

PROMPT ENGINEERING TERMS

A prompt written for reflection behavior, helping the model follow the right role, format, constraint, or task step.

Role Prompting

PROMPT ENGINEERING TERMS

Role Prompting is a prompting technique for role prompting. It shapes the model's instructions, examples, constraints, or output format for more reliable responses.

Self-Consistency

PROMPT ENGINEERING TERMS

A method that samples multiple reasoning paths and chooses the answer that appears most consistently.

Structured Prompt

PROMPT ENGINEERING TERMS

A prompt written for structured behavior, helping the model follow the right role, format, constraint, or task step.

System Instruction

PROMPT ENGINEERING TERMS

System Instruction is a prompting technique for system instruction. It shapes the model's instructions, examples, constraints, or output format for more reliable responses.

Tool-Use Prompt

PROMPT ENGINEERING TERMS

A prompt written for tool-use behavior, helping the model follow the right role, format, constraint, or task step.

Zero-Shot Prompting

PROMPT ENGINEERING TERMS

Prompting a model to perform a task without giving examples in the prompt.

Agentic RAG

RAG TERMS

A retrieval workflow where an agent plans searches, uses tools, evaluates evidence, and iterates before answering.

Answer Relevance

RAG TERMS

How well an answer addresses the user’s question or task.

Approximate Nearest Neighbor

RAG TERMS

A fast search method that finds near-best vector matches efficiently at scale.

BM25

RAG TERMS

A classic keyword ranking algorithm used in sparse retrieval and hybrid search.

Chunk Overlap

RAG TERMS

Repeated text shared between adjacent chunks to preserve context across boundaries.

Chunk Size

RAG TERMS

The amount of text placed into each retrieval unit during document processing.

Chunking

RAG TERMS

Splitting documents into smaller passages so retrieval can return useful, focused context.

Citation Generation

RAG TERMS

Producing references or source links that show where an answer was grounded.

Context Compression

RAG TERMS

Reducing retrieved content to the most relevant facts before sending it to the model.

Context Precision

RAG TERMS

The share of retrieved context that is actually useful for answering the question.

Context Recall

RAG TERMS

How much of the needed evidence the retrieval system successfully found.

Context Relevance

RAG TERMS

How relevant retrieved context is to the query and final answer.

Contextual Retrieval

RAG TERMS

Retrieval that uses surrounding context, metadata, or rewritten queries to improve relevance.

Corrective RAG

RAG TERMS

A RAG pattern that detects weak retrieval or unsupported answers and corrects the retrieval or response path.

Cosine Similarity

RAG TERMS

A measure of vector similarity based on the angle between vectors.

Cross-Encoder Reranker

RAG TERMS

A reranker that jointly reads the query and candidate passage to produce a relevance score.

Document Ingestion

RAG TERMS

The process of loading, parsing, cleaning, and preparing source documents for retrieval.

Document Parsing

RAG TERMS

Extracting text, tables, metadata, and structure from source files or webpages.

Dot Product Similarity

RAG TERMS

A similarity score calculated by multiplying and summing vector components.

Embedding

RAG TERMS

A vector representation of data used for semantic search and similarity matching.

Faithfulness

RAG TERMS

The degree to which an answer is supported by the provided context.

GraphRAG

RAG TERMS

A RAG approach that uses graph relationships among entities, documents, and facts to improve reasoning.

Grounded Response

RAG TERMS

An answer supported by retrieved evidence, source-of-truth data, or citations.

Hallucination Detection

RAG TERMS

Methods for identifying unsupported or fabricated model claims.

Hybrid Search

RAG TERMS

Combining keyword and vector retrieval to improve recall and relevance.

Keyword Search

RAG TERMS

Search based on exact words, phrases, or lexical matching.

Long-Context RAG

RAG TERMS

A retrieval approach that combines large context windows with selected external evidence.

Metadata Filtering

RAG TERMS

Restricting retrieval by document attributes such as source, date, owner, product, or permission.

Multi-Query Retrieval

RAG TERMS

Retrieving with several query variations to improve coverage of relevant context.

Query Expansion

RAG TERMS

Adding related terms or variants to a query to improve retrieval coverage.

Query Rewriting

RAG TERMS

Transforming a user question into a better search query or multiple focused queries.

RAG Pipeline

RAG TERMS

The end-to-end flow for ingestion, chunking, embedding, retrieval, reranking, and answer generation.

Reranking

RAG TERMS

Reordering retrieved results using a stronger relevance model after initial search.

Retrieval-Augmented Generation

RAG TERMS

A pattern that retrieves relevant external knowledge before generation so responses can be grounded in sources.

Retrieval Evaluation

RAG TERMS

Measuring whether the retrieval system returns useful and complete context.

Semantic Search

RAG TERMS

Search based on meaning rather than exact keyword matching.

Similarity Search

RAG TERMS

Finding items closest to a query representation in a vector or feature space.

Vector

RAG TERMS

A numeric representation of meaning, features, or model state.

Vector Database

RAG TERMS

A database optimized for storing embeddings and retrieving semantically similar content.

AI Test Automation

SDET AND QA AI TERMS

Using AI to design, run, analyze, or maintain software tests.

API Testing Agent

SDET AND QA AI TERMS

An agent that validates API requests, responses, schemas, status codes, and business rules.

Autonomous Testing

SDET AND QA AI TERMS

A testing approach where AI can create, execute, and analyze tests with limited human help.

Bug Report Generation

SDET AND QA AI TERMS

Using AI to create clear bug reports from logs, screenshots, traces, and failed test evidence.

CI Test Analysis

SDET AND QA AI TERMS

Using AI to explain failures and trends from tests running in a CI pipeline.

Cypress Agent

SDET AND QA AI TERMS

An agent that creates, runs, or debugs Cypress tests for web applications.

Defect Prediction

SDET AND QA AI TERMS

Using data and AI to estimate where bugs are likely to appear.

Failure Classification

SDET AND QA AI TERMS

Grouping test failures by likely cause, such as product bug, test issue, data issue, or environment issue.

Flaky Test Detection

SDET AND QA AI TERMS

Finding tests that pass and fail inconsistently without reliable product changes.

Locator Healing

SDET AND QA AI TERMS

Automatically updating broken UI selectors when page structure changes.

Mobile Testing Agent

SDET AND QA AI TERMS

An agent that helps test mobile apps across devices, OS versions, screens, and user flows.

Playwright Agent

SDET AND QA AI TERMS

An agent that creates, runs, or debugs Playwright tests for browser automation.

QA Copilot

SDET AND QA AI TERMS

An AI assistant that helps testers write cases, analyze failures, and improve test coverage.

Regression Testing

SDET AND QA AI TERMS

Testing existing functionality again to ensure new changes did not break it.

Release Readiness

SDET AND QA AI TERMS

The confidence that a product is stable, tested, and safe enough to release.

Selenium Agent

SDET AND QA AI TERMS

An agent that creates, runs, or debugs Selenium tests for browser automation.

Self-Healing Test

SDET AND QA AI TERMS

An automated test that can repair simple failures such as changed locators or timing issues.

Test Case Generation

SDET AND QA AI TERMS

Using AI to create step-by-step test cases from requirements, stories, or designs.

Test Data Generation

SDET AND QA AI TERMS

Creating realistic input data needed to test features, APIs, or edge cases.

Test Generation

SDET AND QA AI TERMS

Using AI to create tests, scenarios, or validation checks from requirements or code.

Test Impact Analysis

SDET AND QA AI TERMS

Identifying which tests are most affected by a code or configuration change.

Test Prioritization

SDET AND QA AI TERMS

Choosing the most important tests to run first based on risk, change, or history.

Test Script Generation

SDET AND QA AI TERMS

Using AI to produce executable automation scripts from test intent or manual steps.

Visual Testing

SDET AND QA AI TERMS

Checking visual changes in UI layout, screenshots, colors, and components.

API Integration

TOOL USE AND FUNCTION CALLING TERMS

Connecting an agent to external software systems through application programming interfaces.

API Tool

TOOL USE AND FUNCTION CALLING TERMS

API Tool gives an agent controlled access to api capability. It is usually exposed through a function, API, service, or permissioned tool schema.

Browser Tool

TOOL USE AND FUNCTION CALLING TERMS

Browser Tool gives an agent controlled access to browser capability. It is usually exposed through a function, API, service, or permissioned tool schema.

Calculator Tool

TOOL USE AND FUNCTION CALLING TERMS

Calculator Tool gives an agent controlled access to calculator capability. It is usually exposed through a function, API, service, or permissioned tool schema.

Calendar Tool

TOOL USE AND FUNCTION CALLING TERMS

Calendar Tool gives an agent controlled access to calendar capability. It is usually exposed through a function, API, service, or permissioned tool schema.

Code Interpreter

TOOL USE AND FUNCTION CALLING TERMS

A tool that runs code to calculate, analyze files, transform data, or produce artifacts under controlled conditions.

CRM Integration

TOOL USE AND FUNCTION CALLING TERMS

Connecting an agent to customer relationship data so responses and actions can reflect account, case, and customer history.

CRM Tool

TOOL USE AND FUNCTION CALLING TERMS

CRM Tool gives an agent controlled access to crm capability. It is usually exposed through a function, API, service, or permissioned tool schema.

Database Tool

TOOL USE AND FUNCTION CALLING TERMS

Database Tool gives an agent controlled access to database capability. It is usually exposed through a function, API, service, or permissioned tool schema.

Email Tool

TOOL USE AND FUNCTION CALLING TERMS

Email Tool gives an agent controlled access to email capability. It is usually exposed through a function, API, service, or permissioned tool schema.

File Tool

TOOL USE AND FUNCTION CALLING TERMS

File Tool gives an agent controlled access to file capability. It is usually exposed through a function, API, service, or permissioned tool schema.

Function Calling

TOOL USE AND FUNCTION CALLING TERMS

Structured tool invocation where the model returns validated arguments that match a predefined function schema.

Internal System Tool

TOOL USE AND FUNCTION CALLING TERMS

Internal System Tool gives an agent controlled access to internal system capability. It is usually exposed through a function, API, service, or permissioned tool schema.

JSON Schema

TOOL USE AND FUNCTION CALLING TERMS

A formal structure that defines allowed fields, types, and constraints for tool inputs or structured outputs.

Search Tool

TOOL USE AND FUNCTION CALLING TERMS

Search Tool gives an agent controlled access to search capability. It is usually exposed through a function, API, service, or permissioned tool schema.

Tool Calling

TOOL USE AND FUNCTION CALLING TERMS

A model or agent selecting and invoking external capabilities such as APIs, search, databases, code execution, or business workflows.

Tool Chaining

TOOL USE AND FUNCTION CALLING TERMS

The chaining part of tool use, defining how an agent selects, calls, checks, recovers from, or records external actions.

Tool Description

TOOL USE AND FUNCTION CALLING TERMS

The description part of tool use, defining how an agent selects, calls, checks, recovers from, or records external actions.

Tool Discovery

TOOL USE AND FUNCTION CALLING TERMS

The discovery part of tool use, defining how an agent selects, calls, checks, recovers from, or records external actions.

Tool Error

TOOL USE AND FUNCTION CALLING TERMS

The error part of tool use, defining how an agent selects, calls, checks, recovers from, or records external actions.

Tool Execution

TOOL USE AND FUNCTION CALLING TERMS

The execution part of tool use, defining how an agent selects, calls, checks, recovers from, or records external actions.

Tool Input

TOOL USE AND FUNCTION CALLING TERMS

The input part of tool use, defining how an agent selects, calls, checks, recovers from, or records external actions.

Tool Output

TOOL USE AND FUNCTION CALLING TERMS

The output part of tool use, defining how an agent selects, calls, checks, recovers from, or records external actions.

Tool Permission

TOOL USE AND FUNCTION CALLING TERMS

The permission part of tool use, defining how an agent selects, calls, checks, recovers from, or records external actions.

Tool Registry

TOOL USE AND FUNCTION CALLING TERMS

The registry part of tool use, defining how an agent selects, calls, checks, recovers from, or records external actions.

Tool Result

TOOL USE AND FUNCTION CALLING TERMS

The result part of tool use, defining how an agent selects, calls, checks, recovers from, or records external actions.

Tool Retry

TOOL USE AND FUNCTION CALLING TERMS

The retry part of tool use, defining how an agent selects, calls, checks, recovers from, or records external actions.

Tool Routing

TOOL USE AND FUNCTION CALLING TERMS

The routing part of tool use, defining how an agent selects, calls, checks, recovers from, or records external actions.

Tool Sandbox

TOOL USE AND FUNCTION CALLING TERMS

The sandbox part of tool use, defining how an agent selects, calls, checks, recovers from, or records external actions.

Tool Schema

TOOL USE AND FUNCTION CALLING TERMS

The schema part of tool use, defining how an agent selects, calls, checks, recovers from, or records external actions.

Tool Selection

TOOL USE AND FUNCTION CALLING TERMS

The selection part of tool use, defining how an agent selects, calls, checks, recovers from, or records external actions.

Tool Timeout

TOOL USE AND FUNCTION CALLING TERMS

The timeout part of tool use, defining how an agent selects, calls, checks, recovers from, or records external actions.

Tool Validation

TOOL USE AND FUNCTION CALLING TERMS

The validation part of tool use, defining how an agent selects, calls, checks, recovers from, or records external actions.

ANN Index

VECTOR DATABASE AND SEARCH TERMS

An approximate-nearest-neighbor index that finds similar vectors quickly without comparing every vector exactly.

Chroma

VECTOR DATABASE AND SEARCH TERMS

An open-source vector database commonly used for local RAG prototypes and lightweight embedding search.

Dense Retrieval

VECTOR DATABASE AND SEARCH TERMS

Dense Retrieval is a retrieval method focused on dense. It finds useful information before generation, tool use, grounding, or answer verification.

Elasticsearch

VECTOR DATABASE AND SEARCH TERMS

A distributed search engine used for keyword search, logs, analytics, and hybrid retrieval systems.

Embedding Drift

VECTOR DATABASE AND SEARCH TERMS

The term Embedding Drift means an embedding drift concept used in vector database and search terms for practical AI engineering work.

Embedding Space

VECTOR DATABASE AND SEARCH TERMS

The term Embedding Space means an embedding space concept used in vector database and search terms for practical AI engineering work.

FAISS

VECTOR DATABASE AND SEARCH TERMS

A library from Meta for efficient vector similarity search and clustering.

HNSW

VECTOR DATABASE AND SEARCH TERMS

Hierarchical Navigable Small World, a graph-based index commonly used for fast vector search.

Hybrid Retrieval

VECTOR DATABASE AND SEARCH TERMS

Hybrid Retrieval is a retrieval method focused on hybrid. It finds useful information before generation, tool use, grounding, or answer verification.

IVF

VECTOR DATABASE AND SEARCH TERMS

Inverted file indexing, a vector search method that partitions embeddings into clusters for faster lookup.

Metadata Store

VECTOR DATABASE AND SEARCH TERMS

A storage system for metadata data that an AI application can save, query, or retrieve during execution.

Milvus

VECTOR DATABASE AND SEARCH TERMS

An open-source vector database for storing, indexing, and searching large-scale embedding collections.

MMR

VECTOR DATABASE AND SEARCH TERMS

Maximal Marginal Relevance, a retrieval method that balances relevance with diversity in selected results.

OpenSearch

VECTOR DATABASE AND SEARCH TERMS

An open-source search and analytics engine used for logs, keyword search, and hybrid retrieval.

pgvector

VECTOR DATABASE AND SEARCH TERMS

A PostgreSQL extension that stores embeddings and supports vector similarity search inside Postgres.

Pinecone

VECTOR DATABASE AND SEARCH TERMS

A managed vector database service used to store and search embeddings at scale.

PQ

VECTOR DATABASE AND SEARCH TERMS

Product quantization, a compression method that reduces vector storage size for faster large-scale search.

Precision@k

VECTOR DATABASE AND SEARCH TERMS

The term Precision@k means a precision at k concept used in vector database and search terms for practical AI engineering work.

Qdrant

VECTOR DATABASE AND SEARCH TERMS

A vector database focused on similarity search, filtering, and production-ready embedding retrieval.

Recall@k

VECTOR DATABASE AND SEARCH TERMS

The term Recall@k means a recall at k concept used in vector database and search terms for practical AI engineering work.

Semantic Cache

VECTOR DATABASE AND SEARCH TERMS

Semantic Cache stores reusable semantic results. It reduces repeated work, lowers cost, and improves speed when similar requests appear again.

Similarity Threshold

VECTOR DATABASE AND SEARCH TERMS

The term Similarity Threshold means a similarity threshold concept used in vector database and search terms for practical AI engineering work.

Sparse Retrieval

VECTOR DATABASE AND SEARCH TERMS

Sparse Retrieval is a retrieval method focused on sparse. It finds useful information before generation, tool use, grounding, or answer verification.

Top-k Retrieval

VECTOR DATABASE AND SEARCH TERMS

Top-k Retrieval is a retrieval method focused on top-k. It finds useful information before generation, tool use, grounding, or answer verification.

Vector Index

VECTOR DATABASE AND SEARCH TERMS

An index structure for vector that speeds up lookup, retrieval, or similarity matching.

Vector Store

VECTOR DATABASE AND SEARCH TERMS

A storage system for vector data that an AI application can save, query, or retrieve during execution.

Weaviate

VECTOR DATABASE AND SEARCH TERMS

A vector database that combines semantic search, metadata filtering, and knowledge-oriented data modeling.

AI Technique

CLASSICAL AI AND INTELLIGENT SYSTEMS TERMS

A method for organizing and using large, changing, or incomplete knowledge so an AI system can solve problems efficiently.

Artificial Environment

CLASSICAL AI AND INTELLIGENT SYSTEMS TERMS

A simulated or software-created setting where an agent acts without directly operating in the physical world.

Auditory Learning

CLASSICAL AI AND INTELLIGENT SYSTEMS TERMS

Learning by listening to spoken information, audio examples, or sound-based instruction.

Bodily-Kinesthetic Intelligence

CLASSICAL AI AND INTELLIGENT SYSTEMS TERMS

The ability to use body movement or physical control to solve problems or perform skilled tasks.

Common Sense Verification

CLASSICAL AI AND INTELLIGENT SYSTEMS TERMS

The task of checking whether a statement or conclusion matches everyday knowledge about how the world works.

Deductive Reasoning

CLASSICAL AI AND INTELLIGENT SYSTEMS TERMS

Reasoning that starts from a general rule and applies it to a specific case to reach a logical conclusion.

Episodic Learning

CLASSICAL AI AND INTELLIGENT SYSTEMS TERMS

Learning from remembered sequences of events or experiences that occurred over time.

Expert Tasks

CLASSICAL AI AND INTELLIGENT SYSTEMS TERMS

AI tasks that require specialized domain knowledge, such as diagnosis, financial analysis, or engineering fault finding.

Formal Tasks

CLASSICAL AI AND INTELLIGENT SYSTEMS TERMS

AI tasks based on formal rules or symbols, such as mathematics, logic, games, and theorem proving.

Game AI

CLASSICAL AI AND INTELLIGENT SYSTEMS TERMS

AI used to reason about possible moves, strategies, and outcomes in games such as chess, poker, or tic-tac-toe.

Human Intelligence

CLASSICAL AI AND INTELLIGENT SYSTEMS TERMS

The natural ability of people to learn, reason, communicate, perceive, remember, and adapt to new situations.

Inductive Reasoning

CLASSICAL AI AND INTELLIGENT SYSTEMS TERMS

Reasoning that uses specific observations to form a broader general conclusion that may still be uncertain.

Intelligence

CLASSICAL AI AND INTELLIGENT SYSTEMS TERMS

The ability to reason, learn, solve problems, use language, remember information, and adapt to new situations.

Intelligent Robot

CLASSICAL AI AND INTELLIGENT SYSTEMS TERMS

A robot that can sense its surroundings, process information, learn from experience, and adapt its actions.

Intelligent System

CLASSICAL AI AND INTELLIGENT SYSTEMS TERMS

A computer, robot, or software system that uses AI methods to perceive, reason, learn, or act toward a goal.

Interpersonal Intelligence

CLASSICAL AI AND INTELLIGENT SYSTEMS TERMS

The ability to understand other people's emotions, beliefs, intentions, and social signals.

Intra-personal Intelligence

CLASSICAL AI AND INTELLIGENT SYSTEMS TERMS

The ability to understand one's own feelings, goals, motivations, and internal mental state.

Learning

CLASSICAL AI AND INTELLIGENT SYSTEMS TERMS

The process of gaining knowledge or skill from study, practice, instruction, examples, or experience.

Linguistic Intelligence

CLASSICAL AI AND INTELLIGENT SYSTEMS TERMS

The ability to understand and use spoken or written language effectively.

Logical-Mathematical Intelligence

CLASSICAL AI AND INTELLIGENT SYSTEMS TERMS

The ability to reason with numbers, relationships, abstractions, and logical structures.

Machine Intelligence

CLASSICAL AI AND INTELLIGENT SYSTEMS TERMS

The ability of a machine or software system to perform tasks that normally require human-like reasoning, learning, or perception.

Motor Learning

CLASSICAL AI AND INTELLIGENT SYSTEMS TERMS

Learning physical actions or movement patterns through repeated practice and correction.

Mundane Tasks

CLASSICAL AI AND INTELLIGENT SYSTEMS TERMS

Everyday tasks such as perception, language use, common sense, and movement that are natural for humans but hard for machines.

Musical Intelligence

CLASSICAL AI AND INTELLIGENT SYSTEMS TERMS

The ability to recognize, create, and reason about sound patterns such as pitch, rhythm, and melody.

Observational Learning

CLASSICAL AI AND INTELLIGENT SYSTEMS TERMS

Learning by watching another person, agent, or system perform a behavior and imitating it.

Perceptual Learning

CLASSICAL AI AND INTELLIGENT SYSTEMS TERMS

Learning to recognize, distinguish, or classify stimuli based on previous sensory experience.

Problem Solving

CLASSICAL AI AND INTELLIGENT SYSTEMS TERMS

The process of moving from a current state to a desired goal by choosing actions that overcome obstacles.

Relational Learning

CLASSICAL AI AND INTELLIGENT SYSTEMS TERMS

Learning based on relationships among objects or properties rather than isolated features alone.

Spatial Intelligence

CLASSICAL AI AND INTELLIGENT SYSTEMS TERMS

The ability to understand, transform, and reason about visual or spatial relationships.

Spatial Learning

CLASSICAL AI AND INTELLIGENT SYSTEMS TERMS

Learning from maps, images, layouts, movement paths, or other visual-spatial signals.

Stimulus-Response Learning

CLASSICAL AI AND INTELLIGENT SYSTEMS TERMS

Learning to perform a particular behavior when a specific trigger or stimulus appears.

Task Classification

CLASSICAL AI AND INTELLIGENT SYSTEMS TERMS

A way to group AI work into categories such as mundane tasks, formal tasks, and expert tasks.

Theorem Proving

CLASSICAL AI AND INTELLIGENT SYSTEMS TERMS

The use of logic and search to prove that a mathematical or formal statement follows from known rules or assumptions.

Vision System

CLASSICAL AI AND INTELLIGENT SYSTEMS TERMS

An AI system that interprets visual input such as images, scenes, maps, faces, or objects.

Accessible Environment

AGENT AND ENVIRONMENT TERMS

An environment where the agent’s sensors can access the full state needed to choose actions.

Actuator

AGENT AND ENVIRONMENT TERMS

A physical or software mechanism that turns an agent’s decision into an action in the environment.

Agent

AGENT AND ENVIRONMENT TERMS

A system or software program that perceives an environment and acts toward one or more goals.

Agent Function

AGENT AND ENVIRONMENT TERMS

The mapping from an agent’s percept history to the action it should take next.

Agent Program

AGENT AND ENVIRONMENT TERMS

The implementation that runs on an architecture and produces actions from percepts or state.

Behavior of Agent

AGENT AND ENVIRONMENT TERMS

The actions an agent performs after receiving a particular sequence of percepts.

Condition-Action Rule

AGENT AND ENVIRONMENT TERMS

A rule that maps a detected condition directly to an action, often used in reflex agents.

Continuous Environment

AGENT AND ENVIRONMENT TERMS

An environment whose states or actions can vary across a continuous range rather than fixed options.

Deterministic Environment

AGENT AND ENVIRONMENT TERMS

An environment where the next state is fully determined by the current state and the agent’s action.

Discrete Environment

AGENT AND ENVIRONMENT TERMS

An environment with a limited set of clearly separated states, actions, or time steps.

Dynamic Environment

AGENT AND ENVIRONMENT TERMS

An environment that can change while an agent is deciding or acting.

Effector

AGENT AND ENVIRONMENT TERMS

A body part, motor, actuator, or software output channel that carries out an agent’s action.

Episodic Environment

AGENT AND ENVIRONMENT TERMS

An environment where each decision episode is independent of previous episodes.

Goal-Based Agent

AGENT AND ENVIRONMENT TERMS

An agent that chooses actions by considering which steps will move it closer to a desired goal.

Human Agent

AGENT AND ENVIRONMENT TERMS

A human viewed as an agent with sensory organs for perception and body parts for action.

Ideal Rational Agent

AGENT AND ENVIRONMENT TERMS

An agent that selects the action expected to maximize its performance measure using percepts and knowledge.

Inaccessible Environment

AGENT AND ENVIRONMENT TERMS

An environment where the agent cannot directly sense all information needed for a complete state description.

Internal State

AGENT AND ENVIRONMENT TERMS

The agent’s stored representation of aspects of the world that are not currently observable.

Model-Based Reflex Agent

AGENT AND ENVIRONMENT TERMS

A reflex agent that maintains an internal model of how the world changes and how actions affect it.

Multi-Agent Environment

AGENT AND ENVIRONMENT TERMS

An environment containing more than one agent, where agents may cooperate, compete, or affect each other.

Non-Deterministic Environment

AGENT AND ENVIRONMENT TERMS

An environment where an action may lead to different possible next states even from the same current state.

Non-Episodic Environment

AGENT AND ENVIRONMENT TERMS

An environment where current actions can affect later situations and future rewards.

Observable Environment

AGENT AND ENVIRONMENT TERMS

An environment where percepts reveal the full current state relevant to the agent’s decision.

Partially Observable Environment

AGENT AND ENVIRONMENT TERMS

An environment where the agent receives incomplete information and must reason with uncertainty.

PEAS

AGENT AND ENVIRONMENT TERMS

A task specification using Performance measure, Environment, Actuators, and Sensors to describe an agent problem.

Percept

AGENT AND ENVIRONMENT TERMS

The information an agent receives from its sensors at a particular moment.

Percept Sequence

AGENT AND ENVIRONMENT TERMS

The complete history of percepts an agent has received so far.

Percepts

AGENT AND ENVIRONMENT TERMS

The sensed inputs an agent receives from its environment.

Performance Measure of Agent

AGENT AND ENVIRONMENT TERMS

The success criteria used to judge how well an agent is performing its task.

Rational Agent

AGENT AND ENVIRONMENT TERMS

An agent that chooses actions expected to achieve the best outcome based on its knowledge and percepts.

Rationality

AGENT AND ENVIRONMENT TERMS

The quality of choosing sensible actions that are expected to produce useful results.

Robotic Agent

AGENT AND ENVIRONMENT TERMS

A physical agent that uses sensors such as cameras and effectors such as motors to act in the real world.

Sensor

AGENT AND ENVIRONMENT TERMS

A device or software input channel that lets an agent detect information about its environment.

Simple Reflex Agent

AGENT AND ENVIRONMENT TERMS

An agent that chooses actions using only the current percept and condition-action rules.

Single-Agent Environment

AGENT AND ENVIRONMENT TERMS

An environment in which one agent acts without strategic interaction with other agents.

Softbot

AGENT AND ENVIRONMENT TERMS

A software robot that acts inside digital environments such as websites, file systems, or databases.

Software Agent

AGENT AND ENVIRONMENT TERMS

An agent implemented as software that acts through encoded instructions, APIs, tools, or digital actions.

Static Environment

AGENT AND ENVIRONMENT TERMS

An environment that does not change while an agent is deciding what to do.

Turing Test Environment

AGENT AND ENVIRONMENT TERMS

A test setting where a human judge compares typed responses from a human and a machine.

Utility

AGENT AND ENVIRONMENT TERMS

A numeric preference score that represents how desirable a state or outcome is for an agent.

Utility-Based Agent

AGENT AND ENVIRONMENT TERMS

An agent that chooses actions by comparing the expected utility of possible outcomes.

World Model

AGENT AND ENVIRONMENT TERMS

An agent’s representation of how the environment behaves and how actions change it.

A* Search

SEARCH ALGORITHMS AND PROBLEM-SOLVING TERMS

A best-first search method that combines path cost so far with a heuristic estimate of remaining cost.

Admissibility

SEARCH ALGORITHMS AND PROBLEM-SOLVING TERMS

A property of a search algorithm or heuristic that guarantees an optimal solution when its assumptions hold.

Best First Search

SEARCH ALGORITHMS AND PROBLEM-SOLVING TERMS

A search strategy that expands the most promising node first according to an evaluation function.

Bidirectional Search

SEARCH ALGORITHMS AND PROBLEM-SOLVING TERMS

A search strategy that works forward from the start and backward from the goal until the two searches meet.

Branching Factor

SEARCH ALGORITHMS AND PROBLEM-SOLVING TERMS

The average number of child nodes generated from each node in a search space.

Breadth-First Search

SEARCH ALGORITHMS AND PROBLEM-SOLVING TERMS

A search strategy that explores all nodes at the current depth before moving to the next level.

Brute-Force Search

SEARCH ALGORITHMS AND PROBLEM-SOLVING TERMS

A search approach that explores possible states without using problem-specific guidance.

Closed List

SEARCH ALGORITHMS AND PROBLEM-SOLVING TERMS

A collection of nodes that a search algorithm has already expanded.

Cut-Off Depth

SEARCH ALGORITHMS AND PROBLEM-SOLVING TERMS

A maximum search depth used to prevent depth-first search from exploring endlessly.

Depth-First Search

SEARCH ALGORITHMS AND PROBLEM-SOLVING TERMS

A search strategy that follows one path as deeply as possible before backtracking.

Depth of Problem

SEARCH ALGORITHMS AND PROBLEM-SOLVING TERMS

The length of the shortest sequence of operators from an initial state to a goal state.

Edge

SEARCH ALGORITHMS AND PROBLEM-SOLVING TERMS

A connection in a problem-space graph that represents an operator or transition between states.

FIFO Queue

SEARCH ALGORITHMS AND PROBLEM-SOLVING TERMS

A first-in, first-out data structure commonly used to implement breadth-first search.

Goal State

SEARCH ALGORITHMS AND PROBLEM-SOLVING TERMS

The desired state that a search algorithm is trying to reach.

Greedy Best First Search

SEARCH ALGORITHMS AND PROBLEM-SOLVING TERMS

A search strategy that expands the node estimated to be closest to the goal.

Heuristic Evaluation Function

SEARCH ALGORITHMS AND PROBLEM-SOLVING TERMS

A function that estimates how close a state is to a goal or how costly the remaining path may be.

Heuristic Search

SEARCH ALGORITHMS AND PROBLEM-SOLVING TERMS

A search approach that uses domain knowledge to choose more promising paths first.

Informed Search

SEARCH ALGORITHMS AND PROBLEM-SOLVING TERMS

A search strategy that uses heuristic information about the problem to improve efficiency.

Initial State

SEARCH ALGORITHMS AND PROBLEM-SOLVING TERMS

The starting state from which a search problem begins.

Iterative Deepening Depth-First Search

SEARCH ALGORITHMS AND PROBLEM-SOLVING TERMS

A search method that repeatedly runs depth-first search with increasing depth limits until it finds a solution.

LIFO Stack

SEARCH ALGORITHMS AND PROBLEM-SOLVING TERMS

A last-in, first-out data structure commonly used to implement depth-first search.

Local Beam Search

SEARCH ALGORITHMS AND PROBLEM-SOLVING TERMS

A local search method that keeps several candidate states and repeatedly selects the best successors.

Local Maximum

SEARCH ALGORITHMS AND PROBLEM-SOLVING TERMS

A state that is better than nearby states but may not be the best solution overall.

Local Search Algorithm

SEARCH ALGORITHMS AND PROBLEM-SOLVING TERMS

A search method that improves a candidate solution by moving to neighboring solutions.

Node

SEARCH ALGORITHMS AND PROBLEM-SOLVING TERMS

A point in a search graph that represents a state or partial solution.

Objective Function

SEARCH ALGORITHMS AND PROBLEM-SOLVING TERMS

A scoring function that measures how good a candidate solution is.

Open List

SEARCH ALGORITHMS AND PROBLEM-SOLVING TERMS

A collection of generated nodes that a search algorithm has not expanded yet.

Operator

SEARCH ALGORITHMS AND PROBLEM-SOLVING TERMS

An action or transition rule that changes one problem state into another.

Path Cost

SEARCH ALGORITHMS AND PROBLEM-SOLVING TERMS

The accumulated cost of moving through a sequence of states or actions.

Problem Instance

SEARCH ALGORITHMS AND PROBLEM-SOLVING TERMS

A specific search problem defined by an initial state and a goal state.

Problem Space

SEARCH ALGORITHMS AND PROBLEM-SOLVING TERMS

The set of states and operators in which a search problem is solved.

Problem Space Graph

SEARCH ALGORITHMS AND PROBLEM-SOLVING TERMS

A graph representation of a search problem where nodes are states and edges are operators.

Pure Heuristic Search

SEARCH ALGORITHMS AND PROBLEM-SOLVING TERMS

A search method that expands nodes according to heuristic values and manages open and closed lists.

Search Algorithm

SEARCH ALGORITHMS AND PROBLEM-SOLVING TERMS

A method for exploring states and actions to find a solution to a problem.

Simulated Annealing

SEARCH ALGORITHMS AND PROBLEM-SOLVING TERMS

A local search method that sometimes accepts worse moves early so it can escape poor local solutions.

Single-Agent Pathfinding Problem

SEARCH ALGORITHMS AND PROBLEM-SOLVING TERMS

A problem where one agent must find a sequence of moves from a start state to a goal state.

Space Complexity

SEARCH ALGORITHMS AND PROBLEM-SOLVING TERMS

The maximum amount of memory an algorithm needs while solving a problem.

Successor

SEARCH ALGORITHMS AND PROBLEM-SOLVING TERMS

A new state that results from applying an operator to a current state.

Time Complexity

SEARCH ALGORITHMS AND PROBLEM-SOLVING TERMS

The amount of computation or number of generated nodes an algorithm may require.

Travelling Salesman Problem

SEARCH ALGORITHMS AND PROBLEM-SOLVING TERMS

An optimization problem that seeks the lowest-cost tour visiting each city once and returning to the start.

Uniform Cost Search

SEARCH ALGORITHMS AND PROBLEM-SOLVING TERMS

A search strategy that expands the path with the lowest accumulated cost first.

Crisp Input

Fuzzy Logic Terms

A precise numerical input value before it is converted into fuzzy membership levels.

Crisp Output

Fuzzy Logic Terms

A definite numerical result produced after fuzzy reasoning is converted back into a precise value.

Defuzzification

Fuzzy Logic Terms

The process of converting fuzzy output values into a single crisp output value.

Defuzzification Module

Fuzzy Logic Terms

The fuzzy-logic component that transforms fuzzy inference results into a usable crisp value.

Degree of Membership

Fuzzy Logic Terms

A number between 0 and 1 that shows how strongly an element belongs to a fuzzy set.

Fuzzification

Fuzzy Logic Terms

The process of converting crisp inputs into fuzzy sets using membership functions.

Fuzzification Module

Fuzzy Logic Terms

The fuzzy-logic component that maps precise input values into fuzzy linguistic categories.

Fuzzy Inference

Fuzzy Logic Terms

Reasoning over fuzzy inputs and rules to produce fuzzy conclusions.

Fuzzy Input

Fuzzy Logic Terms

Input represented with degrees of possibility rather than one exact true-or-false value.

Fuzzy Logic

Fuzzy Logic Terms

A reasoning method that handles partial truth and uncertainty instead of only true or false values.

Fuzzy Logic System

Fuzzy Logic Terms

A system that uses fuzzy sets, rules, and inference to produce useful outputs from uncertain inputs.

Fuzzy Output

Fuzzy Logic Terms

A fuzzy conclusion produced by an inference engine before defuzzification.

Fuzzy Rule Base

Fuzzy Logic Terms

A collection of IF-THEN rules used by a fuzzy system to make decisions.

Fuzzy Set

Fuzzy Logic Terms

A set where elements can partially belong with membership values between 0 and 1.

Gaussian Membership Function

Fuzzy Logic Terms

A smooth bell-shaped membership function used to represent gradual fuzzy categories.

IF-THEN Rule

Fuzzy Logic Terms

A rule that connects a condition with an action or conclusion, often used in fuzzy and expert systems.

Linguistic Term

Fuzzy Logic Terms

A word or phrase such as cold, warm, or hot that represents a fuzzy category.

Linguistic Variable

Fuzzy Logic Terms

A variable whose values are words or phrases instead of only numbers.

Membership Function

FUZZY LOGIC TERMS

A function that assigns each input value a degree of membership in a fuzzy set.

Singleton Membership Function

FUZZY LOGIC TERMS

A membership function that assigns membership at a single point or exact value.

Trapezoidal Membership Function

FUZZY LOGIC TERMS

A membership function shaped like a trapezoid, often used for ranges with a flat middle.

Triangular Membership Function

FUZZY LOGIC TERMS

A simple membership function shaped like a triangle, commonly used in fuzzy systems.

Universe of Discourse

FUZZY LOGIC TERMS

The complete range of values over which a fuzzy variable and its membership functions are defined.

Context-Free Grammar

NLP AND LINGUISTICS TERMS

A grammar where each rewrite rule has a single nonterminal symbol on the left side.

Discourse

NLP AND LINGUISTICS TERMS

The way connected sentences or utterances influence the meaning of each other.

Discourse Integration

NLP AND LINGUISTICS TERMS

The NLP step that interprets a sentence using the meaning of surrounding sentences.

Lexical Ambiguity

NLP AND LINGUISTICS TERMS

Uncertainty caused when a word can have multiple meanings or grammatical roles.

Lexical Analysis

NLP AND LINGUISTICS TERMS

The NLP step that breaks text into meaningful units such as paragraphs, sentences, and words.

Lexicon

NLP AND LINGUISTICS TERMS

The collection of words and phrases known in a language or NLP system.

Morpheme

NLP AND LINGUISTICS TERMS

The smallest meaningful unit in a language, such as a root, prefix, or suffix.

Morphology

NLP AND LINGUISTICS TERMS

The study of how words are built from smaller meaningful units.

Natural Language Processing

NLP AND LINGUISTICS TERMS

The AI field that helps computers understand, analyze, generate, or act on human language.

Noun Phrase

NLP AND LINGUISTICS TERMS

A group of words centered on a noun that can act as a subject, object, or complement.

Parse Tree

NLP AND LINGUISTICS TERMS

A tree structure that shows how a sentence is built from grammar rules.

Parsing

NLP AND LINGUISTICS TERMS

The process of analyzing sentence structure according to grammar rules.

Phonology

NLP AND LINGUISTICS TERMS

The study of how speech sounds are organized in a language.

Pragmatic Analysis

NLP AND LINGUISTICS TERMS

The NLP step that interprets what a speaker likely means using situation and world knowledge.

Pragmatics

NLP AND LINGUISTICS TERMS

The study of how context affects the meaning and use of language.

Referential Ambiguity

NLP AND LINGUISTICS TERMS

Uncertainty about what a pronoun or reference points to in a sentence or conversation.

Rewrite Rule

NLP AND LINGUISTICS TERMS

A grammar rule that expands one symbol into a sequence of other symbols.

Semantic Analysis

NLP AND LINGUISTICS TERMS

The NLP step that checks meaning and maps sentence structure to concepts in a domain.

Semantics

NLP AND LINGUISTICS TERMS

The study of word and sentence meaning.

Sentence Planning

NLP AND LINGUISTICS TERMS

The NLG step that selects words, phrasing, tone, and sentence organization.

Speaker-Dependent Speech Recognition

NLP AND LINGUISTICS TERMS

Speech recognition that is trained or tuned for a particular speaker’s voice.

Speaker-Independent Speech Recognition

NLP AND LINGUISTICS TERMS

Speech recognition designed to work across many speakers without speaker-specific training.

Speech

NLP AND LINGUISTICS TERMS

Spoken language used as input to or output from an AI or NLP system.

Speech Recognition

NLP AND LINGUISTICS TERMS

The process of identifying what words were spoken from an audio signal.

Syntactic Ambiguity

NLP AND LINGUISTICS TERMS

Uncertainty caused when a sentence can be parsed into more than one structure.

Syntactic Analysis

NLP AND LINGUISTICS TERMS

The NLP step that checks grammar and the structural relationships among words.

Syntax

NLP AND LINGUISTICS TERMS

The rules for arranging words into valid phrases and sentences.

Terminal Symbol

NLP AND LINGUISTICS TERMS

A grammar symbol that appears in the final sentence and is not expanded further.

Text Planning

NLP AND LINGUISTICS TERMS

The NLG step that decides what content should be included in the generated text.

Text Realization

NLP AND LINGUISTICS TERMS

The NLG step that turns a sentence plan into actual sentence structure and wording.

Top-Down Parser

NLP AND LINGUISTICS TERMS

A parser that starts with a high-level grammar symbol and rewrites it until it matches the input sentence.

Translation Engine

NLP AND LINGUISTICS TERMS

A system component that converts text from a source language into a target language.

Verb Phrase

NLP AND LINGUISTICS TERMS

A group of words centered on a verb that describes an action, state, or event.

Voice Recognition

NLP AND LINGUISTICS TERMS

The process of identifying who is speaking from characteristics of a voice.

World Knowledge

NLP AND LINGUISTICS TERMS

General knowledge about people, objects, events, and situations used to interpret language.

Written Text

NLP AND LINGUISTICS TERMS

Language represented as characters or words rather than speech audio.

Backward Chaining

EXPERT SYSTEMS TERMS

An inference strategy that starts with a possible conclusion and works backward to find supporting facts.

Blackboard

EXPERT SYSTEMS TERMS

A shared memory area used by cooperating expert systems or components to exchange information.

Declarative Knowledge

EXPERT SYSTEMS TERMS

Knowledge that states facts, relationships, or truths about a domain.

Domain Expert

EXPERT SYSTEMS TERMS

A person with deep practical knowledge in the problem area an expert system is designed to solve.

Domain Knowledge

EXPERT SYSTEMS TERMS

Specialized knowledge about a particular field, task, or problem area.

Expert System

EXPERT SYSTEMS TERMS

A computer application that solves complex domain problems using expert-level knowledge and inference rules.

Expert System Development Environment

EXPERT SYSTEMS TERMS

The hardware, programming tools, languages, and databases used to build an expert system.

Explanation Facility

EXPERT SYSTEMS TERMS

An expert-system feature that explains how a recommendation or conclusion was reached.

Factual Knowledge

EXPERT SYSTEMS TERMS

Domain information that is widely accepted as true by experts or scholars.

Forward Chaining

EXPERT SYSTEMS TERMS

An inference strategy that starts from known facts and applies rules forward to reach conclusions.

Heuristic Knowledge

EXPERT SYSTEMS TERMS

Practical rule-of-thumb knowledge based on experience, judgment, evaluation, or experimentation.

Heuristics

EXPERT SYSTEMS TERMS

Experience-based rules or shortcuts that guide problem solving when exact methods are costly or unavailable.

Human Expert

EXPERT SYSTEMS TERMS

A knowledgeable person whose expertise is captured or consulted when building an expert system.

Inference Engine

EXPERT SYSTEMS TERMS

The expert-system component that applies rules to facts and knowledge to derive conclusions.

JESS

EXPERT SYSTEMS TERMS

Java Expert System Shell, a Java-based shell used to build rule-based expert systems.

Knowledge Acquisition

EXPERT SYSTEMS TERMS

The process of gathering domain knowledge from experts, documents, cases, or observations for an expert system.

Knowledge-Based Expert System

EXPERT SYSTEMS TERMS

An expert system that uses a knowledge base and inference engine to solve domain-specific problems.

Knowledge Engineer

EXPERT SYSTEMS TERMS

A specialist who collects, organizes, and encodes expert knowledge for an AI or expert system.

Knowledge Engineering

EXPERT SYSTEMS TERMS

The discipline of acquiring, structuring, and representing expert knowledge for AI systems.

Knowledge Representation

EXPERT SYSTEMS TERMS

The method used to organize knowledge so an AI system can reason over it.

LISP

EXPERT SYSTEMS TERMS

A symbolic programming language historically used for building AI and expert-system applications.

Procedural Knowledge

EXPERT SYSTEMS TERMS

Knowledge about how to perform steps, methods, or procedures to complete a task.

PROLOG

EXPERT SYSTEMS TERMS

A logic programming language historically used for rule-based reasoning and AI applications.

Prototype

EXPERT SYSTEMS TERMS

An early working version of a system used for testing, feedback, and refinement.

Rapid Prototyping

EXPERT SYSTEMS TERMS

A development approach that quickly builds working models so users and engineers can refine requirements.

Rule

EXPERT SYSTEMS TERMS

A knowledge representation pattern that links conditions to conclusions or actions, often written as IF-THEN-ELSE.

Rule-Based Expert System

EXPERT SYSTEMS TERMS

An expert system that reasons by repeatedly applying IF-THEN rules to facts.

Rule Conflict

EXPERT SYSTEMS TERMS

A situation where multiple expert-system rules apply at the same time and the inference engine must choose among them.

Shell

EXPERT SYSTEMS TERMS

An expert-system framework that provides inference, interface, and explanation features without domain knowledge.

User Interface

EXPERT SYSTEMS TERMS

The part of an expert system that lets users ask questions, provide inputs, and receive explanations.

Vidwan

EXPERT SYSTEMS TERMS

An expert-system shell developed in India for encoding knowledge as IF-THEN rules.

Analog Signal

ROBOTICS AND COMPUTER VISION TERMS

A continuous signal, such as speech waveform or image data, used as raw input in robotic systems.

Autonomous Robot

ROBOTICS AND COMPUTER VISION TERMS

A robot that can operate independently without direct external control.

Ball Wheel

ROBOTICS AND COMPUTER VISION TERMS

An omnidirectional spherical wheel that can move in multiple directions.

Biometrics

ROBOTICS AND COMPUTER VISION TERMS

The use of measurable human traits such as face, fingerprint, or voice for identification.

Camera Sensor

ROBOTICS AND COMPUTER VISION TERMS

A vision sensor that captures images or video so a robot can perceive visual information.

Castor Wheel

ROBOTICS AND COMPUTER VISION TERMS

A wheel that rotates around both its axle and an offset steering joint for easier turning.

Computer Vision

ROBOTICS AND COMPUTER VISION TERMS

The AI field that extracts, analyzes, and understands useful information from images or video.

Electric Motor

ROBOTICS AND COMPUTER VISION TERMS

A robotic actuator that converts electrical energy into rotational movement.

Estimating Position

ROBOTICS AND COMPUTER VISION TERMS

A computer-vision task that determines an object's position relative to a camera or scene.

Face Detection

ROBOTICS AND COMPUTER VISION TERMS

A computer-vision task that locates human faces in images or video.

Gait

ROBOTICS AND COMPUTER VISION TERMS

A repeated pattern of leg movements used by a walking robot to move.

Gesture Analysis

ROBOTICS AND COMPUTER VISION TERMS

A computer-vision task that interprets human body or hand movements.

Image Acquisition Device

ROBOTICS AND COMPUTER VISION TERMS

Hardware such as a camera that captures visual input for a computer-vision system.

Legged Locomotion

ROBOTICS AND COMPUTER VISION TERMS

Robot movement using one or more legs, often useful on uneven terrain.

Mechanical Construction

ROBOTICS AND COMPUTER VISION TERMS

The physical form, structure, and mechanical design of a robot.

Muscle Wire

ROBOTICS AND COMPUTER VISION TERMS

A wire actuator that contracts when electric current passes through it.

Object Recognition

ROBOTICS AND COMPUTER VISION TERMS

A computer-vision task that identifies objects in images or video.

OCR

ROBOTICS AND COMPUTER VISION TERMS

Optical character recognition, the process of converting scanned or image-based text into editable text.

Omni-Wheel

ROBOTICS AND COMPUTER VISION TERMS

A wheel with rollers that allows sideways or multidirectional movement.

Piezo Motor

ROBOTICS AND COMPUTER VISION TERMS

A motor that uses piezoelectric materials to create precise motion.

Pneumatic Air Muscle

ROBOTICS AND COMPUTER VISION TERMS

A soft actuator that contracts when air pressure changes.

Power Supply

ROBOTICS AND COMPUTER VISION TERMS

The battery, solar, hydraulic, pneumatic, or electrical source that powers a robot.

Remote Sensing

ROBOTICS AND COMPUTER VISION TERMS

Collecting information about an object or area from a distance using sensors or imaging devices.

Robot

ROBOTICS AND COMPUTER VISION TERMS

An artificial agent that acts in the physical world through sensors, computation, and effectors.

Robot Locomotion

ROBOTICS AND COMPUTER VISION TERMS

The mechanism that enables a robot to move through its environment.

Robot Program

ROBOTICS AND COMPUTER VISION TERMS

The software logic that determines what a robot does, when it acts, and how it performs actions.

Robotics

ROBOTICS AND COMPUTER VISION TERMS

The AI and engineering field focused on designing, building, and controlling robots.

Slip/Skid Locomotion

ROBOTICS AND COMPUTER VISION TERMS

Robot movement using tracks or skid steering, often seen in tank-like vehicles.

Standard Wheel

ROBOTICS AND COMPUTER VISION TERMS

A wheel that rotates around its axle and supports straightforward rolling motion.

Swedish Wheel

ROBOTICS AND COMPUTER VISION TERMS

An omni-wheel design with angled rollers that supports multidirectional movement.

Tactile Sensor

ROBOTICS AND COMPUTER VISION TERMS

A sensor that detects touch, pressure, or contact forces.

Tracked Locomotion

ROBOTICS AND COMPUTER VISION TERMS

Robot movement using continuous tracks for stability and traction.

Ultrasonic Motor

ROBOTICS AND COMPUTER VISION TERMS

A motor that uses ultrasonic vibration to produce motion, often for precise positioning.

Vision Sensor

ROBOTICS AND COMPUTER VISION TERMS

A sensor that captures visual data, allowing a robot or AI system to perceive its surroundings.

Wheeled Locomotion

ROBOTICS AND COMPUTER VISION TERMS

Robot movement using wheels, usually more power-efficient and easier to implement than legged motion.

Activation

NEURAL NETWORK AND PROBABILISTIC AI TERMS

The output value of a neuron or node after it processes its inputs.

Artificial Neural Network (ANN)

NEURAL NETWORK AND PROBABILISTIC AI TERMS

A computing system made of connected processing nodes inspired by biological neural networks.

Axon

NEURAL NETWORK AND PROBABILISTIC AI TERMS

A biological neuron structure that carries signals away from the neuron body to other neurons.

Back Propagation Algorithm

NEURAL NETWORK AND PROBABILISTIC AI TERMS

A supervised learning algorithm that updates neural-network weights by propagating error backward.

Bayes Net

NEURAL NETWORK AND PROBABILISTIC AI TERMS

Another name for a Bayesian network used to model probabilistic relationships.

Bayesian Network

NEURAL NETWORK AND PROBABILISTIC AI TERMS

A directed acyclic graph that represents probabilistic relationships among random variables.

Belief Network

NEURAL NETWORK AND PROBABILISTIC AI TERMS

Another name for a Bayesian network that reasons about uncertain domains.

Biological Neuron

NEURAL NETWORK AND PROBABILISTIC AI TERMS

A nerve cell that inspires the structure of artificial neural-network nodes.

Boolean Node

NEURAL NETWORK AND PROBABILISTIC AI TERMS

A Bayesian-network node that represents a proposition with true or false values.

Child Node

NEURAL NETWORK AND PROBABILISTIC AI TERMS

A node that is directly influenced by another node in a Bayesian network.

Conditional Probability

NEURAL NETWORK AND PROBABILISTIC AI TERMS

The probability of an event given that another related event or condition is known.

Conditional Probability Table (CPT)

NEURAL NETWORK AND PROBABILISTIC AI TERMS

A table that lists the probability of a node's values for each combination of parent-node values.

Content Addressable Memory

NEURAL NETWORK AND PROBABILISTIC AI TERMS

A memory style where information is retrieved by content similarity rather than by an explicit address.

Dendrite

NEURAL NETWORK AND PROBABILISTIC AI TERMS

A biological neuron branch that receives signals from other neurons.

Directed Acyclic Graph (DAG)

NEURAL NETWORK AND PROBABILISTIC AI TERMS

A directed graph with no cycles, often used to represent dependency structures.

Discrete Random Variable

NEURAL NETWORK AND PROBABILISTIC AI TERMS

A random variable that can take one value from a limited set of possible values.

Feedback ANN

NEURAL NETWORK AND PROBABILISTIC AI TERMS

A neural network topology where information can loop back through previous units.

FeedForward ANN

NEURAL NETWORK AND PROBABILISTIC AI TERMS

A neural network topology where information moves in one direction from input to output.

Instantiation

NEURAL NETWORK AND PROBABILISTIC AI TERMS

A specific assignment of values to variables or parent nodes in a probabilistic model.

Integral Value Node

NEURAL NETWORK AND PROBABILISTIC AI TERMS

A Bayesian-network node that represents integer-valued states such as age ranges or counts.

Link

NEURAL NETWORK AND PROBABILISTIC AI TERMS

A connection between neural-network nodes that carries a weighted signal.

Neurocomputer

NEURAL NETWORK AND PROBABILISTIC AI TERMS

A computing system designed around neural-network principles.

Node Value

NEURAL NETWORK AND PROBABILISTIC AI TERMS

The numeric output or activation value produced by a neural-network node.

Ordered Value Node

NEURAL NETWORK AND PROBABILISTIC AI TERMS

A Bayesian-network node whose possible values have a meaningful order, such as low, medium, and high.

Parent Node

NEURAL NETWORK AND PROBABILISTIC AI TERMS

A node that directly influences another node in a Bayesian network.

Pattern Classification

NEURAL NETWORK AND PROBABILISTIC AI TERMS

The task of assigning an input pattern to one of several categories.

Pattern Generation

NEURAL NETWORK AND PROBABILISTIC AI TERMS

The task of producing new examples that follow learned patterns.

Pattern Recognition

NEURAL NETWORK AND PROBABILISTIC AI TERMS

The task of detecting meaningful patterns in data such as images, speech, text, or signals.

Probabilistic Dependency

NEURAL NETWORK AND PROBABILISTIC AI TERMS

A relationship where one variable changes the likelihood of another variable.

Processing Element

NEURAL NETWORK AND PROBABILISTIC AI TERMS

A simple computing unit in a neural network that receives inputs, performs a calculation, and sends output.

Random Variable

NEURAL NETWORK AND PROBABILISTIC AI TERMS

A variable whose value is uncertain and described using probabilities.

Reinforcement Learning

NEURAL NETWORK AND PROBABILISTIC AI TERMS

A learning method where an agent improves decisions through rewards or penalties from interaction.

Supervised Learning

NEURAL NETWORK AND PROBABILISTIC AI TERMS

A learning method that trains from examples paired with known correct answers.

Unsupervised Learning

NEURAL NETWORK AND PROBABILISTIC AI TERMS

A learning method that finds hidden patterns in data without labeled answers.

Self-Improving AI

AI ETHICS, RISK, AND HISTORICAL TERMS

An AI system that can improve its own behavior or capability over time, raising both performance and control concerns.

Threat to Human Dignity

AI ETHICS, RISK, AND HISTORICAL TERMS

The concern that AI may replace people in roles where human judgment, care, or ethical responsibility should remain central.

Threat to Privacy

AI ETHICS, RISK, AND HISTORICAL TERMS

The risk that AI systems may collect, infer, or expose sensitive personal information from speech, text, or behavior.

Threat to Safety

AI ETHICS, RISK, AND HISTORICAL TERMS

The risk that powerful or poorly controlled AI systems may cause unintended harmful consequences.

Turing Test

AI ETHICS, RISK, AND HISTORICAL TERMS

A test of machine intelligence where a judge tries to distinguish a machine's responses from a human's responses.

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